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Chapter 5

Namespace Documentation

5.1 Config Namespace Reference

Namespace for all configuration functions that initialize objects based on configuration file.

Typedefs

- using ConfigCollection = ConfigReader::ConfigCollection
- using SettingError = std::invalid_argument

Functions

- void InitializeScreenOutput (const ConfigCollection &theCfg)

  Initialize screen output options.
- std::unique_ptr< Metric > GetMetric (const ConfigCollection &theCfg)

  Use configuration to create the correct Metric with specified parameters.
- std::unique_ptr< Source > GetSource (const ConfigCollection &theCfg, const Metric *const theMetric)

  Use configuration to create the correct Source with specified parameters.
- void InitializeDiagnostics (const ConfigCollection &theCfg, DiagBitflag &alldiags, DiagBitflag &valdiag, const Metric *const theMetric)

  Use configuration to set the Diagnostics bitflag appropriately; initialize all DiagnosticOptions for all Diagnostics that are turned on; and set bitflag for diagnostic to be used for coarseness evaluating in Mesh.
- void InitializeTerminations (const ConfigCollection &theCfg, TermBitflag &allterms, const Metric *const the↩ Metric)

  Use configuration to set the Termination bitflag appropriately; and initialize all TerminationOptions for all Terminations that are turned on.
- std::unique_ptr< ViewScreen > GetViewScreen (const ConfigCollection &theCfg, DiagBitflag valdiag, const Metric *const theMetric)

  Use configuration to create the ViewScreen object with specified parameters.
- std::unique_ptr< Mesh > GetMesh (const ConfigCollection &theCfg, DiagBitflag valdiag)

  Use configuration to create the Mesh object with specified parameters.
- GeodesicIntegratorFunc GetGeodesicIntegrator (const ConfigCollection &theCfg)

  Use configuration to set the GeodesicIntegratorFunc pointer to the correct integrator function.
- std::unique_ptr< GeodesicOutputHandler > GetOutputHandler (const ConfigCollection &theCfg, DiagBitflag alldiags, DiagBitflag valdiag, std::string FirstLineInfo)

  Use configuration to create the GeodesicOutputHandler object with specified parameters.

Variables

- constexpr auto Output_Important_Default = OutputLevel::Level_0_WARNING
- constexpr auto Output_Other_Default = OutputLevel::Level_1_PROC

5.1.1 Detailed Description

Namespace for all configuration functions that initialize objects based on configuration file.

5.1.2 Typedef Documentation

5.1.2.1 ConfigCollection

```
using Config::ConfigCollection = ConfigReader::ConfigCollection
```

5.1.2.2 SettingError

```
using Config::SettingError = std::invalid_argument
```

5.1.3 Function Documentation

5.1.3.1 GetGeodesicIntegrator()

```
GeodesicIntegratorFunc Config::GetGeodesicIntegrator (
            const ConfigCollection & theCfg)
```

Use configuration to set the GeodesicIntegratorFunc pointer to the correct integrator function.

Parameters

theCfg	Pointer to the configuration object

Returns

GeodesicIntegratorFunc function

5.1.3.2 GetMesh()

```
std::unique_ptr< Mesh > Config::GetMesh (
            const ConfigCollection & theCfg,
            DiagBitflag valdiag)
```

Use configuration to create the Mesh object with specified parameters.

Parameters

theCfg	Pointer to the configuration object
valdiag	Diagnostic bitflag used for coarseness evaluation in the Mesh

Returns

Pointer to the Mesh object

5.1.3.3 GetMetric()

```
std::unique_ptr< Metric > Config::GetMetric (
            const ConfigCollection & theCfg)
```

Use configuration to create the correct Metric with specified parameters.

Parameters

theCfg	Pointer to the configuration object

Returns

Metric pointer

5.1.3.4 GetOutputHandler()

```
std::unique_ptr< GeodesicOutputHandler > Config::GetOutputHandler (
            const ConfigCollection & theCfg,
            DiagBitflag alldiags,
            DiagBitflag valdiag,
            std::string FirstLineInfo)
```

Use configuration to create the GeodesicOutputHandler object with specified parameters.

Parameters

theCfg	Pointer to the configuration object
alldiags	Diagnostics bitflags
valdiag	Diagnostic bitflag used for coarseness evaluation in the Mesh
FirstLineInfo	String to be written as the first line in the output file

Returns

Pointer to the GeodesicOutputHandler object

5.1.3.5 GetSource()

```
std::unique_ptr< Source > Config::GetSource (
            const ConfigCollection & theCfg,
            const Metric *const theMetric)
```

Use configuration to create the correct Source with specified parameters.

Parameters

theCfg	Pointer to the configuration object
theMetric	Pointer to the Metric object

Returns

Pointer to the Source object

5.1.3.6 GetViewScreen()

```
std::unique_ptr< ViewScreen > Config::GetViewScreen (
            const ConfigCollection & theCfg,
            DiagBitflag valdiag,
            const Metric *const theMetric)
```

Use configuration to create the ViewScreen object with specified parameters.

Parameters

theCfg	Pointer to the configuration object
valdiag	Diagnostic bitflag used for coarseness evaluation in the Mesh
theMetric	Pointer to the Metric object

Returns

Pointer to the ViewScreen object

5.1.3.7 InitializeDiagnostics()

```
void Config::InitializeDiagnostics (
            const ConfigCollection & theCfg,
            DiagBitflag & alldiags,
            DiagBitflag & valdiag,
            const Metric *const theMetric)
```

Use configuration to set the Diagnostics bitflag appropriately; initialize all DiagnosticOptions for all Diagnostics that are turned on; and set bitflag for diagnostic to be used for coarseness evaluating in Mesh.

Parameters

theCfg	Pointer to the configuration object
alldiags	Pointer to the diagnostics bitflags
valdiag	Pointer to the diagnostics used for coarseness evaluation in the Mesh
theMetric	Pointer to the Metric object

5.1.3.8 InitializeScreenOutput()

```
void Config::InitializeScreenOutput (
            const ConfigCollection & theCfg)
```

Initialize screen output options.

Parameters

theCfg	Pointer to the configuration object

5.1.3.9 InitializeTerminations()

```
void Config::InitializeTerminations (
            const ConfigCollection & theCfg,
            TermBitflag & allterms,
            const Metric *const theMetric)
```

Use configuration to set the Termination bitflag appropriately; and initialize all TerminationOptions for all Terminations that are turned on.

Parameters

theCfg	Pointer to the configuration object
allterms	Pointer to the termination bitflags
theMetric	Pointer to the Metric object

5.1.4 Variable Documentation

5.1.4.1 Output_Important_Default

```
auto Config::Output_Important_Default = OutputLevel::Level_0_WARNING  [constexpr]
```

5.1.4.2 Output_Other_Default

```
auto Config::Output_Other_Default = OutputLevel::Level_1_PROC  [constexpr]
```

5.2 ConfigReader Namespace Reference

Namespace to put all ConfigReader objects.

Classes

- class ConfigCollection
- class ConfigReaderException

Variables

- const long long MaxStreamSize {std::numeric_limits<std::streamsize>::max()}

5.2.1 Detailed Description

Namespace to put all ConfigReader objects.

5.2.2 Variable Documentation

5.2.2.1 MaxStreamSize

```
const long long ConfigReader::MaxStreamSize {std::numeric_limits<std::streamsize>::max()}
```

5.3 Integrators Namespace Reference

Functions

- std::string GetFullIntegratorDescription ()

  Description string getter for the Integrator.
- real GetAdaptiveStep (Point curpos, OneIndex curvel)

  Determine the (affine parameter) step size to take.
- void IntegrateGeodesicStep_RK4 (Point curpos, OneIndex curvel, Point &nextpos, OneIndex &nextvel, real &stepsize, const Metric *theMetric, const Source *theSource)

  Integrate the geodesic equation by one step using Runge-Kutta-4.
- void IntegrateGeodesicStep_Verlet (Point curpos, OneIndex curvel, Point &nextpos, OneIndex &nextvel, real &stepsize, const Metric *theMetric, const Source *theSource)

  Integrate the geodesic equation by one step using the Verlet algorithm.

Variables

- constexpr real delta_nodiv0 = 1e-20

  This is used to avoid dividing by zero.
- real Derivative_hval {1e-7}

  The spacing used in stencils for the calculation of numerical derivatives (e.g. in Christoffel symbols)
- std::string IntegratorDescription {"Verlet"}

  The name of the integrator selected.
- real epsilon {0.03}

  This is the base step size to be taken (the integrator will adapt this if necessary)
- real SmallestPossibleStepsize {1e-12}

  The affine parameter must always go forward by at least this amount.
- real VerletVelocityTolerance {-1.0}

5.3.1 Function Documentation

5.3.1.1 GetAdaptiveStep()

```
real Integrators::GetAdaptiveStep (
            Point curpos,
            OneIndex curvel)
```

Determine the (affine parameter) step size to take.

algorithm as in Raptor (eqs (21)-(24)), which is taken from Noble et al. (2007) & Dolence et al. (2009)

Parameters

curpos	current position
curvel	current velocity

Returns

real

5.3.1.2 GetFullIntegratorDescription()

```
std::string Integrators::GetFullIntegratorDescription ()
```

Description string getter for the Integrator.

Returns

std::string

5.3.1.3 IntegrateGeodesicStep_RK4()

```
void Integrators::IntegrateGeodesicStep_RK4 (
            Point curpos,
            OneIndex curvel,
            Point & nextpos,
            OneIndex & nextvel,
            real & stepsize,
            const Metric * theMetric,
            const Source * theSource)
```

Integrate the geodesic equation by one step using Runge-Kutta-4.

Parameters

curpos	current position
curvel	current velocity
nextpos	reference to next position
nextvel	reference to next velocity
stepsize	reference to the stepsize
theMetric	pointer to the Metric object
theSource	pointer to the Source object

5.3.1.4 IntegrateGeodesicStep_Verlet()

```
void Integrators::IntegrateGeodesicStep_Verlet (
            Point curpos,
            OneIndex curvel,
            Point & nextpos,
            OneIndex & nextvel,
            real & stepsize,
            const Metric * theMetric,
            const Source * theSource)
```

Integrate the geodesic equation by one step using the Verlet algorithm.

Parameters

curpos	current position
curvel	current velocity
nextpos	reference to next position
nextvel	reference to next velocity
stepsize	reference to the stepsize
theMetric	pointer to the Metric object
theSource	pointer to the Source object

5.3.2 Variable Documentation

5.3.2.1 delta_nodiv0

```
real Integrators::delta_nodiv0 = 1e-20  [constexpr]
```

This is used to avoid dividing by zero.

5.3.2.2 Derivative_hval

```
real Integrators::Derivative_hval {1e-7}  [inline]
```

The spacing used in stencils for the calculation of numerical derivatives (e.g. in Christoffel symbols)

5.3.2.3 epsilon

```
real Integrators::epsilon {0.03}  [inline]
```

This is the base step size to be taken (the integrator will adapt this if necessary)

5.3.2.4 IntegratorDescription

```
std::string Integrators::IntegratorDescription {"Verlet"}  [inline]
```

The name of the integrator selected.

5.3.2.5 SmallestPossibleStepsize

```
real Integrators::SmallestPossibleStepsize {1e-12}  [inline]
```

The affine parameter must always go forward by at least this amount.

5.3.2.6 VerletVelocityTolerance

```
real Integrators::VerletVelocityTolerance {-1.0}  [inline]
```

Tolerance on the change in velocity in the iteration loop of the Verlet integration algorithm (default -1.0: no iteration; any positive value enables the (uncapped) fixed-point iteration)

5.4 tk Namespace Reference

Namespaces

- namespace internal

Classes

- class spline

5.5 tk::internal Namespace Reference

Classes

- class band_matrix

5.6 Utilities Namespace Reference

Namespace that contains our utility functions.

Classes

- class Timer

  Timer class to keep track of elapsed time.

Functions

- std::string GetTimeStampString ()

  Get a string of the current time (in a format that can be used to append to file names)
- std::vector< std::string > GetDiagNameStrings (DiagBitflag alldiags, DiagBitflag valdiag)

  Helper function to get all Diagnostic Names (for outputting to files)
- std::string GetFirstLineInfoString (const Metric *theMetric, const Source *theSource, DiagBitflag alldiags, DiagBitflag valdiag, TermBitflag allterms, const ViewScreen *theView)

  This returns the full string to be written to every output file as its first line.

5.6.1 Detailed Description

Namespace that contains our utility functions.

5.6.2 Function Documentation

5.6.2.1 GetDiagNameStrings()

```
std::vector< std::string > Utilities::GetDiagNameStrings (
            DiagBitflag alldiags,
            DiagBitflag valdiag)
```

Helper function to get all Diagnostic Names (for outputting to files)

Parameters

alldiags	bitflag for all Diagnostics
valdiag	bitflag for the value diagnostic

Returns

std::vector<std::string>

5.6.2.2 GetFirstLineInfoString()

```
std::string Utilities::GetFirstLineInfoString (
            const Metric * theMetric,
            const Source * theSource,
            DiagBitflag alldiags,
            DiagBitflag valdiag,
            TermBitflag allterms,
            const ViewScreen * theView)
```

This returns the full string to be written to every output file as its first line.

Parameters

theMetric	pointer to the Metric object
theSource	pointer to the Source object
alldiags	bitflag for all Diagnostics
valdiag	bitflag for the value diagnostic
allterms	bitflag for all Terminations
theView	pointer to the ViewScreen object

Returns

std::string

5.6.2.3 GetTimeStampString()

```
std::string Utilities::GetTimeStampString ()
```

Get a string of the current time (in a format that can be used to append to file names)

Returns

std::string

Chapter 6

Class Documentation

6.1 tk::internal::band_matrix Class Reference

`#include <Spline.h>`

Public Member Functions

- band_matrix ()
- band_matrix (int dim, int n_u, int n_l)
- ~band_matrix ()
- void resize (int dim, int n_u, int n_l)
- int dim () const
- int num_upper () const
- int num_lower () const
- double & operator() (int i, int j)
- double operator() (int i, int j) const
- double & saved_diag (int i)
- double saved_diag (int i) const
- void lu_decompose ()
- std::vector< double > r_solve (const std::vector< double > &b) const
- std::vector< double > l_solve (const std::vector< double > &b) const
- std::vector< double > lu_solve (const std::vector< double > &b, bool is_lu_decomposed=false)

Private Attributes

- std::vector< std::vector< double > > m_upper
- std::vector< std::vector< double > > m_lower

6.1.1 Constructor & Destructor Documentation

6.1.1.1 band_matrix() [1/2]

`tk::internal::band_matrix::band_matrix () [inline]`

6.1.1.2 band_matrix() [2/2]

```
tk::internal::band_matrix::band_matrix (
            int dim,
            int n_u,
            int n_l)
```

6.1.1.3 ~band_matrix()

```
tk::internal::band_matrix::~band_matrix ()  [inline]
```

6.1.2 Member Function Documentation

6.1.2.1 dim()

```
int tk::internal::band_matrix::dim () const
```

6.1.2.2 l_solve()

```
std::vector< double > tk::internal::band_matrix::l_solve (
            const std::vector< double > & b) const
```

6.1.2.3 lu_decompose()

```
void tk::internal::band_matrix::lu_decompose ()
```

6.1.2.4 lu_solve()

```
std::vector< double > tk::internal::band_matrix::lu_solve (
            const std::vector< double > & b,
            bool is_lu_decomposed = false)
```

6.1.2.5 num_lower()

```
int tk::internal::band_matrix::num_lower () const
```

6.1.2.6 num_upper()

```
int tk::internal::band_matrix::num_upper () const
```

6.1.2.7 operator()() [1/2]

```
double & tk::internal::band_matrix::operator() (
            int i,
            int j)
```

6.1.2.8 operator()() [2/2]

```
double tk::internal::band_matrix::operator() (
            int i,
            int j) const
```

6.1.2.9 r_solve()

```
std::vector< double > tk::internal::band_matrix::r_solve (
            const std::vector< double > & b) const
```

6.1.2.10 resize()

```
void tk::internal::band_matrix::resize (
            int dim,
            int n_u,
            int n_l)
```

6.1.2.11 saved_diag() [1/2]

```
double & tk::internal::band_matrix::saved_diag (
            int i)
```

6.1.2.12 saved_diag() [2/2]

```
double tk::internal::band_matrix::saved_diag (
            int i) const
```

6.1.3 Member Data Documentation

6.1.3.1 m_lower

```
std::vector<std::vector<double> > tk::internal::band_matrix::m_lower  [private]
```

6.1.3.2 m_upper

```
std::vector<std::vector<double> > tk::internal::band_matrix::m_upper  [private]
```

The documentation for this class was generated from the following files:

- FOORT/src/Spline.h
- FOORT/src/Spline.cpp

6.2 BicubicSplineInterpolator Class Reference

`#include <Interpolator.h>`

Public Member Functions

- BicubicSplineInterpolator (const std::string x_file="RotatingBosonStar/x.txt", const std::string th_↩ file="RotatingBosonStar/theta.txt")
- void set_grid (const Grid ∗grid)
- double interpolate (double p_x, double p_theta, bool allow_extrapolation=false) const

Private Member Functions

- void compute_spline_coefficients (const Grid ∗grid)
- double estimate_dx (const Grid ∗grid, size_t j, size_t i) const
- double estimate_dy (const Grid ∗grid, size_t j, size_t i) const
- double estimate_dxdy (const Grid ∗grid, size_t j, size_t i) const
- void compute_cell_coefficients (std::array< double, 16 > &coeffs, double f00, double f10, double f01, double f11, double fx00, double fx10, double fx01, double fx11, double fy00, double fy10, double fy01, double fy11, double fxy00, double fxy10, double fxy01, double fxy11) const

Private Attributes

- std::vector< double > m_x
- std::vector< double > m_theta
- std::vector< std::array< double, 16 > > m_coeffs

6.2.1 Constructor & Destructor Documentation

6.2.1.1 BicubicSplineInterpolator()

```
BicubicSplineInterpolator::BicubicSplineInterpolator (
            const std::string x_file = "RotatingBosonStar/x.txt",
            const std::string th_file = "RotatingBosonStar/theta.txt")  [inline]
```

6.2.2 Member Function Documentation

6.2.2.1 compute_cell_coefficients()

```
void BicubicSplineInterpolator::compute_cell_coefficients (
            std::array< double, 16 > & coeffs,
            double f00,
            double f10,
            double f01,
            double f11,
            double fx00,
            double fx10,
            double fx01,
            double fx11,
            double fy00,
            double fy10,
            double fy01,
            double fy11,
            double fxy00,
            double fxy10,
            double fxy01,
            double fxy11) const  [private]
```

6.2.2.2 compute_spline_coefficients()

```
void BicubicSplineInterpolator::compute_spline_coefficients (
            const Grid * grid)  [private]
```

6.2.2.3 estimate_dx()

```
double BicubicSplineInterpolator::estimate_dx (
            const Grid * grid,
            size_t j,
            size_t i) const  [private]
```

6.2.2.4 estimate_dxdy()

```
double BicubicSplineInterpolator::estimate_dxdy (
            const Grid * grid,
            size_t j,
            size_t i) const  [private]
```

6.2.2.5 estimate_dy()

```
double BicubicSplineInterpolator::estimate_dy (
            const Grid * grid,
            size_t j,
            size_t i) const  [private]
```

6.2.2.6 interpolate()

```
double BicubicSplineInterpolator::interpolate (
            double p_x,
            double p_theta,
            bool allow_extrapolation = false) const
```

6.2.2.7 set_grid()

```
void BicubicSplineInterpolator::set_grid (
            const Grid * grid)  [inline]
```

6.2.3 Member Data Documentation

6.2.3.1 m_coeffs

```
std::vector<std::array<double, 16> > BicubicSplineInterpolator::m_coeffs  [private]
```

6.2.3.2 m_theta

```
std::vector<double> BicubicSplineInterpolator::m_theta  [private]
```

6.2.3.3 m_x

```
std::vector<double> BicubicSplineInterpolator::m_x  [private]
```

The documentation for this class was generated from the following files:

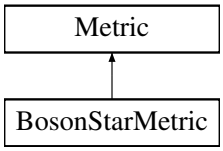
- FOORT/src/Interpolator.h
- FOORT/src/Interpolator.cpp

6.3 BosonStarMetric Class Reference

Boson star metric with solitonic potential.

`#include <Metric.h>`

Inheritance diagram for BosonStarMetric:



Public Member Functions

- BosonStarMetric (double Phi_infinity, int num_lines, bool rLogScale=false, std::string Phi_filename="Phi.dat", std::string m_filename="m.dat")

  Construct a new Boson Star Metric object.
- TwoIndex getMetric_dd (const Point &p) const final

  BosonStar metric getter, indices down.
- TwoIndex getMetric_uu (const Point &p) const final

  BosonStar metric getter, indices up.
- std::string getFullDescriptionStr () const final

  BosonStar metric description string getter.

Public Member Functions inherited from Metric

- virtual ~Metric ()=default
- Metric (bool rlogscale=false)

  Construct a new Metric object.
- virtual ThreeIndex getChristoffel_udd (const Point &p) const

  Return the Christoffel symbols of the metric (indices up, down, down)
- virtual FourIndex getRiemann_uddd (const Point &p) const

  Riemann tensor of the metric (indices up, down, down, down)
- virtual real getKretschmann (const Point &p) const

  Get the Kretschmann scalar at a point.
- bool getrLogScale () const

  Get whether we are using a logarithmic radial scale.

Protected Member Functions

- void read_data ()

  function to read the data

Protected Attributes

- const double m_Phi_infinity

  The value of Phi at infinity, before rescaling.
- const int m_num_lines
- std::string m_Phi_filename

  filename with Phi data
- std::string m_m_filename

  filename with m data
- tk::spline m_PhiSpline

  The spline interpolator for Phi.
- tk::spline m_mSpline

  The spline interpolator for m.

Protected Attributes inherited from Metric

- std::vector< int > m_Symmetries {}

  The symmetries (coordinate Killing vectors) of the metric. Should be set by descendant constructor.
- const bool m_rLogScale

  Are we using a logarithmic r coordinate?

6.3.1 Detailed Description

Boson star metric with solitonic potential.

Note

Implementation by Seppe Staelens

The default files correspond to sigma = 0.06 and phi_c = 0.044

6.3.2 Constructor & Destructor Documentation

6.3.2.1 BosonStarMetric()

```
BosonStarMetric::BosonStarMetric (
            double Phi_infinity,
            int num_lines,
            bool rLogScale = false,
            std::string Phi_filename = "Phi.dat",
            std::string m_filename = "m.dat")
```

Construct a new Boson Star Metric object.

Parameters

Phi_infinity	value of Phi = log(lapse) at infinity
num_lines	number of lines in the data files
rLogScale	whether we are using a logarithmic radial scale
Phi_filename	filename for the Phi data
m_filename	filename for the mass aspect data

6.3.3 Member Function Documentation

6.3.3.1 getFullDescriptionStr()

```
std::string BosonStarMetric::getFullDescriptionStr () const  [final], [virtual]
```

BosonStar metric description string getter.

Returns

std::string

Reimplemented from Metric.

6.3.3.2 getMetric_dd()

```
TwoIndex BosonStarMetric::getMetric_dd (
            const Point & p) const  [final], [virtual]
```

BosonStar metric getter, indices down.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.3.3.3 getMetric_uu()

```
TwoIndex BosonStarMetric::getMetric_uu (
            const Point & p) const  [final], [virtual]
```

BosonStar metric getter, indices up.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.3.3.4 read_data()

```
void BosonStarMetric::read_data ()  [protected]
```

function to read the data

Read the numerically obtained boson star metric data from the files.

6.3.4 Member Data Documentation

6.3.4.1 m_m_filename

```
std::string BosonStarMetric::m_m_filename  [protected]
```

filename with m data

6.3.4.2 m_mSpline

```
tk::spline BosonStarMetric::m_mSpline  [protected]
```

The spline interpolator for m.

6.3.4.3 m_num_lines

```
const int BosonStarMetric::m_num_lines  [protected]
```

Number of lines to be read from the data files. Mainly to avoid reading the data at extreme distances, which could upset the spline interpolation.

6.3.4.4 m_Phi_filename

```
std::string BosonStarMetric::m_Phi_filename  [protected]
```

filename with Phi data

6.3.4.5 m_Phi_infinity

```
const double BosonStarMetric::m_Phi_infinity  [protected]
```

The value of Phi at infinity, before rescaling.

6.3.4.6 m_PhiSpline

```
tk::spline BosonStarMetric::m_PhiSpline  [protected]
```

The spline interpolator for Phi.

The documentation for this class was generated from the following files:

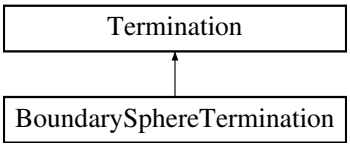
- FOORT/src/Metric.h
- FOORT/src/Metric.cpp

6.4 BoundarySphereTermination Class Reference

BoundarySphereTermination: Terminate geodesics if they reach outside of a boundary sphere.

```
#include <Terminations.h>
```

Inheritance diagram for BoundarySphereTermination:



Public Member Functions

- BoundarySphereTermination (Geodesic *const theGeodesic)
- Term CheckTermination () final

  Check if the boundary sphere has been reached.
- std::string getFullDescriptionStr () const final

  Description string getter for the Boundary Sphere Termination.

Public Member Functions inherited from Termination

- Termination ()=delete
- Termination (Geodesic *const theGeodesic)
- virtual void Reset ()

  Reset to 0 to start integrating a new geodesic.
- virtual ~Termination ()=default

Static Public Attributes

- static std::unique_ptr< BoundarySphereTermOptions > TermOptions

  The options that the BoundarySphereTermination keeps (contains the radius of the boundary sphere)

Additional Inherited Members

Protected Member Functions inherited from Termination

- bool DecideUpdate (largecounter UpdateNSteps)

  Returns true if the Termination should update its internal status.

Protected Attributes inherited from Termination

- Geodesic *const m_OwnerGeodesic

  The geodesic that owns the Termination (a const pointer to the Geodesic)
- largecounter m_StepsSinceUpdated {}

  The termination is itself in charge of keeping track of how many steps it has been since it has been updated. The Termination's TerminationOptions struct tells it how many steps it needs to wait between updates.

6.4.1 Detailed Description

BoundarySphereTermination: Terminate geodesics if they reach outside of a boundary sphere.

6.4.2 Constructor & Destructor Documentation

6.4.2.1 BoundarySphereTermination()

```
BoundarySphereTermination::BoundarySphereTermination (
            Geodesic *const theGeodesic)  [inline]
```

6.4.3 Member Function Documentation

6.4.3.1 CheckTermination()

```
Term BoundarySphereTermination::CheckTermination ()  [final], [virtual]
```

Check if the boundary sphere has been reached.

Returns

Term

Implements Termination.

6.4.3.2 getFullDescriptionStr()

```
std::string BoundarySphereTermination::getFullDescriptionStr () const  [final], [virtual]
```

Description string getter for the Boundary Sphere Termination.

Returns

std::string

Implements Termination.

6.4.4 Member Data Documentation

6.4.4.1 TermOptions

```
std::unique_ptr< BoundarySphereTermOptions > BoundarySphereTermination::TermOptions  [static]
```

The options that the BoundarySphereTermination keeps (contains the radius of the boundary sphere)

The documentation for this class was generated from the following files:

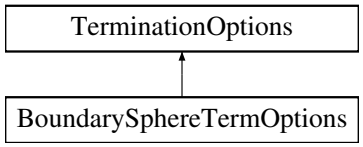
- FOORT/src/Terminations.h
- FOORT/src/Config.cpp
- FOORT/src/Terminations.cpp

6.5 BoundarySphereTermOptions Struct Reference

Options for BoundarySphereTermination.

```
#include <Terminations.h>
```

Inheritance diagram for BoundarySphereTermOptions:



Public Member Functions

- BoundarySphereTermOptions (real theRadius, bool therLogScale, largecounter Nsteps)

Public Member Functions inherited from TerminationOptions

- TerminationOptions (largecounter Nsteps)
- virtual ~TerminationOptions ()=default

Public Attributes

- const real SphereRadius

  Radius of the boundary sphere.
- const bool rLogScale

  Whether we are using a logarithmic r coordinate.

Public Attributes inherited from TerminationOptions

- const largecounter UpdateEveryNSteps

  Number of steps between updates.

6.5.1 Detailed Description

Options for BoundarySphereTermination.

Keeps track of the boundary sphere radius, whether we are using a logarithmic r coordinate.

6.5.2 Constructor & Destructor Documentation

6.5.2.1 BoundarySphereTermOptions()

```
BoundarySphereTermOptions::BoundarySphereTermOptions (
            real theRadius,
            bool therLogScale,
            largecounter Nsteps)  [inline]
```

6.5.3 Member Data Documentation

6.5.3.1 rLogScale

```
const bool BoundarySphereTermOptions::rLogScale
```

Whether we are using a logarithmic r coordinate.

6.5.3.2 SphereRadius

```
const real BoundarySphereTermOptions::SphereRadius
```

Radius of the boundary sphere.

The documentation for this struct was generated from the following file:

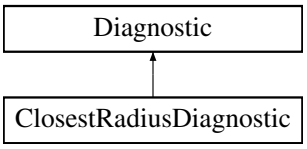
- FOORT/src/Terminations.h

6.6 ClosestRadiusDiagnostic Class Reference

Diagnostic that keeps track of the closest (true, not log) radius that the geodesic passes through.

`#include <Diagnostics.h>`

Inheritance diagram for ClosestRadiusDiagnostic:



Public Member Functions

- ClosestRadiusDiagnostic (Geodesic *const theGeodesic)
- void Reset () final

  ClosestRadiusDiagnostic functions.
- void UpdateData () final

  Check to see if we have travelled closer to the center and update the closest radius.
- std::string getFullDataStr () const final

  Return the closest radius as a string.
- std::vector< real > getFinalDataVal () const final

  Return the closest radius as a vector of size one.
- real FinalDataValDistance (const std::vector< real > &val1, const std::vector< real > &val2) const final

  Return the absolute value of the difference between the closest radii.
- std::string getNameStr () const final

  Simple name without spaces.
- std::string getFullDescriptionStr () const final

  More descriptive string (with spaces)

Public Member Functions inherited from Diagnostic

- Diagnostic ()=delete
- Diagnostic (Geodesic *const theGeodesic)
- virtual ~Diagnostic ()=default

Static Public Attributes

- static std::unique_ptr< ClosestRadiusOptions > DiagOptions

Private Attributes

- real m_ClosestRadius {-1}

Additional Inherited Members

Protected Member Functions inherited from Diagnostic

- bool DecideUpdate (const UpdateFrequency &myUpdateFrequency)

  Returns true if the Diagnostic should update its initial status.

Protected Attributes inherited from Diagnostic

- Geodesic *const m_OwnerGeodesic
- largecounter m_StepsSinceUpdated {}

6.6.1 Detailed Description

Diagnostic that keeps track of the closest (true, not log) radius that the geodesic passes through.

6.6.2 Constructor & Destructor Documentation

6.6.2.1 ClosestRadiusDiagnostic()

```
ClosestRadiusDiagnostic::ClosestRadiusDiagnostic (
            Geodesic *const theGeodesic)  [inline]
```

6.6.3 Member Function Documentation

6.6.3.1 FinalDataValDistance()

```
real ClosestRadiusDiagnostic::FinalDataValDistance (
            const std::vector< real > & val1,
            const std::vector< real > & val2) const  [final], [virtual]
```

Return the absolute value of the difference between the closest radii.

Parameters

val1	first closest radius
val2	second closest radius

Returns

real

Implements Diagnostic.

6.6.3.2 getFinalDataVal()

```
std::vector< real > ClosestRadiusDiagnostic::getFinalDataVal () const  [final], [virtual]
```

Return the closest radius as a vector of size one.

Returns

std::vector<real>

Implements Diagnostic.

6.6.3.3 getFullDataStr()

```
std::string ClosestRadiusDiagnostic::getFullDataStr () const  [final], [virtual]
```

Return the closest radius as a string.

Returns

std::string

Implements Diagnostic.

6.6.3.4 getFullDescriptionStr()

```
std::string ClosestRadiusDiagnostic::getFullDescriptionStr () const  [final], [virtual]
```

More descriptive string (with spaces)

Returns

std::string

Reimplemented from Diagnostic.

6.6.3.5 getNameStr()

```
std::string ClosestRadiusDiagnostic::getNameStr () const  [final], [virtual]
```

Simple name without spaces.

Returns

std::string

Implements Diagnostic.

6.6.3.6 Reset()

```
void ClosestRadiusDiagnostic::Reset ()  [final], [virtual]
```

ClosestRadiusDiagnostic functions.

Reset the closest radius to -1 and call base class Reset function

Reimplemented from Diagnostic.

6.6.3.7 UpdateData()

```
void ClosestRadiusDiagnostic::UpdateData ()  [final], [virtual]
```

Check to see if we have travelled closer to the center and update the closest radius.

Implements Diagnostic.

6.6.4 Member Data Documentation

6.6.4.1 DiagOptions

```
std::unique_ptr< ClosestRadiusOptions > ClosestRadiusDiagnostic::DiagOptions  [static]
```

6.6.4.2 m_ClosestRadius

```
real ClosestRadiusDiagnostic::m_ClosestRadius {-1}  [private]
```

The documentation for this class was generated from the following files:

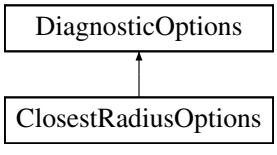
- FOORT/src/Diagnostics.h
- FOORT/src/Config.cpp
- FOORT/src/Diagnostics.cpp

6.7 ClosestRadiusOptions Struct Reference

Options for ClosestRadiusDiagnostic.

```
#include <Diagnostics.h>
```

Inheritance diagram for ClosestRadiusOptions:



Public Member Functions

- ClosestRadiusOptions (bool rlog, UpdateFrequency thefrequency)

Public Member Functions inherited from DiagnosticOptions

- DiagnosticOptions (UpdateFrequency thefrequency)
- virtual ~DiagnosticOptions ()=default

Public Attributes

- const bool RLogScale

Public Attributes inherited from DiagnosticOptions

- const UpdateFrequency theUpdateFrequency

6.7.1 Detailed Description

Options for ClosestRadiusDiagnostic.

Inherits from DiagnosticOptions, and adds the flag for using log(r) scale.

6.7.2 Constructor & Destructor Documentation

6.7.2.1 ClosestRadiusOptions()

```
ClosestRadiusOptions::ClosestRadiusOptions (
            bool rlog,
            UpdateFrequency thefrequency)  [inline]
```

6.7.3 Member Data Documentation

6.7.3.1 RLogScale

```
const bool ClosestRadiusOptions::RLogScale
```

The documentation for this struct was generated from the following file:

- FOORT/src/Diagnostics.h

6.8 ConfigReader::ConfigCollection Class Reference

```
#include <ConfigReader.h>
```

Classes

- struct ConfigSetting

Public Member Functions

- bool Exists (std::string_view SettingName) const

  Check if a setting with the given name exists in the collection.
- int NrSettings () const

  Return the total number of settings in the collection.
- bool IsCollection (std::string_view SettingName) const

  Check if a setting with the given name is a collection.
- bool IsCollection (int SettingIndex) const

  Check if a setting with the given index is a collection.
- const ConfigCollection & operator[ ] (std::string_view CollectionName) const

  Access a subcollection by name.
- const ConfigCollection & operator[ ] (int CollectionIndex) const

  Access a subcollection by index.
- template<class OutputType>
  bool LookupValue (std::string_view SettingName, OutputType &theOutput) const

  Looks up the value of a setting with the given name and puts it in the output variable.
- template<class OutputType>
  bool LookupValue (int SettingIndex, OutputType &theOutput) const

  Looks up the value of a setting with the given index and puts it in the output variable.
- bool LookupValueInteger (std::string_view SettingName, int &theOutput) const

  If the output is int, we only look up to see if the value is an int.
- bool LookupValueInteger (std::string_view SettingName, long &theOutput) const

  If the output is long, the setting value can either be a long or an int, so we check for both.
- bool LookupValueInteger (std::string_view SettingName, long long &theOutput) const

  If the output is long long, the setting value can either be a long long, a long, or an int, so we check for all.
- bool LookupValueInteger (std::string_view SettingName, unsigned int &theOutput) const

  If the output is unsigned int, the setting value can either be an unsigned int, an int, a long, or a long long, so we check for all.
- bool LookupValueInteger (std::string_view SettingName, unsigned long &theOutput) const

  If the output is unsigned long, the setting value can either be an unsigned long, an unsigned int, an int, a long, or a long long, so we check for all.
- bool LookupValueInteger (std::string_view SettingName, unsigned long long &theOutput) const

  If the output is unsigned long long, the setting value can either be an unsigned long long, an unsigned long, an unsigned int, an int, a long, or a long long, so we check for all.
- template<class OutputType>
  bool LookupValueInteger (int SettingIndex, OutputType &theOutput) const

  This simply defers to the setting-name-based implementation of the function with given output type.
- void DisplayCollection (std::ostream &OutputStream, int Indent=0) const

  Output the entire collection to a given output stream.
- bool ReadFile (const std::string &FileName)

  Read in a configuration file.

Private Types

- using ConfigSettingValue

Private Member Functions

- int GetSettingIndex (std::string_view SettingName) const

  Return the index of a specific setting in the collection.
- void DisplaySetting (std::ostream &OutputStream, int SettingIndex, int Indent) const

  Output a single setting to the output stream.
- void DisplayTabs (std::ostream &OutputStream, int NrTabs) const

  Output a given number of tabs to the output stream.
- void ReadCollection (std::ifstream &InputFile)

  Read in an entire collection (including subcollections). Pre: stream is positioned right after '{' (or at beginning of file). Post: stream is positioned right after '}' (or at EOF).
- void ReadSettingName (std::ifstream &InputFile, std::string &theName)

  Read in a setting name. Pre: next non-whitespace character in stream is beginning of name. Post: stream has just read in all characters of name (but no more); returns "" if there is no more setting to read in the current collection, in this case '}' has been read in (for subcollections).
- void ReadSettingSpecificChar (std::ifstream &InputFile, char theChar) const

  Read in one specific character only (not '/'). Pre: next non-whitespace character in stream is this character. Post: stream is just after character.
- void ReadSettingValue (std::ifstream &InputFile, ConfigSettingValue &theValue)

  Read in setting value (could be subcollection). Pre: next non-whitespace character in stream is beginning of value (can be '{' for subcollection, digit or '-' or '.' or 'e' for number, "" for string, or 't'/'f' for boolean). Post: Value has entirely been read in (i.e. next character should be ';').

Private Attributes

- std::vector< ConfigSetting > m_Settings {}

6.8.1 Member Typedef Documentation

6.8.1.1 ConfigSettingValue

```
using ConfigReader::ConfigCollection::ConfigSettingValue  [private]
```

Initial value:

```
 std::variant<bool,
                                                int, long, long long,
                                                double,
                                                std::string,
                                                std::unique_ptr<ConfigCollection»
```

6.8.2 Member Function Documentation

6.8.2.1 DisplayCollection()

```
void ConfigCollection::DisplayCollection (
            std::ostream & OutputStream,
            int Indent = 0) const
```

Output the entire collection to a given output stream.

Parameters

OutputStream	pointer to an outputstream
Indent	indentation level

6.8.2.2 DisplaySetting()

```
void ConfigCollection::DisplaySetting (
            std::ostream & OutputStream,
            int SettingIndex,
            int Indent) const  [private]
```

Output a single setting to the output stream.

Parameters

OutputStream	pointer to an outputstream
SettingIndex	index of the setting to output
Indent	indentation level

6.8.2.3 DisplayTabs()

```
void ConfigCollection::DisplayTabs (
            std::ostream & OutputStream,
            int NrTabs) const  [private]
```

Output a given number of tabs to the output stream.

Parameters

OutputStream	pointer to an outputstream
NrTabs	number of tabs to output

6.8.2.4 Exists()

```
bool ConfigCollection::Exists (
            std::string_view SettingName) const
```

Check if a setting with the given name exists in the collection.

Parameters

SettingName	The name of the setting to check

Returns

true

false

6.8.2.5 GetSettingIndex()

```
int ConfigCollection::GetSettingIndex (
            std::string_view SettingName) const  [private]
```

Return the index of a specific setting in the collection.

Parameters

SettingName	The name of the setting to find

Returns

int

6.8.2.6 IsCollection() [1/2]

```
bool ConfigCollection::IsCollection (
            int SettingIndex) const
```

Check if a setting with the given index is a collection.

Parameters

SettingIndex	The index of the setting to check

Returns

true

false

6.8.2.7 IsCollection() [2/2]

```
bool ConfigCollection::IsCollection (
            std::string_view SettingName) const
```

Check if a setting with the given name is a collection.

Parameters

SettingName	The name of the setting to check

Returns

true

false

6.8.2.8 LookupValue() [1/2]

```
template<class OutputType>
bool ConfigReader::ConfigCollection::LookupValue (
            int SettingIndex,
            OutputType & theOutput) const
```

Looks up the value of a setting with the given index and puts it in the output variable.

Template Parameters

OutputType	

Parameters

SettingIndex	Setting index
theOutput	Pointer to the output variable

Returns

true

false

6.8.2.9 LookupValue() [2/2]

```
template<class OutputType>
bool ConfigReader::ConfigCollection::LookupValue (
            std::string_view SettingName,
            OutputType & theOutput) const
```

Looks up the value of a setting with the given name and puts it in the output variable.

Template Parameters

OutputType	

Parameters

SettingName	Setting name
theOutput	Pointer to the output variable

Returns

true

false

6.8.2.10 LookupValueInteger() [1/7]

```
template<class OutputType>
bool ConfigReader::ConfigCollection::LookupValueInteger (
            int SettingIndex,
            OutputType & theOutput) const
```

This simply defers to the setting-name-based implementation of the function with given output type.

Template Parameters

OutputType	

Parameters

SettingIndex	Setting index
theOutput	Pointer to the output variable

Returns

true

false

6.8.2.11 LookupValueInteger() [2/7]

```
bool ConfigCollection::LookupValueInteger (
            std::string_view SettingName,
            int & theOutput) const
```

If the output is int, we only look up to see if the value is an int.

Parameters

SettingName	Name of the setting to look up
theOutput	integer ouput

Returns

true

false

6.8.2.12 LookupValueInteger() [3/7]

```
bool ConfigCollection::LookupValueInteger (
            std::string_view SettingName,
            long & theOutput) const
```

If the output is long, the setting value can either be a long or an int, so we check for both.

Parameters

SettingName	Name of the setting to look up
theOutput	long output

Returns

true

false

6.8.2.13 LookupValueInteger() [4/7]

```
bool ConfigCollection::LookupValueInteger (
            std::string_view SettingName,
            long long & theOutput) const
```

If the output is long long, the setting value can either be a long long, a long, or an int, so we check for all.

Parameters

SettingName	Name of the setting to look up
theOutput	long long output

Returns

true

false

6.8.2.14 LookupValueInteger() [5/7]

```
bool ConfigCollection::LookupValueInteger (
            std::string_view SettingName,
            unsigned int & theOutput) const
```

If the output is unsigned int, the setting value can either be an unsigned int, an int, a long, or a long long, so we check for all.

Parameters

SettingName	Name of the setting to look up
theOutput	unsigned int output

Returns

true

false

6.8.2.15 LookupValueInteger() [6/7]

```
bool ConfigCollection::LookupValueInteger (
            std::string_view SettingName,
            unsigned long & theOutput) const
```

If the output is unsigned long, the setting value can either be an unsigned long, an unsigned int, an int, a long, or a long long, so we check for all.

Parameters

SettingName	Name of the setting to look up
theOutput	unsigned long output

Returns

true

false

6.8.2.16 LookupValueInteger() [7/7]

```
bool ConfigCollection::LookupValueInteger (
            std::string_view SettingName,
            unsigned long long & theOutput) const
```

If the output is unsigned long long, the setting value can either be an unsigned long long, an unsigned long, an unsigned int, an int, a long, or a long long, so we check for all.

Parameters

SettingName	Name of the setting to look up
theOutput	unsigned long long output

Returns

true

false

6.8.2.17 NrSettings()

```
int ConfigCollection::NrSettings () const
```

Return the total number of settings in the collection.

Returns

int

6.8.2.18 operator[]() [1/2]

```
const ConfigCollection & ConfigCollection::operator[] (
            int CollectionIndex) const
```

Access a subcollection by index.

Parameters

CollectionIndex	The index of the subcollection to access

Returns

const ConfigCollection&

6.8.2.19 operator[]() [2/2]

```
const ConfigCollection & ConfigCollection::operator[] (
            std::string_view CollectionName) const
```

Access a subcollection by name.

Parameters

CollectionName	The name of the subcollection to access

Returns

const ConfigCollection&

6.8.2.20 ReadCollection()

```
void ConfigCollection::ReadCollection (
            std::ifstream & InputFile)  [private]
```

Read in an entire collection (including subcollections). Pre: stream is positioned right after '{' (or at beginning of file). Post: stream is positioned right after '}' (or at EOF).

Parameters

InputFile	pointer to an input file stream

6.8.2.21 ReadFile()

```
bool ConfigCollection::ReadFile (
            const std::string & FileName)
```

Read in a configuration file.

Parameters

FileName	File name of the configuration file

Returns

true if succesful

false if the file does not exist, or if there is a parse/syntax error

6.8.2.22 ReadSettingName()

```
void ConfigCollection::ReadSettingName (
            std::ifstream & InputFile,
            std::string & theName)  [private]
```

Read in a setting name. Pre: next non-whitespace character in stream is beginning of name. Post: stream has just read in all characters of name (but no more); returns "" if there is no more setting to read in the current collection, in this case '}' has been read in (for subcollections).

Parameters

InputFile	Pointer to the input file stream
theName	Pointer to the name of the setting

6.8.2.23 ReadSettingSpecificChar()

```
void ConfigCollection::ReadSettingSpecificChar (
            std::ifstream & InputFile,
            char theChar) const  [private]
```

Read in one specific character only (not '/'). Pre: next non-whitespace character in stream is this character. Post: stream is just after character.

Parameters

InputFile	Pointer to the input file stream
theChar	The character to read in

6.8.2.24 ReadSettingValue()

```
void ConfigCollection::ReadSettingValue (
            std::ifstream & InputFile,
            ConfigSettingValue & theValue)  [private]
```

Read in setting value (could be subcollection). Pre: next non-whitespace character in stream is beginning of value (can be '{' for subcollection, digit or '-' or '.' or 'e' for number, "" for string, or 't'/'f' for boolean). Post: Value has entirely been read in (i.e. next character should be ';').

Parameters

InputFile	Pointer to the input file stream
theValue	Pointer to the setting value

6.8.3 Member Data Documentation

6.8.3.1 m_Settings

```
std::vector<ConfigSetting> ConfigReader::ConfigCollection::m_Settings {}  [private]
```

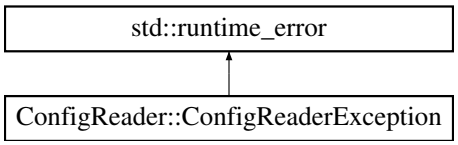
The documentation for this class was generated from the following files:

- FOORT/src/ConfigReader.h
- FOORT/src/ConfigReader.cpp

6.9 ConfigReader::ConfigReaderException Class Reference

`#include <ConfigReader.h>`

Inheritance diagram for ConfigReader::ConfigReaderException:



Public Member Functions

- ConfigReaderException (const std::string &error, std::vector< int > settingtrace={})
- std::vector< int > trace () const

Private Attributes

- std::vector< int > m_settingtrace

6.9.1 Constructor & Destructor Documentation

6.9.1.1 ConfigReaderException()

```
ConfigReader::ConfigReaderException::ConfigReaderException (
            const std::string & error,
            std::vector< int > settingtrace = {})  [inline]
```

6.9.2 Member Function Documentation

6.9.2.1 trace()

```
std::vector< int > ConfigReader::ConfigReaderException::trace () const  [inline]
```

6.9.3 Member Data Documentation

6.9.3.1 m_settingtrace

```
std::vector<int> ConfigReader::ConfigReaderException::m_settingtrace  [private]
```

The documentation for this class was generated from the following file:

- FOORT/src/ConfigReader.h

6.10 ConfigReader::ConfigCollection::ConfigSetting Struct Reference

Public Attributes

- std::string SettingName
- ConfigSettingValue SettingValue

6.10.1 Member Data Documentation

6.10.1.1 SettingName

```
std::string ConfigReader::ConfigCollection::ConfigSetting::SettingName
```

6.10.1.2 SettingValue

```
ConfigSettingValue ConfigReader::ConfigCollection::ConfigSetting::SettingValue
```

The documentation for this struct was generated from the following file:

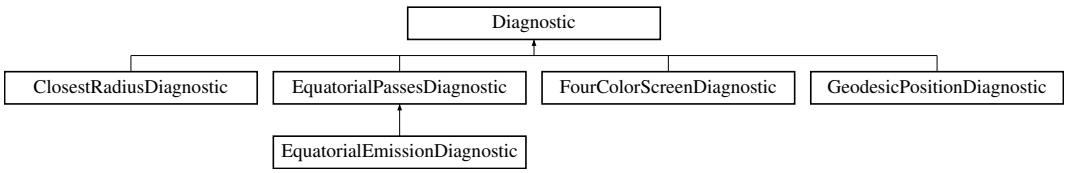
- FOORT/src/ConfigReader.h

6.11 Diagnostic Class Reference

Abstract base class for all diagnostics.

```
#include <Diagnostics.h>
```

Inheritance diagram for Diagnostic:



Public Member Functions

- Diagnostic ()=delete
- Diagnostic (Geodesic *const theGeodesic)
- virtual void Reset ()

  Diagnostic (abstract base class) functions.
- virtual ~Diagnostic ()=default
- virtual void UpdateData ()=0
- virtual std::string getFullDataStr () const =0
- virtual std::vector< real > getFinalDataVal () const =0
- virtual real FinalDataValDistance (const std::vector< real > &val1, const std::vector< real > &val2) const =0
- virtual std::string getNameStr () const =0
- virtual std::string getFullDescriptionStr () const

  Base class just returns the short name of the Diagnostic.

Protected Member Functions

- bool DecideUpdate (const UpdateFrequency &myUpdateFrequency)

  Returns true if the Diagnostic should update its initial status.

Protected Attributes

- Geodesic ∗const m_OwnerGeodesic
- largecounter m_StepsSinceUpdated {}

6.11.1 Detailed Description

Abstract base class for all diagnostics.

6.11.2 Constructor & Destructor Documentation

6.11.2.1 Diagnostic() [1/2]

```
Diagnostic::Diagnostic ()  [delete]
```

6.11.2.2 Diagnostic() [2/2]

```
Diagnostic::Diagnostic (
            Geodesic *const theGeodesic)  [inline]
```

6.11.2.3 ∼Diagnostic()

```
virtual Diagnostic::∼Diagnostic ()  [virtual], [default]
```

6.11.3 Member Function Documentation

6.11.3.1 DecideUpdate()

```
bool Diagnostic::DecideUpdate (
            const UpdateFrequency & myUpdateFrequency)  [protected]
```

Returns true if the Diagnostic should update its initial status.

Should be called from within UpdateData() with the appropriate DiagnosticOptions::theUpdateFrequency.

Parameters

myUpdateFrequency	UpdateFrequency struct

Returns

true

false

6.11.3.2 FinalDataValDistance()

```
virtual real Diagnostic::FinalDataValDistance (
            const std::vector< real > & val1,
            const std::vector< real > & val2) const  [pure virtual]
```

Implemented in ClosestRadiusDiagnostic, EquatorialEmissionDiagnostic, EquatorialPassesDiagnostic, FourColorScreenDiagnostic, and GeodesicPositionDiagnostic.

6.11.3.3 getFinalDataVal()

```
virtual std::vector< real > Diagnostic::getFinalDataVal () const  [pure virtual]
```

Implemented in ClosestRadiusDiagnostic, EquatorialEmissionDiagnostic, EquatorialPassesDiagnostic, FourColorScreenDiagnostic, and GeodesicPositionDiagnostic.

6.11.3.4 getFullDataStr()

```
virtual std::string Diagnostic::getFullDataStr () const  [pure virtual]
```

Implemented in ClosestRadiusDiagnostic, EquatorialEmissionDiagnostic, EquatorialPassesDiagnostic, FourColorScreenDiagnostic, and GeodesicPositionDiagnostic.

6.11.3.5 getFullDescriptionStr()

```
std::string Diagnostic::getFullDescriptionStr () const  [virtual]
```

Base class just returns the short name of the Diagnostic.

This function is pure virtual in the base class.

Returns

std::string

Reimplemented in ClosestRadiusDiagnostic, EquatorialEmissionDiagnostic, EquatorialPassesDiagnostic, FourColorScreenDiagnostic, and GeodesicPositionDiagnostic.

6.11.3.6 getNameStr()

```
virtual std::string Diagnostic::getNameStr () const  [pure virtual]
```

Implemented in ClosestRadiusDiagnostic, EquatorialEmissionDiagnostic, EquatorialPassesDiagnostic, FourColorScreenDiagnostic, and GeodesicPositionDiagnostic.

6.11.3.7 Reset()

```
void Diagnostic::Reset ()  [virtual]
```

Diagnostic (abstract base class) functions.

Reset Diagnostic object steps to zero.

Reimplemented in ClosestRadiusDiagnostic, EquatorialEmissionDiagnostic, EquatorialPassesDiagnostic, FourColorScreenDiagnostic, and GeodesicPositionDiagnostic.

6.11.3.8 UpdateData()

```
virtual void Diagnostic::UpdateData ()  [pure virtual]
```

Implemented in ClosestRadiusDiagnostic, EquatorialEmissionDiagnostic, EquatorialPassesDiagnostic, FourColorScreenDiagnostic, and GeodesicPositionDiagnostic.

6.11.4 Member Data Documentation

6.11.4.1 m_OwnerGeodesic

```
Geodesic* const Diagnostic::m_OwnerGeodesic  [protected]
```

6.11.4.2 m_StepsSinceUpdated

```
largecounter Diagnostic::m_StepsSinceUpdated {}  [protected]
```

The documentation for this class was generated from the following files:

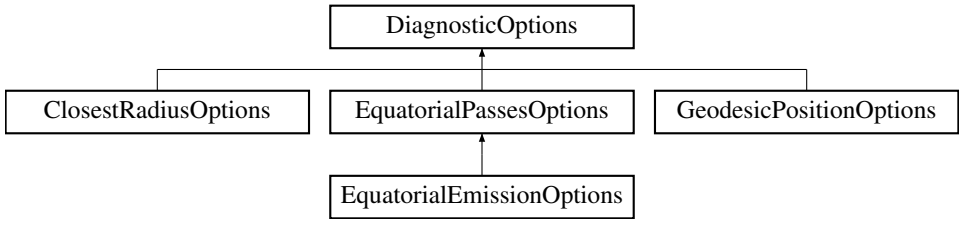
- FOORT/src/Diagnostics.h
- FOORT/src/Diagnostics.cpp

6.12 DiagnosticOptions Struct Reference

Base class for DiagnosticOptions. Other Diagnostics can inherit from here if they require more options.

```
#include <Diagnostics.h>
```

Inheritance diagram for DiagnosticOptions:



Public Member Functions

- DiagnosticOptions (UpdateFrequency thefrequency)
- virtual ~DiagnosticOptions ()=default

Public Attributes

- const UpdateFrequency theUpdateFrequency

6.12.1 Detailed Description

Base class for DiagnosticOptions. Other Diagnostics can inherit from here if they require more options.

6.12.2 Constructor & Destructor Documentation

6.12.2.1 DiagnosticOptions()

```
DiagnosticOptions::DiagnosticOptions (
            UpdateFrequency thefrequency)  [inline]
```

6.12.2.2 ~DiagnosticOptions()

```
virtual DiagnosticOptions::~DiagnosticOptions ()  [virtual], [default]
```

6.12.3 Member Data Documentation

6.12.3.1 theUpdateFrequency

```
const UpdateFrequency DiagnosticOptions::theUpdateFrequency
```

The documentation for this struct was generated from the following file:

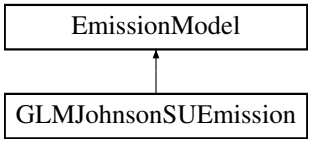
- FOORT/src/Diagnostics.h

6.13 EmissionModel Struct Reference

Emmision model abstract base class.

```
#include <DiagnosticsEmission.h>
```

Inheritance diagram for EmissionModel:



Public Member Functions

- virtual ∼EmissionModel ()=default
- virtual real GetEmission (const Point &p) const =0
- virtual std::string getFullDescriptionStr () const

  Emission model functions.

6.13.1 Detailed Description

Emmision model abstract base class.

6.13.2 Constructor & Destructor Documentation

6.13.2.1 ∼EmissionModel()

```
virtual EmissionModel::~EmissionModel ()  [virtual], [default]
```

6.13.3 Member Function Documentation

6.13.3.1 GetEmission()

```
virtual real EmissionModel::GetEmission (
            const Point & p) const  [pure virtual]
```

Implemented in GLMJohnsonSUEmission.

6.13.3.2 getFullDescriptionStr()

```
std::string EmissionModel::getFullDescriptionStr () const  [virtual]
```

Emission model functions.

Base class default string getter

Returns

std::string

Reimplemented in GLMJohnsonSUEmission.

The documentation for this struct was generated from the following files:

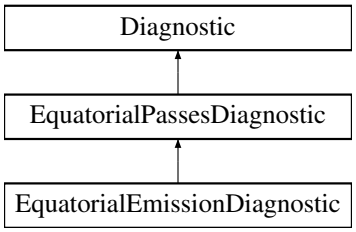
- FOORT/src/DiagnosticsEmission.h
- FOORT/src/DiagnosticsEmission.cpp

6.14 EquatorialEmissionDiagnostic Class Reference

Diagnostic that calculates brightness intensity for the geodesic, based on a specified equatorial disc emission model.

`#include <Diagnostics.h>`

Inheritance diagram for EquatorialEmissionDiagnostic:



Public Member Functions

- EquatorialEmissionDiagnostic (Geodesic *const theGeodesic)
- void Reset () final

  EquatorialEmissionDiagnostic functions.
- void UpdateData () final

  Update the intensity based on the emission model and the number of equatorial passes.
- std::string getFullDataStr () const final

  Return the intensity and number of equatorial passes as a string.
- std::vector< real > getFinalDataVal () const final

  Return the intensity and number of equatorial passes as a vector of size two.
- real FinalDataValDistance (const std::vector< real > &val1, const std::vector< real > &val2) const final

  Return the distance between two geodesics based on their intensity and number of equatorial passes.
- std::string getNameStr () const final

  Simple name without spaces.
- std::string getFullDescriptionStr () const final

  More descriptive string (with spaces)

Public Member Functions inherited from EquatorialPassesDiagnostic

- EquatorialPassesDiagnostic (Geodesic *const theGeodesic)
- void Reset () override

  EquatorialPassesDiagnostic functions.
- void UpdateData () override

  Check to see if we have crossed the equatorial plane and update the number of passes.
- std::string getFullDataStr () const override

  Return the number of equatorial passes as a string.
- std::vector< real > getFinalDataVal () const override

  Return the number of equatorial passes as a vector of size one.
- real FinalDataValDistance (const std::vector< real > &val1, const std::vector< real > &val2) const override

  Return the absolute value of the difference between the number of equatorial passes.
- std::string getNameStr () const override

  Simple name without spaces.
- std::string getFullDescriptionStr () const override

  More descriptive string (with spaces)

Public Member Functions inherited from Diagnostic

- Diagnostic ()=delete
- Diagnostic (Geodesic ∗const theGeodesic)
- virtual ∼Diagnostic ()=default

Static Public Attributes

- static std::unique_ptr< EquatorialEmissionOptions > DiagOptions

Static Public Attributes inherited from EquatorialPassesDiagnostic

- static std::unique_ptr< EquatorialPassesOptions > DiagOptions

  Needs the extra option of a threshold.

Private Attributes

- real m_Intensity {0.0}

Additional Inherited Members

Protected Member Functions inherited from Diagnostic

- bool DecideUpdate (const UpdateFrequency &myUpdateFrequency)

  Returns true if the Diagnostic should update its initial status.

Protected Attributes inherited from EquatorialPassesDiagnostic

- int m_EquatPasses {0}

  Keeps track of how many passes have been made.

Protected Attributes inherited from Diagnostic

- Geodesic ∗const m_OwnerGeodesic
- largecounter m_StepsSinceUpdated {}

6.14.1 Detailed Description

Diagnostic that calculates brightness intensity for the geodesic, based on a specified equatorial disc emission model.

Inherits from EquatorialPassesDiagnostic, since it needs to keep track of when it passes through the equatorial plane. As a result it also keeps track of the number of equatorial passes. Both the intensity profile and fluid velocity profile are specified in the options struct.

6.14.2 Constructor & Destructor Documentation

6.14.2.1 EquatorialEmissionDiagnostic()

```
EquatorialEmissionDiagnostic::EquatorialEmissionDiagnostic (
            Geodesic *const theGeodesic)  [inline]
```

6.14.3 Member Function Documentation

6.14.3.1 FinalDataValDistance()

```
real EquatorialEmissionDiagnostic::FinalDataValDistance (
            const std::vector< real > & val1,
            const std::vector< real > & val2) const  [final], [virtual]
```

Return the distance between two geodesics based on their intensity and number of equatorial passes.

Parameters

val1	first geodesic intensity and number of equatorial passes
val2	second geodesic intensity and number of equatorial passes

Returns

real

Implements Diagnostic.

6.14.3.2 getFinalDataVal()

```
std::vector< real > EquatorialEmissionDiagnostic::getFinalDataVal () const  [final], [virtual]
```

Return the intensity and number of equatorial passes as a vector of size two.

Returns

std::vector<real>

Implements Diagnostic.

6.14.3.3 getFullDataStr()

```
std::string EquatorialEmissionDiagnostic::getFullDataStr () const  [final], [virtual]
```

Return the intensity and number of equatorial passes as a string.

Returns

std::string

Implements Diagnostic.

6.14.3.4 getFullDescriptionStr()

```
std::string EquatorialEmissionDiagnostic::getFullDescriptionStr () const  [final], [virtual]
```

More descriptive string (with spaces)

Returns

std::string

Reimplemented from Diagnostic.

6.14.3.5 getNameStr()

```
std::string EquatorialEmissionDiagnostic::getNameStr () const  [final], [virtual]
```

Simple name without spaces.

Returns

std::string

Implements Diagnostic.

6.14.3.6 Reset()

```
void EquatorialEmissionDiagnostic::Reset ()  [final], [virtual]
```

EquatorialEmissionDiagnostic functions.

Reset the intensity and number of equatorial passes to zero and call base class Reset function

Reimplemented from Diagnostic.

6.14.3.7 UpdateData()

```
void EquatorialEmissionDiagnostic::UpdateData ()  [final], [virtual]
```

Update the intensity based on the emission model and the number of equatorial passes.

Implements Diagnostic.

6.14.4 Member Data Documentation

6.14.4.1 DiagOptions

```
std::unique_ptr< EquatorialEmissionOptions > EquatorialEmissionDiagnostic::DiagOptions  [static]
```

6.14.4.2 m_Intensity

```
real EquatorialEmissionDiagnostic::m_Intensity {0.0}  [private]
```

The documentation for this class was generated from the following files:

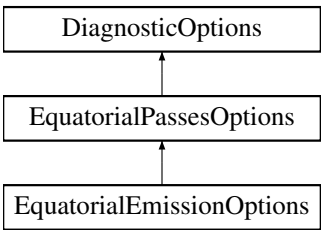
- FOORT/src/Diagnostics.h
- FOORT/src/Config.cpp
- FOORT/src/Diagnostics.cpp

6.15 EquatorialEmissionOptions Struct Reference

Options for EquatorialEmissionDiagnostic.

```
#include <Diagnostics.h>
```

Inheritance diagram for EquatorialEmissionOptions:



Public Member Functions

- EquatorialEmissionOptions (real thefudgefactor, int equatupper, std::unique_ptr< EmissionModel > theemission, std::unique_ptr< FluidVelocityModel > thefluidmodel, bool rlog, int theredshiftpower, real thethreshold, UpdateFrequency thefrequency)

Public Member Functions inherited from EquatorialPassesOptions

- EquatorialPassesOptions (real thethreshold, UpdateFrequency thefrequency)

Public Member Functions inherited from DiagnosticOptions

- DiagnosticOptions (UpdateFrequency thefrequency)
- virtual ~DiagnosticOptions ()=default

Public Attributes

- const real GeometricFudgeFactor
- const int EquatPassUpperBound
- const bool RLogScale
- const int RedShiftPower
- const std::unique_ptr< EmissionModel > TheEmissionModel
- const std::unique_ptr< FluidVelocityModel > TheFluidVelocityModel

Public Attributes inherited from EquatorialPassesOptions

- const real Threshold

Public Attributes inherited from DiagnosticOptions

- const UpdateFrequency theUpdateFrequency

6.15.1 Detailed Description

Options for EquatorialEmissionDiagnostic.

Inherits from EquatorialPassesOptions, and adds the fudge factor, upper bound, emission model, fluid velocity model, redshift power, and threshold.

6.15.2 Constructor & Destructor Documentation

6.15.2.1 EquatorialEmissionOptions()

```
EquatorialEmissionOptions::EquatorialEmissionOptions (
            real thefudgefactor,
            int equatupper,
            std::unique_ptr< EmissionModel > theemission,
            std::unique_ptr< FluidVelocityModel > thefluidmodel,
            bool rlog,
            int theredshiftpower,
            real thethreshold,
            UpdateFrequency thefrequency)  [inline]
```

6.15.3 Member Data Documentation

6.15.3.1 EquatPassUpperBound

```
const int EquatorialEmissionOptions::EquatPassUpperBound
```

6.15.3.2 GeometricFudgeFactor

```
const real EquatorialEmissionOptions::GeometricFudgeFactor
```

6.15.3.3 RedShiftPower

```
const int EquatorialEmissionOptions::RedShiftPower
```

6.15.3.4 RLogScale

```
const bool EquatorialEmissionOptions::RLogScale
```

6.15.3.5 TheEmissionModel

```
const std::unique_ptr<EmissionModel> EquatorialEmissionOptions::TheEmissionModel
```

6.15.3.6 TheFluidVelocityModel

```
const std::unique_ptr<FluidVelocityModel> EquatorialEmissionOptions::TheFluidVelocityModel
```

The documentation for this struct was generated from the following file:

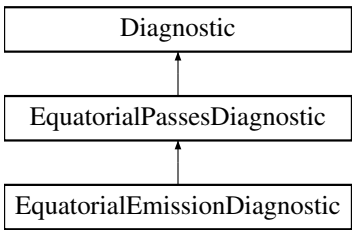
- FOORT/src/Diagnostics.h

6.16 EquatorialPassesDiagnostic Class Reference

Diagnostic that counts the number of passes through the equatorial plane.

```
#include <Diagnostics.h>
```

Inheritance diagram for EquatorialPassesDiagnostic:



Public Member Functions

- EquatorialPassesDiagnostic (Geodesic *const theGeodesic)
- void Reset () override

  EquatorialPassesDiagnostic functions.
- void UpdateData () override

  Check to see if we have crossed the equatorial plane and update the number of passes.
- std::string getFullDataStr () const override

  Return the number of equatorial passes as a string.
- std::vector< real > getFinalDataVal () const override

  Return the number of equatorial passes as a vector of size one.
- real FinalDataValDistance (const std::vector< real > &val1, const std::vector< real > &val2) const override

  Return the absolute value of the difference between the number of equatorial passes.
- std::string getNameStr () const override

  Simple name without spaces.
- std::string getFullDescriptionStr () const override

  More descriptive string (with spaces)

Public Member Functions inherited from Diagnostic

- Diagnostic ()=delete
- Diagnostic (Geodesic *const theGeodesic)
- virtual ~Diagnostic ()=default

Static Public Attributes

- static std::unique_ptr< EquatorialPassesOptions > DiagOptions

  Needs the extra option of a threshold.

Protected Attributes

- int m_EquatPasses {0}

  Keeps track of how many passes have been made.

Protected Attributes inherited from Diagnostic

- Geodesic *const m_OwnerGeodesic
- largecounter m_StepsSinceUpdated {}

Private Attributes

- real m_PrevTheta {-1}

  Keeps track of the previous theta angle, so that we can compare with current theta angle.

Additional Inherited Members

Protected Member Functions inherited from Diagnostic

- bool DecideUpdate (const UpdateFrequency &myUpdateFrequency)

  Returns true if the Diagnostic should update its initial status.

6.16.1 Detailed Description

Diagnostic that counts the number of passes through the equatorial plane.

6.16.2 Constructor & Destructor Documentation

6.16.2.1 EquatorialPassesDiagnostic()

```
EquatorialPassesDiagnostic::EquatorialPassesDiagnostic (
            Geodesic *const theGeodesic)  [inline]
```

6.16.3 Member Function Documentation

6.16.3.1 FinalDataValDistance()

```
real EquatorialPassesDiagnostic::FinalDataValDistance (
            const std::vector< real > & val1,
            const std::vector< real > & val2) const  [override], [virtual]
```

Return the absolute value of the difference between the number of equatorial passes.

Parameters

val1	first number of equatorial passes
val2	second number of equatorial passes

Returns

real

Implements Diagnostic.

6.16.3.2 getFinalDataVal()

```
std::vector< real > EquatorialPassesDiagnostic::getFinalDataVal () const  [override], [virtual]
```

Return the number of equatorial passes as a vector of size one.

Returns

std::vector<real>

Implements Diagnostic.

6.16.3.3 getFullDataStr()

```
std::string EquatorialPassesDiagnostic::getFullDataStr () const  [override], [virtual]
```

Return the number of equatorial passes as a string.

Returns

std::string

Implements Diagnostic.

6.16.3.4 getFullDescriptionStr()

```
std::string EquatorialPassesDiagnostic::getFullDescriptionStr () const  [override], [virtual]
```

More descriptive string (with spaces)

Returns

std::string

Reimplemented from Diagnostic.

6.16.3.5 getNameStr()

```
std::string EquatorialPassesDiagnostic::getNameStr () const  [override], [virtual]
```

Simple name without spaces.

Returns

std::string

Implements Diagnostic.

6.16.3.6 Reset()

```
void EquatorialPassesDiagnostic::Reset ()  [override], [virtual]
```

EquatorialPassesDiagnostic functions.

Reset the number of equatorial passes and the previous theta angle and call base class Reset function

Reimplemented from Diagnostic.

6.16.3.7 UpdateData()

```
void EquatorialPassesDiagnostic::UpdateData ()  [override], [virtual]
```

Check to see if we have crossed the equatorial plane and update the number of passes.

Implements Diagnostic.

6.16.4 Member Data Documentation

6.16.4.1 DiagOptions

```
std::unique_ptr< EquatorialPassesOptions > EquatorialPassesDiagnostic::DiagOptions  [static]
```

Needs the extra option of a threshold.

6.16.4.2 m_EquatPasses

```
int EquatorialPassesDiagnostic::m_EquatPasses {0}  [protected]
```

Keeps track of how many passes have been made.

6.16.4.3 m_PrevTheta

```
real EquatorialPassesDiagnostic::m_PrevTheta {-1}  [private]
```

Keeps track of the previous theta angle, so that we can compare with current theta angle.

The documentation for this class was generated from the following files:

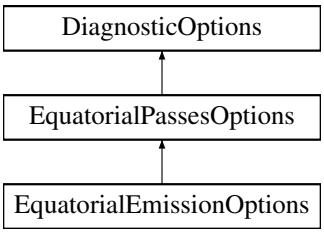
- FOORT/src/Diagnostics.h
- FOORT/src/Config.cpp
- FOORT/src/Diagnostics.cpp

6.17 EquatorialPassesOptions Struct Reference

Options for EquatorialPassesDiagnostic.

```
#include <Diagnostics.h>
```

Inheritance diagram for EquatorialPassesOptions:



Public Member Functions

- EquatorialPassesOptions (real thethreshold, UpdateFrequency thefrequency)

Public Member Functions inherited from DiagnosticOptions

- DiagnosticOptions (UpdateFrequency thefrequency)
- virtual ~DiagnosticOptions ()=default

Public Attributes

- const real Threshold

Public Attributes inherited from DiagnosticOptions

- const UpdateFrequency theUpdateFrequency

6.17.1 Detailed Description

Options for EquatorialPassesDiagnostic.

Inherits from DiagnosticOptions, and adds the threshold for counting passes.

6.17.2 Constructor & Destructor Documentation

6.17.2.1 EquatorialPassesOptions()

```
EquatorialPassesOptions::EquatorialPassesOptions (
            real thethreshold,
            UpdateFrequency thefrequency)  [inline]
```

6.17.3 Member Data Documentation

6.17.3.1 Threshold

```
const real EquatorialPassesOptions::Threshold
```

The documentation for this struct was generated from the following file:

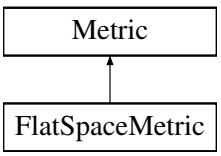
- FOORT/src/Diagnostics.h

6.18 FlatSpaceMetric Class Reference

Flat space metric in 4D.

```
#include <Metric.h>
```

Inheritance diagram for FlatSpaceMetric:



Public Member Functions

- FlatSpaceMetric (bool rlogscale=false)

  Construct a new Flat Space Metric object.
- TwoIndex getMetric_dd (const Point &p) const final

  Flat metric getter, indices down.
- TwoIndex getMetric_uu (const Point &p) const final

  Flat metric getter, indices up.
- std::string getFullDescriptionStr () const final

  Flat metric description string getter.

Public Member Functions inherited from Metric

- virtual ~Metric ()=default
- Metric (bool rlogscale=false)

  Construct a new Metric object.
- virtual ThreeIndex getChristoffel_udd (const Point &p) const

  Return the Christoffel symbols of the metric (indices up, down, down)
- virtual FourIndex getRiemann_uddd (const Point &p) const

  Riemann tensor of the metric (indices up, down, down, down)
- virtual real getKretschmann (const Point &p) const

  Get the Kretschmann scalar at a point.
- bool getrLogScale () const

  Get whether we are using a logarithmic radial scale.

Additional Inherited Members

Protected Attributes inherited from Metric

- std::vector< int > m_Symmetries {}

  The symmetries (coordinate Killing vectors) of the metric. Should be set by descendant constructor.
- const bool m_rLogScale

  Are we using a logarithmic r coordinate?

6.18.1 Detailed Description

Flat space metric in 4D.

6.18.2 Constructor & Destructor Documentation

6.18.2.1 FlatSpaceMetric()

```
FlatSpaceMetric::FlatSpaceMetric (
            bool rlogscale = false)
```

Construct a new Flat Space Metric object.

Parameters

rlogscale	

6.18.3 Member Function Documentation

6.18.3.1 getFullDescriptionStr()

```
std::string FlatSpaceMetric::getFullDescriptionStr () const  [final], [virtual]
```

Flat metric description string getter.

Returns

std::string

Reimplemented from Metric.

6.18.3.2 getMetric_dd()

```
TwoIndex FlatSpaceMetric::getMetric_dd (
            const Point & p) const  [final], [virtual]
```

Flat metric getter, indices down.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.18.3.3 getMetric_uu()

```
TwoIndex FlatSpaceMetric::getMetric_uu (
            const Point & p) const  [final], [virtual]
```

Flat metric getter, indices up.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

The documentation for this class was generated from the following files:

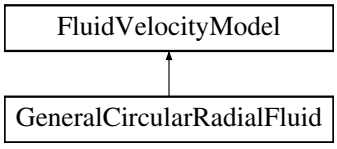
- FOORT/src/Metric.h
- FOORT/src/Metric.cpp

6.19 FluidVelocityModel Struct Reference

Fluid velocity model abstract base class.

`#include <DiagnosticsEmission.h>`

Inheritance diagram for FluidVelocityModel:



Public Member Functions

- FluidVelocityModel (const Metric ∗const theMetric)
- virtual ∼FluidVelocityModel ()=default
- virtual OneIndex GetFourVelocityd (const Point &p) const =0
- virtual std::string getFullDescriptionStr () const

  Base class default string getter.

Protected Attributes

- const Metric ∗const m_theMetric

6.19.1 Detailed Description

Fluid velocity model abstract base class.

6.19.2 Constructor & Destructor Documentation

6.19.2.1 FluidVelocityModel()

```
FluidVelocityModel::FluidVelocityModel (
            const Metric *const theMetric)  [inline]
```

6.19.2.2 ∼FluidVelocityModel()

```
virtual FluidVelocityModel::~FluidVelocityModel ()  [virtual], [default]
```

6.19.3 Member Function Documentation

6.19.3.1 GetFourVelocityd()

```
virtual OneIndex FluidVelocityModel::GetFourVelocityd (
            const Point & p) const  [pure virtual]
```

Implemented in GeneralCircularRadialFluid.

6.19.3.2 getFullDescriptionStr()

```
std::string FluidVelocityModel::getFullDescriptionStr () const  [virtual]
```

Base class default string getter.

Returns

std::string

Reimplemented in GeneralCircularRadialFluid.

6.19.4 Member Data Documentation

6.19.4.1 m_theMetric

```
const Metric* const FluidVelocityModel::m_theMetric  [protected]
```

The documentation for this struct was generated from the following files:

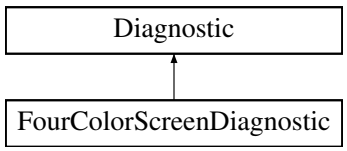
- FOORT/src/DiagnosticsEmission.h
- FOORT/src/DiagnosticsEmission.cpp

6.20 FourColorScreenDiagnostic Class Reference

The four color screen diagnostic.

```
#include <Diagnostics.h>
```

Inheritance diagram for FourColorScreenDiagnostic:



Public Member Functions

- FourColorScreenDiagnostic (Geodesic *const theGeodesic)
- void Reset () final

  FourColorScreen functions.
- void UpdateData () override

  Update the quadrant based on the geodesic's final position.
- std::string getFullDataStr () const final

  Return the value of the quadrant as a string.
- std::vector< real > getFinalDataVal () const final

  Return the value of the quadrant as a vector of size one.
- real FinalDataValDistance (const std::vector< real > &val1, const std::vector< real > &val2) const final

  Discrete metric for distance: return 0 if the quadrants are the same, 1 if they are not.
- std::string getNameStr () const final

  Simple name without spaces.
- std::string getFullDescriptionStr () const final

  Return full name, possibly with spaces.

Public Member Functions inherited from Diagnostic

- Diagnostic ()=delete
- Diagnostic (Geodesic *const theGeodesic)
- virtual ~Diagnostic ()=default

Private Attributes

- int m_quadrant {0}

  Note initialization to 0; this means the default value returned will be 0 (e.g. if Term::BoundarySphere is not reached)

Additional Inherited Members

Protected Member Functions inherited from Diagnostic

- bool DecideUpdate (const UpdateFrequency &myUpdateFrequency)

  Returns true if the Diagnostic should update its initial status.

Protected Attributes inherited from Diagnostic

- Geodesic *const m_OwnerGeodesic
- largecounter m_StepsSinceUpdated {}

6.20.1 Detailed Description

The four color screen diagnostic.

This diagnostic associates one of four colors based on the quadrant that the geodesic finishes in. Will ONLY return a color if the geodesic indeed finishes because it passes through the boundary sphere!

6.20.2 Constructor & Destructor Documentation

6.20.2.1 FourColorScreenDiagnostic()

```
FourColorScreenDiagnostic::FourColorScreenDiagnostic (
            Geodesic *const theGeodesic)  [inline]
```

6.20.3 Member Function Documentation

6.20.3.1 FinalDataValDistance()

```
real FourColorScreenDiagnostic::FinalDataValDistance (
            const std::vector< real > & val1,
            const std::vector< real > & val2) const  [final], [virtual]
```

Discrete metric for distance: return 0 if the quadrants are the same, 1 if they are not.

Parameters

val1	first quadrant value
val2	second quadrant value

Returns

real

Implements Diagnostic.

6.20.3.2 getFinalDataVal()

```
std::vector< real > FourColorScreenDiagnostic::getFinalDataVal () const  [final], [virtual]
```

Return the value of the quadrant as a vector of size one.

Returns

std::vector<real>

Implements Diagnostic.

6.20.3.3 getFullDataStr()

```
std::string FourColorScreenDiagnostic::getFullDataStr () const  [final], [virtual]
```

Return the value of the quadrant as a string.

Returns

std::string

Implements Diagnostic.

6.20.3.4 getFullDescriptionStr()

```
std::string FourColorScreenDiagnostic::getFullDescriptionStr () const  [final], [virtual]
```

Return full name, possibly with spaces.

Returns

std::string

Reimplemented from Diagnostic.

6.20.3.5 getNameStr()

```
std::string FourColorScreenDiagnostic::getNameStr () const  [final], [virtual]
```

Simple name without spaces.

Returns

std::string

Implements Diagnostic.

6.20.3.6 Reset()

```
void FourColorScreenDiagnostic::Reset ()  [final], [virtual]
```

FourColorScreen functions.

Reset the quadrant to its default option; also call base class Reset function

Reimplemented from Diagnostic.

6.20.3.7 UpdateData()

```
void FourColorScreenDiagnostic::UpdateData ()  [override], [virtual]
```

Update the quadrant based on the geodesic's final position.

Implements Diagnostic.

6.20.4 Member Data Documentation

6.20.4.1 m_quadrant

```
int FourColorScreenDiagnostic::m_quadrant {0}  [private]
```

Note initialization to 0; this means the default value returned will be 0 (e.g. if Term::BoundarySphere is not reached)

The documentation for this class was generated from the following files:

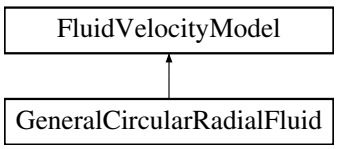
- FOORT/src/Diagnostics.h
- FOORT/src/Diagnostics.cpp

6.21 GeneralCircularRadialFluid Struct Reference

General circular radial fluid velocity model.

```
#include <DiagnosticsEmission.h>
```

Inheritance diagram for GeneralCircularRadialFluid:



Public Member Functions

- GeneralCircularRadialFluid (real subKeplerParam, real betar, real betaphi, const Metric *const theMetric, real ISCO_lowerbound, real ISCO_upperbound)
- OneIndex GetFourVelocityd (const Point &p) const final

  Get the local four-velocity of the fluid according to the GeneralCircularRadial model.
- std::string getFullDescriptionStr () const final

  Description string getter for the GeneralCircularRadial fluid velocity model.

Public Member Functions inherited from FluidVelocityModel

- FluidVelocityModel (const Metric *const theMetric)
- virtual ~FluidVelocityModel ()=default

Private Member Functions

- OneIndex GetCircularVelocityd (const Point &p, bool subKeplerianOn=true) const

  Get the circular velocity for a point, with optional subKeplerian rescaling.
- OneIndex GetInsideISCOCircularVelocityd (const Point &p) const

  Returns velocity for non-geodetic motion where E,L = (E,L)_ISCO and radially falling inward.
- OneIndex GetRadialVelocityd (const Point &p) const

  Returns radial velocity for pure infalling matter with E = 1 and L = 0 at infinity.
- void FindISCO ()

  Helper function to find the ISCO of the spacetime, if it exists.
- TwoIndex GetChristrRaisedDer (real r) const

  Helper function to get \partial_r(g^{ab}\Gamma^r_{bc}g^{cd})

Private Attributes

- const real m_subKeplerParam

  Sub-Keplerian parameter (0 = Keplerian, 1 = sub-Keplerian)
- const real m_betaR

  Radial velocity parameter.
- const real m_betaPhi

  Azimuthal velocity parameter.
- bool m_ISCOexists {false}

  Flag to indicate if ISCO exists for this Metric.
- real m_ISCOr {-1.0}

  ISCO radius.
- real m_ISCOlowerbound

  Lower bound for ISCO radius search.
- real m_ISCOupperbound

  Upper bound for ISCO radius search.
- real m_ISCOpt {}

  ISCO t momentum.
- real m_ISCOpphi {}

  ISCO phi momentum.

Additional Inherited Members

Protected Attributes inherited from FluidVelocityModel

- const Metric *const m_theMetric

6.21.1 Detailed Description

General circular radial fluid velocity model.

This fluid velocity model has three tuneable parameters and represents fluid travelling at a mix of (sub)Keplerian circular orbits and radially infalling orbits in the equatorial plane. Details can be found in e.g. https://doi.org/10.1103/PhysRevD.107.043030 .

6.21.2 Constructor & Destructor Documentation

6.21.2.1 GeneralCircularRadialFluid()

```
GeneralCircularRadialFluid::GeneralCircularRadialFluid (
            real subKeplerParam,
            real betar,
            real betaphi,
            const Metric *const theMetric,
            real ISCO_lowerbound,
            real ISCO_upperbound)  [inline]
```

6.21.3 Member Function Documentation

6.21.3.1 FindISCO()

```
void GeneralCircularRadialFluid::FindISCO ()  [private]
```

Helper function to find the ISCO of the spacetime, if it exists.

This helper function finds the ISCO. We use a binary search to converge on the ISCO value; the initial outer bounds for a metric with horizon are the horizon radius and 10*(horizon radius) (in true radii, not log(r) coordinates)

6.21.3.2 GetChristrRaisedDer()

```
TwoIndex GeneralCircularRadialFluid::GetChristrRaisedDer (
            real r) const  [private]
```

Helper function to get \partial_r(g^{ab}\Gamma^r_{bc}g^{cd})

This function calculates \partial_r(g^ { ab }\Gamma ^ r_{ bc }g^ { cd }). We do so by calculating the central difference with O(h^4) accuracy, but taking the sqrt of the usual h used for derivatives

Parameters

r	Radius at which to calculate the derivative

Returns

TwoIndex

6.21.3.3 GetCircularVelocityd()

```
OneIndex GeneralCircularRadialFluid::GetCircularVelocityd (
            const Point & p,
            bool subKeplerianOn = true) const  [private]
```

Get the circular velocity for a point, with optional subKeplerian rescaling.

Returns the velocity for circular (sub)Keplerian motion (outside the ISCO). First the circular geodesic is calculated, then if subKeplerianOn == true, the orbit is reparametrized to non-geodetic subKeplerian circular motion according to the subKeplerian parameter

Parameters

p	Point at which to calculate the velocity
subKeplerianOn	Boolean to indicate whether to rescale the angular momentum for sub-Keplerian motion

Returns

OneIndex circular velocity with index down. If no solution, returns all zeros

6.21.3.4 GetFourVelocityd()

```
OneIndex GeneralCircularRadialFluid::GetFourVelocityd (
            const Point & p) const  [final], [virtual]
```

Get the local four-velocity of the fluid according to the GeneralCircularRadial model.

Parameters

p	Point at which to calculate the four-velocity

Returns

OneIndex Local four-velocity with index down

Implements FluidVelocityModel.

6.21.3.5 getFullDescriptionStr()

```
std::string GeneralCircularRadialFluid::getFullDescriptionStr () const  [final], [virtual]
```

Description string getter for the GeneralCircularRadial fluid velocity model.

Returns

std::string

Reimplemented from FluidVelocityModel.

6.21.3.6 GetInsideISCOCircularVelocityd()

```
OneIndex GeneralCircularRadialFluid::GetInsideISCOCircularVelocityd (
            const Point & p) const  [private]
```

Returns velocity for non-geodetic motion where E,L = (E,L)_ISCO and radially falling inward.

Parameters

p	Point at which to calculate the velocity

Returns

OneIndex

6.21.3.7 GetRadialVelocityd()

```
OneIndex GeneralCircularRadialFluid::GetRadialVelocityd (
            const Point & p) const  [private]
```

Returns radial velocity for pure infalling matter with E = 1 and L = 0 at infinity.

Parameters

p	Point at which to calculate the velocity

Returns

OneIndex

6.21.4 Member Data Documentation

6.21.4.1 m_betaPhi

```
const real GeneralCircularRadialFluid::m_betaPhi  [private]
```

Azimuthal velocity parameter.

6.21.4.2 m_betaR

```
const real GeneralCircularRadialFluid::m_betaR  [private]
```

Radial velocity parameter.

6.21.4.3 m_ISCOexists

```
bool GeneralCircularRadialFluid::m_ISCOexists {false}  [private]
```

Flag to indicate if ISCO exists for this Metric.

6.21.4.4 m_ISCOlowerbound

```
real GeneralCircularRadialFluid::m_ISCOlowerbound  [private]
```

Lower bound for ISCO radius search.

6.21.4.5 m_ISCOpphi

```
real GeneralCircularRadialFluid::m_ISCOpphi {}  [private]
```

ISCO phi momentum.

6.21.4.6 m_ISCOpt

```
real GeneralCircularRadialFluid::m_ISCOpt {}  [private]
```

ISCO t momentum.

6.21.4.7 m_ISCOr

```
real GeneralCircularRadialFluid::m_ISCOr {-1.0}  [private]
```

ISCO radius.

6.21.4.8 m_ISCOupperbound

```
real GeneralCircularRadialFluid::m_ISCOupperbound  [private]
```

Upper bound for ISCO radius search.

6.21.4.9 m_subKeplerParam

```
const real GeneralCircularRadialFluid::m_subKeplerParam  [private]
```

Sub-Keplerian parameter (0 = Keplerian, 1 = sub-Keplerian)

The documentation for this struct was generated from the following files:

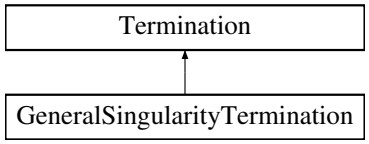
- FOORT/src/DiagnosticsEmission.h
- FOORT/src/DiagnosticsEmission.cpp

6.22 GeneralSingularityTermination Class Reference

GeneralSingularityTermination: Terminate geodesics if they get too close to one of a number of singularities (of arbitrary codimension)

```
#include <Terminations.h>
```

Inheritance diagram for GeneralSingularityTermination:



Public Member Functions

- GeneralSingularityTermination (Geodesic *const theGeodesic)
- Term CheckTermination () final

  Check if the geodesic is too close to a singularity.
- std::string getFullDescriptionStr () const final

  Description string getter for the General Singularity Termination.

Public Member Functions inherited from Termination

- Termination ()=delete
- Termination (Geodesic *const theGeodesic)
- virtual void Reset ()

  Reset to 0 to start integrating a new geodesic.
- virtual ~Termination ()=default

Static Public Attributes

- static std::unique_ptr< GeneralSingularityTermOptions > TermOptions

  Options (contains horizon radius, if we are using logarithmic r coordinate, and distance allowed from the horizon)

Private Member Functions

- std::string SingularityToString (int singnr) const

  List the singularities in a human-readable format.

Additional Inherited Members

Protected Member Functions inherited from Termination

- bool DecideUpdate (largecounter UpdateNSteps)

  Returns true if the Termination should update its internal status.

Protected Attributes inherited from Termination

- Geodesic *const m_OwnerGeodesic

  The geodesic that owns the Termination (a const pointer to the Geodesic)
- largecounter m_StepsSinceUpdated {}

  The termination is itself in charge of keeping track of how many steps it has been since it has been updated. The Termination's TerminationOptions struct tells it how many steps it needs to wait between updates.

6.22.1 Detailed Description

GeneralSingularityTermination: Terminate geodesics if they get too close to one of a number of singularities (of arbitrary codimension)

6.22.2 Constructor & Destructor Documentation

6.22.2.1 GeneralSingularityTermination()

```
GeneralSingularityTermination::GeneralSingularityTermination (
            Geodesic *const theGeodesic)  [inline]
```

6.22.3 Member Function Documentation

6.22.3.1 CheckTermination()

```
Term GeneralSingularityTermination::CheckTermination ()  [final], [virtual]
```

Check if the geodesic is too close to a singularity.

Returns

Term

Implements Termination.

6.22.3.2 getFullDescriptionStr()

```
std::string GeneralSingularityTermination::getFullDescriptionStr () const  [final], [virtual]
```

Description string getter for the General Singularity Termination.

Returns

std::string

Implements Termination.

6.22.3.3 SingularityToString()

```
std::string GeneralSingularityTermination::SingularityToString (
            int singnr) const  [private]
```

List the singularities in a human-readable format.

Returns

std::string

6.22.4 Member Data Documentation

6.22.4.1 TermOptions

```
std::unique_ptr< GeneralSingularityTermOptions > GeneralSingularityTermination::TermOptions
[static]
```

Options (contains horizon radius, if we are using logarithmic r coordinate, and distance allowed from the horizon)

The documentation for this class was generated from the following files:

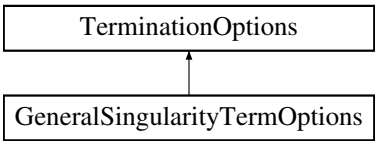
- FOORT/src/Terminations.h
- FOORT/src/Config.cpp
- FOORT/src/Terminations.cpp

6.23 GeneralSingularityTermOptions Struct Reference

Options for GeneralSingularityTermination.

```
#include <Terminations.h>
```

Inheritance diagram for GeneralSingularityTermOptions:



Public Member Functions

- GeneralSingularityTermOptions (std::vector< Singularity > sings, real eps, bool consoleoutputon, bool therlogscale, largecounter Nsteps)

Public Member Functions inherited from TerminationOptions

- TerminationOptions (largecounter Nsteps)
- virtual ~TerminationOptions ()=default

Public Attributes

- const std::vector< Singularity > Singularities

  Vector of singularities.
- const real Epsilon

  Minimum flat distance the geodesic is allowed to be from any singularity.
- const bool OutputToConsole

  Whether to output a message to the console when a singularity is encountered.
- const bool rLogScale

  Whether we are using a logarithmic r coordinate or not.

Public Attributes inherited from TerminationOptions

- const largecounter UpdateEveryNSteps

  Number of steps between updates.

6.23.1 Detailed Description

Options for GeneralSingularityTermination.

Keeps track of the singularities, allowed distance and whether we are using a logarithmic r coordinate.

6.23.2 Constructor & Destructor Documentation

6.23.2.1 GeneralSingularityTermOptions()

```
GeneralSingularityTermOptions::GeneralSingularityTermOptions (
            std::vector< Singularity > sings,
            real eps,
            bool consoleoutputon,
            bool therlogscale,
            largecounter Nsteps)  [inline]
```

6.23.3 Member Data Documentation

6.23.3.1 Epsilon

```
const real GeneralSingularityTermOptions::Epsilon
```

Minimum flat distance the geodesic is allowed to be from any singularity.

6.23.3.2 OutputToConsole

```
const bool GeneralSingularityTermOptions::OutputToConsole
```

Whether to output a message to the console when a singularity is encountered.

6.23.3.3 rLogScale

```
const bool GeneralSingularityTermOptions::rLogScale
```

Whether we are using a logarithmic r coordinate or not.

6.23.3.4 Singularities

```
const std::vector<Singularity> GeneralSingularityTermOptions::Singularities
```

Vector of singularities.

The documentation for this struct was generated from the following file:

- FOORT/src/Terminations.h

6.24 Geodesic Class Reference

Geodesic class: an instance of this class is created for each Geodesic that is integrated.

```
#include <Geodesic.h>
```

Public Member Functions

- Geodesic ()=delete
- Geodesic (const Geodesic &)=delete
- Geodesic & operator= (const Geodesic &)=delete
- Geodesic (const Metric *const theMetric, const Source *const theSource, DiagBitflag diagbit, DiagBitflag valdiagbit, TermBitflag termbit, GeodesicIntegratorFunc theIntegrator)

  Construct a new Geodesic object. Arguments initialize private member variables that remain the same over all geodesics.
- void Reset (ScreenIndex scrindex, Point initpos, OneIndex initvel)

  Geodesic (and descendant classes) functions.
- Term Update ()

  This makes the Geodesic integrate itself one step; then the Geodesic loops through all Terminations and Diagnostics to update.
- Term getTermCondition () const

  Get the current termination condition.
- Point getCurrentPos () const

  Get the current position.
- OneIndex getCurrentVel () const

  Get the current velocity.
- real getCurrentLambda () const

  Get the current value of the affine parameter.
- ScreenIndex getScreenIndex () const

  Get the screen index.
- std::vector< std::string > getAllOutputStr () const

  Get the complete output that should be written to the output files.
- std::vector< real > getDiagnosticFinalValue () const

  Get the "value" (from the Diagnostic that was set to the value Diagnostic) that is associated to the Geodesic.

Private Attributes

- Term m_TermCond {Term::Uninitialized}

  As long as this is Term::Continue, not done integrating yet.
- Point m_CurrentPos {}

  Current position.
- OneIndex m_CurrentVel {}

  Current proper velocity.
- real m_curLambda {0.0}

  Current value of affine parameter (starts at 0.0)
- ScreenIndex m_ScreenIndex {}
- const Metric *const m_theMetric

  const pointer to the Metric
- const Source *const m_theSource

  const pointer to the Source for the rhs of the geodesic equation
- const DiagnosticUniqueVector m_AllDiagnostics

  Vector of unique pointers to Diagnostics. An instance of each Diagnostic is created for the Geodesic, so the Geodesic is the owner of these objects.
- const TerminationUniqueVector m_AllTerminations

  Vector of unique pointers to Terminations. An instance of each Termination is created for the Geodesic, so the Geodesic is the owner of these objects.
- const GeodesicIntegratorFunc m_theIntegrator

  This is the function that will integrate the geodesic equation one step.

6.24.1 Detailed Description

Geodesic class: an instance of this class is created for each Geodesic that is integrated.

The Geodesic is in charge of integrating itself until termination, updating its Diagnostics accordingly, and (after termination) returing the appropriate output

6.24.2 Constructor & Destructor Documentation

6.24.2.1 Geodesic() [1/3]

```
Geodesic::Geodesic ()  [delete]
```

6.24.2.2 Geodesic() [2/3]

```
Geodesic::Geodesic (
            const Geodesic & )  [delete]
```

6.24.2.3 Geodesic() [3/3]

```
Geodesic::Geodesic (
            const Metric *const theMetric,
            const Source *const theSource,
            DiagBitflag diagbit,
            DiagBitflag valdiagbit,
            TermBitflag termbit,
            GeodesicIntegratorFunc theIntegrator)  [inline]
```

Construct a new Geodesic object. Arguments initialize private member variables that remain the same over all geodesics.

Parameters

theMetric	Pointer to the Metric object.
theSource	Pointer to the Source object.
diagbit	Diagnostic bitflag.
valdiagbit	Value diagnostic bitflag.
termbit	Termination bitflag.
theIntegrator	Geoedesic integrator function to use for integrating geodesic equation.

6.24.3 Member Function Documentation

6.24.3.1 getAllOutputStr()

```
std::vector< std::string > Geodesic::getAllOutputStr () const
```

Get the complete output that should be written to the output files.

There is one string more than the count of Diagnostics: one string per Diagnostic, PLUS the first string is the screen index.

Returns

std::vector<std::string>

6.24.3.2 getCurrentLambda()

```
real Geodesic::getCurrentLambda () const
```

Get the current value of the affine parameter.

Returns

real

6.24.3.3 getCurrentPos()

```
Point Geodesic::getCurrentPos () const
```

Get the current position.

Returns

Point

6.24.3.4 getCurrentVel()

```
OneIndex Geodesic::getCurrentVel () const
```

Get the current velocity.

Returns

OneIndex

6.24.3.5 getDiagnosticFinalValue()

```
std::vector< real > Geodesic::getDiagnosticFinalValue () const
```

Get the "value" (from the Diagnostic that was set to the value Diagnostic) that is associated to the Geodesic.

Will be used to determine "distance" between Geodesics which is used in Mesh refinement.

Returns

std::vector<real>

6.24.3.6 getScreenIndex()

```
ScreenIndex Geodesic::getScreenIndex () const
```

Get the screen index.

Returns

ScreenIndex

6.24.3.7 getTermCondition()

```
Term Geodesic::getTermCondition () const
```

Get the current termination condition.

Returns

Term

6.24.3.8 operator=()

```
Geodesic & Geodesic::operator= (
            const Geodesic & )  [delete]
```

6.24.3.9 Reset()

```
void Geodesic::Reset (
            ScreenIndex scrindex,
            Point initpos,
            OneIndex initvel)
```

Geodesic (and descendant classes) functions.

This initializes/resets the geodesic with a given ScreenIndex, initial position, and initial velocity

Also resets all Diagnostics and Terminations, resets the TermCondition to Term::Continue, and puts the Geodesic back to lambda = 0.0.

Parameters

scrindex	ScreenIndex
initpos	Initial position
initvel	Initial velocity

6.24.3.10 Update()

```
Term Geodesic::Update ()
```

This makes the Geodesic integrate itself one step; then the Geodesic loops through all Terminations and Diagnostics to update.

Returns

Term

6.24.4 Member Data Documentation

6.24.4.1 m_AllDiagnostics

```
const DiagnosticUniqueVector Geodesic::m_AllDiagnostics  [private]
```

Vector of unique pointers to Diagnostics. An instance of each Diagnostic is created for the Geodesic, so the Geodesic is the owner of these objects.

6.24.4.2 m_AllTerminations

```
const TerminationUniqueVector Geodesic::m_AllTerminations  [private]
```

Vector of unique pointers to Terminations. An instance of each Termination is created for the Geodesic, so the Geodesic is the owner of these objects.

6.24.4.3 m_curLambda

```
real Geodesic::m_curLambda {0.0}  [private]
```

Current value of affine parameter (starts at 0.0)

6.24.4.4 m_CurrentPos

```
Point Geodesic::m_CurrentPos {}  [private]
```

Current position.

6.24.4.5 m_CurrentVel

```
OneIndex Geodesic::m_CurrentVel {}  [private]
```

Current proper velocity.

6.24.4.6 m_ScreenIndex

```
ScreenIndex Geodesic::m_ScreenIndex {}  [private]
```

The Geodesic keeps track of what index it has been assigned; it outputs this information in its final output string

6.24.4.7 m_TermCond

```
Term Geodesic::m_TermCond {Term::Uninitialized}  [private]
```

As long as this is Term::Continue, not done integrating yet.

6.24.4.8 m_theIntegrator

```
const GeodesicIntegratorFunc Geodesic::m_theIntegrator  [private]
```

This is the function that will integrate the geodesic equation one step.

6.24.4.9 m_theMetric

```
const Metric* const Geodesic::m_theMetric  [private]
```

const pointer to the Metric

6.24.4.10 m_theSource

```
const Source* const Geodesic::m_theSource  [private]
```

const pointer to the Source for the rhs of the geodesic equation

The documentation for this class was generated from the following files:

- FOORT/src/Geodesic.h
- FOORT/src/Geodesic.cpp

6.25 GeodesicOutputHandler Class Reference

// GeodesicOutputHandler handles all of the output to file. It gets passed all of the output strings for every Geodesic, it then stores this data until it eventually writes all data to the appropriate files

```
#include <InputOutput.h>
```

Public Member Functions

- GeodesicOutputHandler ()=delete
- GeodesicOutputHandler (std::string FilePrefix, std::string TimeStamp, std::string FileExtension, std::vector< std::string > DiagNames, largecounter nroutputstocache=LARGECOUNTER_MAX - 1, largecounter geodperfile=LARGECOUNTER_MAX, std::string firstlineinfo="")

  Construct a new Geodesic Output Handler object and initialize all const member variables.
- void PrepareForOutput (largecounter nrOutputToCome)

  prepare the cache for output
- void NewGeodesicOutput (largecounter index, std::vector< std::string > theOutput)
- void OutputFinished ()

  There is no more output, so we write anything that is cached to file to clean up and finalize.
- std::string getFullDescriptionStr () const

  full description string getter for the geodesic output handler

Private Member Functions

- void WriteCachedOutputToFile ()

  Write cached output to a file.
- void OpenForFirstTime (const std::string &filename)

  We check here to see if the files are being put in a (sub)directory; if so, we create the directory/directories first to make sure creating/opening the file will succeed.
- std::string GetFileName (int diagnr, unsigned short filenr) const

  Get the file name based on the various parts of the file name stored in the member variables.

Private Attributes

- const std::string m_FilePrefix

  Const string for file prefix.
- const std::string m_TimeStamp

  Const string for timestamp.
- const std::string m_FileExtension

  Const string for file extension.
- const std::vector< std::string > m_DiagNames

  Const string for the diagnostic names.
- const bool m_PrintFirstLineInfo

  Const variable that control whether we write a descriptive first line in every output file or not.
- const std::string m_FirstLineInfoString

  Const string for the first line info.
- const largecounter m_nrOutputsToCache {}
- const largecounter m_nrGeodesicsPerFile {}

  Const setting the max number of geodesics per file.
- bool m_WriteToConsole {false}

  If this is false, then we are writing to file(s). At any time, if a file I/O error occurs, the output handler switches to outputting everything to the console.
- largecounter m_PrevCached {0}

  How many outputs are already cached in m_nrOutputsToCache before the current iteration of output.
- largecounter m_CurrentGeodesicsInFile {0}

  The number of geodesics already written to the current file (once this hits m_nrGeodesicsPerFile, this file is full)
- unsigned short m_CurrentFullFiles {0}

  The current counter of completely full files (this is kept track of so that it knows what the next file to write output to is). (We had better not have more than 60k files!)
- std::vector< std::vector< std::string > > m_AllCachedData {}

  Cached data that has not been written to a file yet (once this hits a size of > m_nrOutputsToCache, this must be written to file(s))

6.25.1 Detailed Description

// GeodesicOutputHandler handles all of the output to file. It gets passed all of the output strings for every Geodesic, it then stores this data until it eventually writes all data to the appropriate files

6.25.2 Constructor & Destructor Documentation

6.25.2.1 GeodesicOutputHandler() [1/2]

```
GeodesicOutputHandler::GeodesicOutputHandler ()  [delete]
```

6.25.2.2 GeodesicOutputHandler() [2/2]

```
GeodesicOutputHandler::GeodesicOutputHandler (
            std::string FilePrefix,
            std::string TimeStamp,
            std::string FileExtension,
            std::vector< std::string > DiagNames,
            largecounter nroutputstocache = LARGECOUNTER_MAX - 1,
            largecounter geodperfile = LARGECOUNTER_MAX,
            std::string firstlineinfo = "")
```

Construct a new Geodesic Output Handler object and initialize all const member variables.

Parameters

FilePrefix	file prefix
TimeStamp	time stamp
FileExtension	file extensions
DiagNames	diagnostics names
nroutputstocache	number of geodesics to put to the cache
geodperfile	max number of geodesics per output file
firstlineinfo	the info to print on the first line

6.25.3 Member Function Documentation

6.25.3.1 GetFileName()

```
std::string GeodesicOutputHandler::GetFileName (
            int diagnr,
            unsigned short filenr) const  [private]
```

Get the file name based on the various parts of the file name stored in the member variables.

Parameters

diagnr	
filenr	

Returns

std::string

6.25.3.2 getFullDescriptionStr()

```
std::string GeodesicOutputHandler::getFullDescriptionStr () const
```

full description string getter for the geodesic output handler

Returns

std::string

6.25.3.3 NewGeodesicOutput()

```
void GeodesicOutputHandler::NewGeodesicOutput (
            largecounter index,
            std::vector< std::string > theOutput)
```

Parameters

index	
theOutput	

6.25.3.4 OpenForFirstTime()

```
void GeodesicOutputHandler::OpenForFirstTime (
            const std::string & filename)  [private]
```

We check here to see if the files are being put in a (sub)directory; if so, we create the directory/directories first to make sure creating/opening the file will succeed.

Parameters

filename	Filename to be found / created

6.25.3.5 OutputFinished()

```
void GeodesicOutputHandler::OutputFinished ()
```

There is no more output, so we write anything that is cached to file to clean up and finalize.

6.25.3.6 PrepareForOutput()

```
void GeodesicOutputHandler::PrepareForOutput (
            largecounter nrOutputToCome)
```

prepare the cache for output

Parameters

nrOutputToCome	number of outputs to come into the cache

6.25.3.7 WriteCachedOutputToFile()

```
void GeodesicOutputHandler::WriteCachedOutputToFile ()  [private]
```

Write cached output to a file.

6.25.4 Member Data Documentation

6.25.4.1 m_AllCachedData

```
std::vector<std::vector<std::string> > GeodesicOutputHandler::m_AllCachedData {}  [private]
```

Cached data that has not been written to a file yet (once this hits a size of > m_nrOutputsToCache, this must be written to file(s))

6.25.4.2 m_CurrentFullFiles

```
unsigned short GeodesicOutputHandler::m_CurrentFullFiles {0}  [private]
```

The current counter of completely full files (this is kept track of so that it knows what the next file to write output to is). (We had better not have more than 60k files!)

6.25.4.3 m_CurrentGeodesicsInFile

```
largecounter GeodesicOutputHandler::m_CurrentGeodesicsInFile {0}  [private]
```

The number of geodesics already written to the current file (once this hits m_nrGeodesicsPerFile, this file is full)

6.25.4.4 m_DiagNames

```
const std::vector<std::string> GeodesicOutputHandler::m_DiagNames  [private]
```

Const string for the diagnostic names.

6.25.4.5 m_FileExtension

```
const std::string GeodesicOutputHandler::m_FileExtension  [private]
```

Const string for file extension.

6.25.4.6 m_FilePrefix

```
const std::string GeodesicOutputHandler::m_FilePrefix  [private]
```

Const string for file prefix.

6.25.4.7 m_FirstLineInfoString

```
const std::string GeodesicOutputHandler::m_FirstLineInfoString  [private]
```

Const string for the first line info.

6.25.4.8 m_nrGeodesicsPerFile

```
const largecounter GeodesicOutputHandler::m_nrGeodesicsPerFile {}  [private]
```

Const setting the max number of geodesics per file.

6.25.4.9 m_nrOutputsToCache

```
const largecounter GeodesicOutputHandler::m_nrOutputsToCache {}  [private]
```

6.25.4.10 m_PrevCached

```
largecounter GeodesicOutputHandler::m_PrevCached {0}  [private]
```

How many outputs are already cached in m_nrOutputsToCache before the current iteration of output.

6.25.4.11 m_PrintFirstLineInfo

```
const bool GeodesicOutputHandler::m_PrintFirstLineInfo  [private]
```

Const variable that control whether we write a descriptive first line in every output file or not.

6.25.4.12 m_TimeStamp

```
const std::string GeodesicOutputHandler::m_TimeStamp  [private]
```

Const string for timestamp.

6.25.4.13 m_WriteToConsole

```
bool GeodesicOutputHandler::m_WriteToConsole {false}  [private]
```

If this is false, then we are writing to file(s). At any time, if a file I/O error occurs, the output handler switches to outputting everything to the console.

The documentation for this class was generated from the following files:

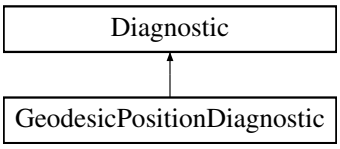
- FOORT/src/InputOutput.h
- FOORT/src/InputOutput.cpp

6.26 GeodesicPositionDiagnostic Class Reference

Geodesic position tracker.

```
#include <Diagnostics.h>
```

Inheritance diagram for GeodesicPositionDiagnostic:



Public Member Functions

- GeodesicPositionDiagnostic (Geodesic *const theGeodesic)
- void Reset () final

  GeodesicPositionDiagnostic functions.
- void UpdateData () final

  Update the saved points vector with the current position of the geodesic.
- std::string getFullDataStr () const final

  Return full data string of all saved points.
- std::vector< real > getFinalDataVal () const final

  Return the final (theta, phi) value of the geodesic.
- real FinalDataValDistance (const std::vector< real > &val1, const std::vector< real > &val2) const final

  Return the angular distance between two geodesics.
- std::string getNameStr () const final

  Simple name without spaces.
- std::string getFullDescriptionStr () const final

  More descriptive string (with spaces)

Public Member Functions inherited from Diagnostic

- Diagnostic ()=delete
- Diagnostic (Geodesic *const theGeodesic)
- virtual ~Diagnostic ()=default

Static Public Attributes

- static std::unique_ptr< GeodesicPositionOptions > DiagOptions

Private Attributes

- std::vector< Point > m_AllSavedPoints {}

Additional Inherited Members

Protected Member Functions inherited from Diagnostic

- bool DecideUpdate (const UpdateFrequency &myUpdateFrequency)

  Returns true if the Diagnostic should update its initial status.

Protected Attributes inherited from Diagnostic

- Geodesic *const m_OwnerGeodesic
- largecounter m_StepsSinceUpdated {}

6.26.1 Detailed Description

Geodesic position tracker.

6.26.2 Constructor & Destructor Documentation

6.26.2.1 GeodesicPositionDiagnostic()

```
GeodesicPositionDiagnostic::GeodesicPositionDiagnostic (
            Geodesic *const theGeodesic)  [inline]
```

6.26.3 Member Function Documentation

6.26.3.1 FinalDataValDistance()

```
real GeodesicPositionDiagnostic::FinalDataValDistance (
            const std::vector< real > & val1,
            const std::vector< real > & val2) const  [final], [virtual]
```

Return the angular distance between two geodesics.

This is based on their final angles on the boundary sphere (theta, phi)

Parameters

val1	first geodesic final position
val2	second geodesic final position

Returns

real

Implements Diagnostic.

6.26.3.2 getFinalDataVal()

```
std::vector< real > GeodesicPositionDiagnostic::getFinalDataVal () const  [final], [virtual]
```

Return the final (theta, phi) value of the geodesic.

Returns

std::vector<real>

Implements Diagnostic.

6.26.3.3 getFullDataStr()

```
std::string GeodesicPositionDiagnostic::getFullDataStr () const  [final], [virtual]
```

Return full data string of all saved points.

The full output string looks like this: "(total nr steps) ;; (step 1) (step 2) (step 3) ..." where each step is a (space-separated) output of the geodesic coordinates at that step

Returns

std::string

Implements Diagnostic.

6.26.3.4 getFullDescriptionStr()

```
std::string GeodesicPositionDiagnostic::getFullDescriptionStr () const  [final], [virtual]
```

More descriptive string (with spaces)

Returns

std::string

Reimplemented from Diagnostic.

6.26.3.5 getNameStr()

```
std::string GeodesicPositionDiagnostic::getNameStr () const  [final], [virtual]
```

Simple name without spaces.

Returns

std::string

Implements Diagnostic.

6.26.3.6 Reset()

```
void GeodesicPositionDiagnostic::Reset ()  [final], [virtual]
```

GeodesicPositionDiagnostic functions.

Empty out vector of points and call base Reset function

Reimplemented from Diagnostic.

6.26.3.7 UpdateData()

```
void GeodesicPositionDiagnostic::UpdateData ()  [final], [virtual]
```

Update the saved points vector with the current position of the geodesic.

Implements Diagnostic.

6.26.4 Member Data Documentation

6.26.4.1 DiagOptions

```
std::unique_ptr< GeodesicPositionOptions > GeodesicPositionDiagnostic::DiagOptions  [static]
```

6.26.4.2 m_AllSavedPoints

```
std::vector<Point> GeodesicPositionDiagnostic::m_AllSavedPoints {}  [private]
```

The documentation for this class was generated from the following files:

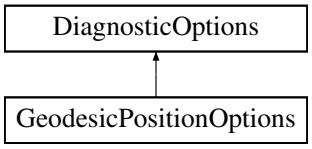
- FOORT/src/Diagnostics.h
- FOORT/src/Config.cpp
- FOORT/src/Diagnostics.cpp

6.27 GeodesicPositionOptions Struct Reference

Options for GeodesicPositionDiagnostic.

`#include <Diagnostics.h>`

Inheritance diagram for GeodesicPositionOptions:



Public Member Functions

- GeodesicPositionOptions (largecounter outputsteps, UpdateFrequency thefrequency)

Public Member Functions inherited from DiagnosticOptions

- DiagnosticOptions (UpdateFrequency thefrequency)
- virtual ~DiagnosticOptions ()=default

Public Attributes

- const largecounter OutputNrSteps

Public Attributes inherited from DiagnosticOptions

- const UpdateFrequency theUpdateFrequency

6.27.1 Detailed Description

Options for GeodesicPositionDiagnostic.

Inherits from DiagnosticOptions, and adds the number of steps to output.

6.27.2 Constructor & Destructor Documentation

6.27.2.1 GeodesicPositionOptions()

```
GeodesicPositionOptions::GeodesicPositionOptions (
            largecounter outputsteps,
            UpdateFrequency thefrequency)  [inline]
```

6.27.3 Member Data Documentation

6.27.3.1 OutputNrSteps

```
const largecounter GeodesicPositionOptions::OutputNrSteps
```

The documentation for this struct was generated from the following file:

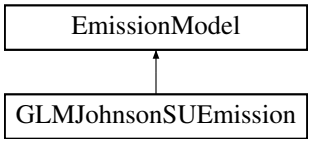
- FOORT/src/Diagnostics.h

6.28 GLMJohnsonSUEmission Struct Reference

The Johnson SU emission model used in GLM 2020, PRD 102(12):124004.

```
#include <DiagnosticsEmission.h>
```

Inheritance diagram for GLMJohnsonSUEmission:



Public Member Functions

- GLMJohnsonSUEmission (real mu, real gamma, real sigma)
- real GetEmission (const Point &p) const final

  Get the emission at a point according to the Johnson SU model used in GLM.
- std::string getFullDescriptionStr () const final

  Description string getter for the Johnson SU emission model.

Public Member Functions inherited from EmissionModel

- virtual ~EmissionModel ()=default

Private Attributes

- const real m_mu

  mu parameter
- const real m_gamma

  gamma parameter
- const real m_sigma

  sigma parameter

6.28.1 Detailed Description

The Johnson SU emission model used in GLM 2020, PRD 102(12):124004.

6.28.2 Constructor & Destructor Documentation

6.28.2.1 GLMJohnsonSUEmission()

```
GLMJohnsonSUEmission::GLMJohnsonSUEmission (
            real mu,
            real gamma,
            real sigma)  [inline]
```

6.28.3 Member Function Documentation

6.28.3.1 GetEmission()

```
real GLMJohnsonSUEmission::GetEmission (
            const Point & p) const  [final], [virtual]
```

Get the emission at a point according to the Johnson SU model used in GLM.

Parameters

p	Point at which to calculate the emission

Returns

real Emission at the point

Implements EmissionModel.

6.28.3.2 getFullDescriptionStr()

```
std::string GLMJohnsonSUEmission::getFullDescriptionStr () const  [final], [virtual]
```

Description string getter for the Johnson SU emission model.

Returns

std::string

Reimplemented from EmissionModel.

6.28.4 Member Data Documentation

6.28.4.1 m_gamma

```
const real GLMJohnsonSUEmission::m_gamma  [private]
```

gamma parameter

6.28.4.2 m_mu

```
const real GLMJohnsonSUEmission::m_mu  [private]
```

mu parameter

6.28.4.3 m_sigma

```
const real GLMJohnsonSUEmission::m_sigma  [private]
```

sigma parameter

The documentation for this struct was generated from the following files:

- FOORT/src/DiagnosticsEmission.h
- FOORT/src/DiagnosticsEmission.cpp

6.29 Grid Class Reference

A class for a 2D grid that contains the values of the necessary functions.

```
#include <Grid.h>
```

Public Member Functions

- Grid (int N_row, int N_col)
  A class for a 2D grid that contains the values of the necessary functions.
- ~Grid ()
  Destructor.
- double & operator() (int i, int j)
  Overload the () operator to access the data.
- const double & operator() (int i, int j) const
  Overload the () operator to access the data (const version)
- void initialize_from_file (std::string file)

Public Attributes

- int N_row

  Number of rows in the grid.
- int N_col

  Number of columns in the grid.
- double * data

  Pointer to the data.
- int size

  Number of rows times the number of columns.

6.29.1 Detailed Description

A class for a 2D grid that contains the values of the necessary functions.

6.29.2 Constructor & Destructor Documentation

6.29.2.1 Grid()

```
Grid::Grid (
            int N_row,
            int N_col)
```

A class for a 2D grid that contains the values of the necessary functions.

6.29.2.2 ~Grid()

```
Grid::~Grid ()  [inline]
```

Destructor.

6.29.3 Member Function Documentation

6.29.3.1 initialize_from_file()

```
void Grid::initialize_from_file (
            std::string file)
```

6.29.3.2 operator()() [1/2]

```
double & Grid::operator() (
            int i,
            int j)  [inline]
```

Overload the () operator to access the data.

6.29.3.3 operator()() [2/2]

```
const double & Grid::operator() (
            int i,
            int j) const  [inline]
```

Overload the () operator to access the data (const version)

6.29.4 Member Data Documentation

6.29.4.1 data

```
double* Grid::data
```

Pointer to the data.

6.29.4.2 N_col

```
int Grid::N_col
```

Number of columns in the grid.

6.29.4.3 N_row

```
int Grid::N_row
```

Number of rows in the grid.

6.29.4.4 size

```
int Grid::size
```

Number of rows times the number of columns.

The documentation for this class was generated from the following files:

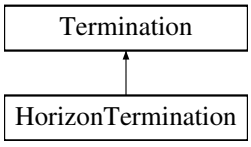
- FOORT/src/Grid.h
- FOORT/src/Grid.cpp

6.30 HorizonTermination Class Reference

HorizonTermination: Terminate geodesics if they get too close to the horizon.

```
#include <Terminations.h>
```

Inheritance diagram for HorizonTermination:



Public Member Functions

- HorizonTermination (Geodesic ∗const theGeodesic)
- Term CheckTermination () final

  Termination checker for the horizon termination.
- std::string getFullDescriptionStr () const final

  Description string getter for the Horizon Termination.

Public Member Functions inherited from Termination

- Termination ()=delete
- Termination (Geodesic ∗const theGeodesic)
- virtual void Reset ()

  Reset to 0 to start integrating a new geodesic.
- virtual ∼Termination ()=default

Static Public Attributes

- static std::unique_ptr< HorizonTermOptions > TermOptions

  Options (contains horizon radius, if we are using logarithmic r coordinate, and distance allowed from the horizon)

Additional Inherited Members

Protected Member Functions inherited from Termination

- bool DecideUpdate (largecounter UpdateNSteps)

  Returns true if the Termination should update its internal status.

Protected Attributes inherited from Termination

- Geodesic ∗const m_OwnerGeodesic

  The geodesic that owns the Termination (a const pointer to the Geodesic)
- largecounter m_StepsSinceUpdated {}

  The termination is itself in charge of keeping track of how many steps it has been since it has been updated. The Termination's TerminationOptions struct tells it how many steps it needs to wait between updates.

6.30.1 Detailed Description

HorizonTermination: Terminate geodesics if they get too close to the horizon.

6.30.2 Constructor & Destructor Documentation

6.30.2.1 HorizonTermination()

```
HorizonTermination::HorizonTermination (
            Geodesic *const theGeodesic)  [inline]
```

6.30.3 Member Function Documentation

6.30.3.1 CheckTermination()

```
Term HorizonTermination::CheckTermination ()  [final], [virtual]
```

Termination checker for the horizon termination.

Returns

Term

Implements Termination.

6.30.3.2 getFullDescriptionStr()

```
std::string HorizonTermination::getFullDescriptionStr () const  [final], [virtual]
```

Description string getter for the Horizon Termination.

Returns

std::string

Implements Termination.

6.30.4 Member Data Documentation

6.30.4.1 TermOptions

```
std::unique_ptr< HorizonTermOptions > HorizonTermination::TermOptions  [static]
```

Options (contains horizon radius, if we are using logarithmic r coordinate, and distance allowed from the horizon)

The documentation for this class was generated from the following files:

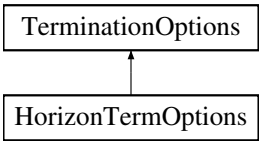
- FOORT/src/Terminations.h
- FOORT/src/Config.cpp
- FOORT/src/Terminations.cpp

6.31 HorizonTermOptions Struct Reference

Options for HorizonTermination.

```
#include <Terminations.h>
```

Inheritance diagram for HorizonTermOptions:



Public Member Functions

- HorizonTermOptions (real theHorizonRadius, bool therLogScale, real theAtHorizonEps, largecounter Nsteps)

Public Member Functions inherited from TerminationOptions

- TerminationOptions (largecounter Nsteps)
- virtual ~TerminationOptions ()=default

Public Attributes

- const real HorizonRadius

  The radius of the horizon.
- const real AtHorizonEps

  The distance allowed from the horizon.
- const bool rLogScale

  Whether we are using a logarithmic r coordinate.

Public Attributes inherited from TerminationOptions

- const largecounter UpdateEveryNSteps

  Number of steps between updates.

6.31.1 Detailed Description

Options for HorizonTermination.

Keeps track of the horizon radius, whether we are using a logarithmic r coordinate, and the distance allowed from the horizon.

6.31.2 Constructor & Destructor Documentation

6.31.2.1 HorizonTermOptions()

```
HorizonTermOptions::HorizonTermOptions (
            real theHorizonRadius,
            bool therLogScale,
            real theAtHorizonEps,
            largecounter Nsteps)  [inline]
```

6.31.3 Member Data Documentation

6.31.3.1 AtHorizonEps

```
const real HorizonTermOptions::AtHorizonEps
```

The distance allowed from the horizon.

6.31.3.2 HorizonRadius

```
const real HorizonTermOptions::HorizonRadius
```

The radius of the horizon.

6.31.3.3 rLogScale

```
const bool HorizonTermOptions::rLogScale
```

Whether we are using a logarithmic r coordinate.

The documentation for this struct was generated from the following file:

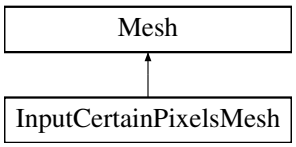
- FOORT/src/Terminations.h

6.32 InputCertainPixelsMesh Class Reference

Mesh which integrates only certain user-inputted pixels.

```
#include <Mesh.h>
```

Inheritance diagram for InputCertainPixelsMesh:



Public Member Functions

- InputCertainPixelsMesh ()=delete
- InputCertainPixelsMesh (const InputCertainPixelsMesh &)=delete
- InputCertainPixelsMesh (largecounter totalPixels, DiagBitflag valdiag)

  Constructor for InputCertainPixelsMesh.
- largecounter getCurNrGeodesics () const final

  Get the current number of geodesics to be integrated.
- void getNewInitConds (largecounter index, ScreenPoint &newunitpoint, ScreenIndex &newscreenindex) const final

  Get the initial conditions for the next pixel to be integrated.
- void GeodesicFinished (largecounter index, std::vector< real > finalValues) final

  Function that is called when a geodesic is finished integrating.
- void EndCurrentLoop () final

  End the current loop of integrating pixels.
- bool IsFinished () const final

  Check whether the Mesh is finished integrating.
- std::string getFullDescriptionStr () const final

  Description string getter.

Public Member Functions inherited from Mesh

- Mesh (DiagBitflag valdiag)
- virtual ~Mesh ()=default

Private Attributes

- const pixelcoord m_RowColumnSize

  Total pixel size of the screen (square grid)
- largecounter m_TotalPixels {0}

  How many pixels have been inputted in total, i.e. need integrating.
- std::vector< ScreenIndex > m_PixelsToIntegrate {}

  All pixels' location.
- bool m_Finished {false}

  Are we finished integrating?

Additional Inherited Members

Protected Attributes inherited from Mesh

- const std::unique_ptr< const Diagnostic > m_DistanceDiagnostic

  The Diagnostic (a const pointer to a const Diagnostic object) that is used to calculate distances (using FinalData↩ ValDistance()) between the "values" that are assigned to Geodesics.

6.32.1 Detailed Description

Mesh which integrates only certain user-inputted pixels.

6.32.2 Constructor & Destructor Documentation

6.32.2.1 InputCertainPixelsMesh() [1/3]

```
InputCertainPixelsMesh::InputCertainPixelsMesh ()  [delete]
```

6.32.2.2 InputCertainPixelsMesh() [2/3]

```
InputCertainPixelsMesh::InputCertainPixelsMesh (
            const InputCertainPixelsMesh & )  [delete]
```

6.32.2.3 InputCertainPixelsMesh() [3/3]

```
InputCertainPixelsMesh::InputCertainPixelsMesh (
            largecounter totalPixels,
            DiagBitflag valdiag)
```

Constructor for InputCertainPixelsMesh.

Constructor will ask for pixels to be inputted by the user through the console

Parameters

totalPixels	total number of pixels in the grid
valdiag	value diagnostic bitflag

6.32.3 Member Function Documentation

6.32.3.1 EndCurrentLoop()

```
void InputCertainPixelsMesh::EndCurrentLoop ()  [final], [virtual]
```

End the current loop of integrating pixels.

This Mesh only has one loop; we are now finished integrating

Implements Mesh.

6.32.3.2 GeodesicFinished()

```
void InputCertainPixelsMesh::GeodesicFinished (
            largecounter index,
            std::vector< real > finalValues)  [final], [virtual]
```

Function that is called when a geodesic is finished integrating.

This Mesh doesn't actually have to do anything with the geodesic (values) when it is done!

Parameters

index	index of the geodesic that is finished
finalValues	final values of the geodesic

Implements Mesh.

6.32.3.3 getCurNrGeodesics()

```
largecounter InputCertainPixelsMesh::getCurNrGeodesics () const  [final], [virtual]
```

Get the current number of geodesics to be integrated.

This is the total number of pixels in the grid

Returns

largecounter

Implements Mesh.

6.32.3.4 getFullDescriptionStr()

```
std::string InputCertainPixelsMesh::getFullDescriptionStr () const  [final], [virtual]
```

Description string getter.

Returns

std::string

Reimplemented from Mesh.

6.32.3.5 getNewInitConds()

```
void InputCertainPixelsMesh::getNewInitConds (
            largecounter index,
            ScreenPoint & newunitpoint,
            ScreenIndex & newscreenindex) const  [final], [virtual]
```

Get the initial conditions for the next pixel to be integrated.

The next pixel is the one with index m_CurrentPixel

Parameters

index	index of the pixel to be initialized
newunitpoint	new unit point to be initialized
newscreenindex	new screen index to be initialized

Implements Mesh.

6.32.3.6 IsFinished()

```
bool InputCertainPixelsMesh::IsFinished () const  [final], [virtual]
```

Check whether the Mesh is finished integrating.

Returns

true

false

Implements Mesh.

6.32.4 Member Data Documentation

6.32.4.1 m_Finished

```
bool InputCertainPixelsMesh::m_Finished {false}  [private]
```

Are we finished integrating?

6.32.4.2 m_PixelsToIntegrate

```
std::vector<ScreenIndex> InputCertainPixelsMesh::m_PixelsToIntegrate {}  [private]
```

All pixels' location.

6.32.4.3 m_RowColumnSize

```
const pixelcoord InputCertainPixelsMesh::m_RowColumnSize  [private]
```

Total pixel size of the screen (square grid)

6.32.4.4 m_TotalPixels

```
largecounter InputCertainPixelsMesh::m_TotalPixels {0}  [private]
```

How many pixels have been inputted in total, i.e. need integrating.

The documentation for this class was generated from the following files:

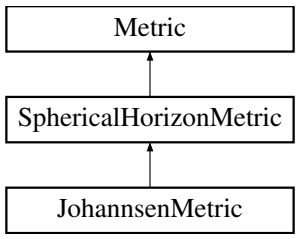
- FOORT/src/Mesh.h
- FOORT/src/Mesh.cpp

6.33 JohannsenMetric Class Reference

Johannsen black hole metric.

`#include <Metric.h>`

Inheritance diagram for JohannsenMetric:



Public Member Functions

- JohannsenMetric ()=delete
- JohannsenMetric (real aParam, real alpha13Param, real alpha22Param, real alpha52Param, real eps3Param, bool rLogScale=false)

  Construct a new Johannsen Metric object. We always have a M = 1 BH.
- TwoIndex getMetric_dd (const Point &p) const final

  Johannsen metric getter, indices down.
- TwoIndex getMetric_uu (const Point &p) const final

  Johannsen metric getter, indices up.
- std::string getFullDescriptionStr () const final

  Johannsen metric description string getter.

Public Member Functions inherited from SphericalHorizonMetric

- SphericalHorizonMetric ()=delete
- SphericalHorizonMetric (real HorizonRadius, bool rLogScale)

  Construct a new Spherical Horizon Metric object.
- real getHorizonRadius () const

  Get the radius of the horizon.

Public Member Functions inherited from Metric

- virtual ~Metric ()=default
- Metric (bool rlogscale=false)

  Construct a new Metric object.
- virtual ThreeIndex getChristoffel_udd (const Point &p) const

  Return the Christoffel symbols of the metric (indices up, down, down)
- virtual FourIndex getRiemann_uddd (const Point &p) const

  Riemann tensor of the metric (indices up, down, down, down)
- virtual real getKretschmann (const Point &p) const

  Get the Kretschmann scalar at a point.
- bool getrLogScale () const

  Get whether we are using a logarithmic radial scale.

Private Attributes

- const real m_aParam

  a parameter for the Johannsen metric
- const real m_alpha13Param

  alpha13 parameter for the Johannsen metric
- const real m_alpha22Param

  alpha22 parameter for the Johannsen metric
- const real m_alpha52Param

  alpha52 parameter for the Johannsen metric
- const real m_eps3Param

  eps3 parameter for the Johannsen metric

Additional Inherited Members

Protected Attributes inherited from SphericalHorizonMetric

- const real m_HorizonRadius

  Radius of the horizon.

Protected Attributes inherited from Metric

- std::vector< int > m_Symmetries {}

  The symmetries (coordinate Killing vectors) of the metric. Should be set by descendant constructor.
- const bool m_rLogScale

  Are we using a logarithmic r coordinate?

6.33.1 Detailed Description

Johannsen black hole metric.

Note

Implementation by Seppe Staelens

6.33.2 Constructor & Destructor Documentation

6.33.2.1 JohannsenMetric() [1/2]

```
JohannsenMetric::JohannsenMetric ()  [delete]
```

6.33.2.2 JohannsenMetric() [2/2]

```
JohannsenMetric::JohannsenMetric (
            real aParam,
            real alpha13Param,
            real alpha22Param,
            real alpha52Param,
            real eps3Param,
            bool rLogScale = false)
```

Construct a new Johannsen Metric object. We always have a M = 1 BH.

Parameters

aParam	a parameter for the Johannsen metric
alpha13Param	alpha13 parameter for the Johannsen metric
alpha22Param	alpha22 parameter for the Johannsen metric
alpha52Param	alpha52 parameter for the Johannsen metric
eps3Param	eps3 parameter for the Johannsen metric
rLogScale	whether we are using a logarithmic radial scale

Note

Implementation by Seppe Staelens

6.33.3 Member Function Documentation

6.33.3.1 getFullDescriptionStr()

```
std::string JohannsenMetric::getFullDescriptionStr () const  [final], [virtual]
```

Johannsen metric description string getter.

Returns

std::string

Reimplemented from Metric.

6.33.3.2 getMetric_dd()

```
TwoIndex JohannsenMetric::getMetric_dd (
            const Point & p) const  [final], [virtual]
```

Johannsen metric getter, indices down.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.33.3.3 getMetric_uu()

```
TwoIndex JohannsenMetric::getMetric_uu (
            const Point & p) const  [final], [virtual]
```

Johannsen metric getter, indices up.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.33.4 Member Data Documentation

6.33.4.1 m_alpha13Param

```
const real JohannsenMetric::m_alpha13Param  [private]
```

alpha13 parameter for the Johannsen metric

6.33.4.2 m_alpha22Param

```
const real JohannsenMetric::m_alpha22Param  [private]
```

alpha22 parameter for the Johannsen metric

6.33.4.3 m_alpha52Param

```
const real JohannsenMetric::m_alpha52Param  [private]
```

alpha52 parameter for the Johannsen metric

6.33.4.4 m_aParam

```
const real JohannsenMetric::m_aParam  [private]
```

a parameter for the Johannsen metric

6.33.4.5 m_eps3Param

```
const real JohannsenMetric::m_eps3Param  [private]
```

eps3 parameter for the Johannsen metric

The documentation for this class was generated from the following files:

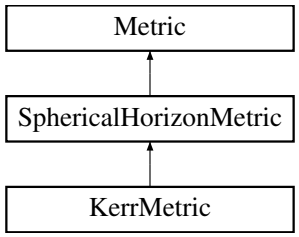
- FOORT/src/Metric.h
- FOORT/src/Metric.cpp

6.34 KerrMetric Class Reference

The Kerr metric.

```
#include <Metric.h>
```

Inheritance diagram for KerrMetric:



Public Member Functions

- KerrMetric ()=delete
- KerrMetric (real aParam, bool rLogScale=false, real mParam=1.)

  Construct a new Kerr Metric object.
- TwoIndex getMetric_dd (const Point &p) const final

  Kerr metric getter, indices down.
- TwoIndex getMetric_uu (const Point &p) const final

  Kerr metric getter, indices up.
- std::string getFullDescriptionStr () const final

  Kerr metric description string getter.

Public Member Functions inherited from SphericalHorizonMetric

- SphericalHorizonMetric ()=delete
- SphericalHorizonMetric (real HorizonRadius, bool rLogScale)

  Construct a new Spherical Horizon Metric object.
- real getHorizonRadius () const

  Get the radius of the horizon.

Public Member Functions inherited from Metric

- virtual ~Metric ()=default
- Metric (bool rlogscale=false)

  Construct a new Metric object.
- virtual ThreeIndex getChristoffel_udd (const Point &p) const

  Return the Christoffel symbols of the metric (indices up, down, down)
- virtual FourIndex getRiemann_uddd (const Point &p) const

  Riemann tensor of the metric (indices up, down, down, down)
- virtual real getKretschmann (const Point &p) const

  Get the Kretschmann scalar at a point.
- bool getrLogScale () const

  Get whether we are using a logarithmic radial scale.

Private Attributes

- const real m_aParam

  Mass-rescaled rotation parameter for Kerr. Note that this should be between -1 and 1.
- const real m_mParam

  Mass parameter for Kerr. Default is 1.

Additional Inherited Members

Protected Attributes inherited from SphericalHorizonMetric

- const real m_HorizonRadius

  Radius of the horizon.

Protected Attributes inherited from Metric

- std::vector< int > m_Symmetries {}

  The symmetries (coordinate Killing vectors) of the metric. Should be set by descendant constructor.
- const bool m_rLogScale

  Are we using a logarithmic r coordinate?

6.34.1 Detailed Description

The Kerr metric.

6.34.2 Constructor & Destructor Documentation

6.34.2.1 KerrMetric() [1/2]

```
KerrMetric::KerrMetric ()  [delete]
```

6.34.2.2 KerrMetric() [2/2]

```
KerrMetric::KerrMetric (
            real aParam,
            bool rLogScale = false,
            real mParam = 1.)
```

Construct a new Kerr Metric object.

Parameters

aParam	a parameter for the Kerr metric
rLogScale	Whether we are using a logarithmic radial scale
mParam	mass parameter for the Kerr metric

6.34.3 Member Function Documentation

6.34.3.1 getFullDescriptionStr()

```
std::string KerrMetric::getFullDescriptionStr () const  [final], [virtual]
```

Kerr metric description string getter.

Returns

std::string

Reimplemented from Metric.

6.34.3.2 getMetric_dd()

```
TwoIndex KerrMetric::getMetric_dd (
            const Point & p) const  [final], [virtual]
```

Kerr metric getter, indices down.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.34.3.3 getMetric_uu()

```
TwoIndex KerrMetric::getMetric_uu (
            const Point & p) const  [final], [virtual]
```

Kerr metric getter, indices up.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.34.4 Member Data Documentation

6.34.4.1 m_aParam

```
const real KerrMetric::m_aParam  [private]
```

Mass-rescaled rotation parameter for Kerr. Note that this should be between -1 and 1.

6.34.4.2 m_mParam

```
const real KerrMetric::m_mParam  [private]
```

Mass parameter for Kerr. Default is 1.

The documentation for this class was generated from the following files:

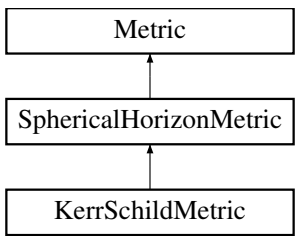
- FOORT/src/Metric.h
- FOORT/src/Metric.cpp

6.35 KerrSchildMetric Class Reference

Kerr metric in Kerr-Schild coordinates.

```
#include <Metric.h>
```

Inheritance diagram for KerrSchildMetric:



Public Member Functions

- KerrSchildMetric ()=delete
- KerrSchildMetric (real aParam, bool rLogScale=false)

  Construct a new Kerr Schild Metric object. We always have a M = 1 BH.
- TwoIndex getMetric_dd (const Point &p) const final

  Kerr-Schild metric getter, indices down.
- TwoIndex getMetric_uu (const Point &p) const final

  Kerr-Schild metric getter, indices up.
- std::string getFullDescriptionStr () const final

  Kerr-Schild metric description string getter.

Public Member Functions inherited from SphericalHorizonMetric

- SphericalHorizonMetric ()=delete
- SphericalHorizonMetric (real HorizonRadius, bool rLogScale)

  Construct a new Spherical Horizon Metric object.
- real getHorizonRadius () const

  Get the radius of the horizon.

Public Member Functions inherited from Metric

- virtual ~Metric ()=default
- Metric (bool rlogscale=false)

  Construct a new Metric object.
- virtual ThreeIndex getChristoffel_udd (const Point &p) const

  Return the Christoffel symbols of the metric (indices up, down, down)
- virtual FourIndex getRiemann_uddd (const Point &p) const

  Riemann tensor of the metric (indices up, down, down, down)
- virtual real getKretschmann (const Point &p) const

  Get the Kretschmann scalar at a point.
- bool getrLogScale () const

  Get whether we are using a logarithmic radial scale.

Private Attributes

- const real m_aParam

  Rotation parameter for Kerr. Note that this should be between -1 and 1 since M=1.

Additional Inherited Members

Protected Attributes inherited from SphericalHorizonMetric

- const real m_HorizonRadius

  Radius of the horizon.

Protected Attributes inherited from Metric

- std::vector< int > m_Symmetries {}

  The symmetries (coordinate Killing vectors) of the metric. Should be set by descendant constructor.
- const bool m_rLogScale

  Are we using a logarithmic r coordinate?

6.35.1 Detailed Description

Kerr metric in Kerr-Schild coordinates.

Note

Normalized so that M = 1

6.35.2 Constructor & Destructor Documentation

6.35.2.1 KerrSchildMetric() [1/2]

```
KerrSchildMetric::KerrSchildMetric ()  [delete]
```

6.35.2.2 KerrSchildMetric() [2/2]

```
KerrSchildMetric::KerrSchildMetric (
            real aParam,
            bool rLogScale = false)
```

Construct a new Kerr Schild Metric object. We always have a M = 1 BH.

Parameters

aParam	a parameter for the Kerr-Schild metric
rLogScale	whether we are using a logarithmic radial scale

6.35.3 Member Function Documentation

6.35.3.1 getFullDescriptionStr()

```
std::string KerrSchildMetric::getFullDescriptionStr () const  [final], [virtual]
```

Kerr-Schild metric description string getter.

Returns

std::string

Reimplemented from Metric.

6.35.3.2 getMetric_dd()

```
TwoIndex KerrSchildMetric::getMetric_dd (
            const Point & p) const  [final], [virtual]
```

Kerr-Schild metric getter, indices down.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.35.3.3 getMetric_uu()

```
TwoIndex KerrSchildMetric::getMetric_uu (
            const Point & p) const  [final], [virtual]
```

Kerr-Schild metric getter, indices up.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.35.4 Member Data Documentation

6.35.4.1 m_aParam

```
const real KerrSchildMetric::m_aParam  [private]
```

Rotation parameter for Kerr. Note that this should be between -1 and 1 since M=1.

The documentation for this class was generated from the following files:

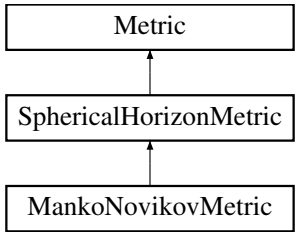
- FOORT/src/Metric.h
- FOORT/src/Metric.cpp

6.36 MankoNovikovMetric Class Reference

Mano-Novikov metric (with angular momentum and M3 parameter turned on)

`#include <Metric.h>`

Inheritance diagram for MankoNovikovMetric:



Public Member Functions

- MankoNovikovMetric ()=delete
- MankoNovikovMetric (real aParam, real alpha3Param, bool rLogScale=false)

  Construct a new Manko Novikov Metric object. We always have a M = 1 BH.
- TwoIndex getMetric_dd (const Point &p) const final

  Manko-Novikov metric getter, indices down.
- TwoIndex getMetric_uu (const Point &p) const final

  Manko-Novikov metric getter, indices up.
- std::string getFullDescriptionStr () const final

  Manko-Novikov metric description string getter.

Public Member Functions inherited from SphericalHorizonMetric

- SphericalHorizonMetric ()=delete
- SphericalHorizonMetric (real HorizonRadius, bool rLogScale)

  Construct a new Spherical Horizon Metric object.
- real getHorizonRadius () const

  Get the radius of the horizon.

Public Member Functions inherited from Metric

- virtual ~Metric ()=default
- Metric (bool rlogscale=false)

  Construct a new Metric object.
- virtual ThreeIndex getChristoffel_udd (const Point &p) const

  Return the Christoffel symbols of the metric (indices up, down, down)
- virtual FourIndex getRiemann_uddd (const Point &p) const

  Riemann tensor of the metric (indices up, down, down, down)
- virtual real getKretschmann (const Point &p) const

  Get the Kretschmann scalar at a point.
- bool getrLogScale () const

  Get whether we are using a logarithmic radial scale.

Private Attributes

- const real m_aParam

  a parameter for the Manko-Novikov metric
- const real m_alpha3Param

  alpha3 parameter for the Manko-Novikov metric
- const real m_alphaParam

  Derived alpha parameter.
- const real m_kParam

  Derived k parameter.

Additional Inherited Members

Protected Attributes inherited from SphericalHorizonMetric

- const real m_HorizonRadius

  Radius of the horizon.

Protected Attributes inherited from Metric

- std::vector< int > m_Symmetries {}

  The symmetries (coordinate Killing vectors) of the metric. Should be set by descendant constructor.
- const bool m_rLogScale

  Are we using a logarithmic r coordinate?

6.36.1 Detailed Description

Mano-Novikov metric (with angular momentum and M3 parameter turned on)

Note

Implementation by Seppe Staelens

6.36.2 Constructor & Destructor Documentation

6.36.2.1 MankoNovikovMetric() [1/2]

```
MankoNovikovMetric::MankoNovikovMetric ()  [delete]
```

6.36.2.2 MankoNovikovMetric() [2/2]

```
MankoNovikovMetric::MankoNovikovMetric (
            real aParam,
            real alpha3Param,
            bool rLogScale = false)
```

Construct a new Manko Novikov Metric object. We always have a M = 1 BH.

The horizon is at r = M + k

Parameters

aParam	a parameter for the Manko-Novikov metric. The angular momentum a = J/M^2
alpha3Param	alpha3 parameter for the Manko-Novikov metric
rLogScale	whether we are using a logarithmic radial scale

6.36.3 Member Function Documentation

6.36.3.1 getFullDescriptionStr()

```
std::string MankoNovikovMetric::getFullDescriptionStr () const  [final], [virtual]
```

Manko-Novikov metric description string getter.

Returns

std::string

Reimplemented from Metric.

6.36.3.2 getMetric_dd()

```
TwoIndex MankoNovikovMetric::getMetric_dd (
            const Point & p) const  [final], [virtual]
```

Manko-Novikov metric getter, indices down.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.36.3.3 getMetric_uu()

```
TwoIndex MankoNovikovMetric::getMetric_uu (
            const Point & p) const  [final], [virtual]
```

Manko-Novikov metric getter, indices up.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.36.4 Member Data Documentation

6.36.4.1 m_alpha3Param

```
const real MankoNovikovMetric::m_alpha3Param  [private]
```

alpha3 parameter for the Manko-Novikov metric

6.36.4.2 m_alphaParam

```
const real MankoNovikovMetric::m_alphaParam  [private]
```

Derived alpha parameter.

6.36.4.3 m_aParam

```
const real MankoNovikovMetric::m_aParam  [private]
```

a parameter for the Manko-Novikov metric

6.36.4.4 m_kParam

```
const real MankoNovikovMetric::m_kParam  [private]
```

Derived k parameter.

The documentation for this class was generated from the following files:

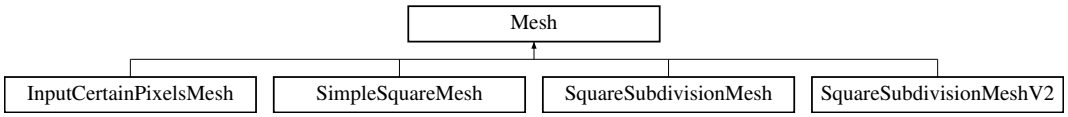
- FOORT/src/Metric.h
- FOORT/src/Metric.cpp

6.37 Mesh Class Reference

Abstract base Mesh class.

```
#include <Mesh.h>
```

Inheritance diagram for Mesh:



Public Member Functions

- Mesh (DiagBitflag valdiag)
- virtual ~Mesh ()=default
- virtual largecounter getCurNrGeodesics () const =0
- virtual void getNewInitConds (largecounter index, ScreenPoint &newunitpoint, ScreenIndex &newscreenindex) const =0
- virtual void GeodesicFinished (largecounter index, std::vector< real > finalValues)=0
- virtual void EndCurrentLoop ()=0
- virtual bool IsFinished () const =0
- virtual std::string getFullDescriptionStr () const

  Basic description string getter.

Protected Attributes

- const std::unique_ptr< const Diagnostic > m_DistanceDiagnostic

  The Diagnostic (a const pointer to a const Diagnostic object) that is used to calculate distances (using FinalData↩ ValDistance()) between the "values" that are assigned to Geodesics.

6.37.1 Detailed Description

Abstract base Mesh class.

6.37.2 Constructor & Destructor Documentation

6.37.2.1 Mesh()

```
Mesh::Mesh (
            DiagBitflag valdiag)  [inline]
```

6.37.2.2 ~Mesh()

```
virtual Mesh::~Mesh ()  [virtual], [default]
```

6.37.3 Member Function Documentation

6.37.3.1 EndCurrentLoop()

```
virtual void Mesh::EndCurrentLoop ()  [pure virtual]
```

Implemented in InputCertainPixelsMesh, SimpleSquareMesh, SquareSubdivisionMesh, and SquareSubdivisionMeshV2.

6.37.3.2 GeodesicFinished()

```
virtual void Mesh::GeodesicFinished (
            largecounter index,
            std::vector< real > finalValues)  [pure virtual]
```

Implemented in InputCertainPixelsMesh, SimpleSquareMesh, SquareSubdivisionMesh, and SquareSubdivisionMeshV2.

6.37.3.3 getCurNrGeodesics()

```
virtual largecounter Mesh::getCurNrGeodesics () const  [pure virtual]
```

Implemented in InputCertainPixelsMesh, SimpleSquareMesh, SquareSubdivisionMesh, and SquareSubdivisionMeshV2.

6.37.3.4 getFullDescriptionStr()

```
std::string Mesh::getFullDescriptionStr () const  [virtual]
```

Basic description string getter.

Returns

std::string

Reimplemented in InputCertainPixelsMesh, SimpleSquareMesh, SquareSubdivisionMesh, and SquareSubdivisionMeshV2.

6.37.3.5 getNewInitConds()

```
virtual void Mesh::getNewInitConds (
            largecounter index,
            ScreenPoint & newunitpoint,
            ScreenIndex & newscreenindex) const  [pure virtual]
```

Implemented in InputCertainPixelsMesh, SimpleSquareMesh, SquareSubdivisionMesh, and SquareSubdivisionMeshV2.

6.37.3.6 IsFinished()

```
virtual bool Mesh::IsFinished () const  [pure virtual]
```

Implemented in InputCertainPixelsMesh, SimpleSquareMesh, SquareSubdivisionMesh, and SquareSubdivisionMeshV2.

6.37.4 Member Data Documentation

6.37.4.1 m_DistanceDiagnostic

```
const std::unique_ptr<const Diagnostic> Mesh::m_DistanceDiagnostic  [protected]
```

The Diagnostic (a const pointer to a const Diagnostic object) that is used to calculate distances (using FinalData↩ ValDistance()) between the "values" that are assigned to Geodesics.

The documentation for this class was generated from the following files:

- FOORT/src/Mesh.h
- FOORT/src/Mesh.cpp

6.38 Metric Class Reference

The abstract base class for all Metrics.

```
#include <Metric.h>
```

Inheritance diagram for Metric:



Public Member Functions

- virtual ~Metric ()=default
- Metric (bool rlogscale=false)

  Construct a new Metric object.
- virtual TwoIndex getMetric_dd (const Point &p) const =0
- virtual TwoIndex getMetric_uu (const Point &p) const =0
- virtual ThreeIndex getChristoffel_udd (const Point &p) const

  Return the Christoffel symbols of the metric (indices up, down, down)
- virtual FourIndex getRiemann_uddd (const Point &p) const

  Riemann tensor of the metric (indices up, down, down, down)
- virtual real getKretschmann (const Point &p) const

  Get the Kretschmann scalar at a point.
- virtual std::string getFullDescriptionStr () const

  Base class description string getter.
- bool getrLogScale () const

  Get whether we are using a logarithmic radial scale.

Protected Attributes

- std::vector< int > m_Symmetries {}

  The symmetries (coordinate Killing vectors) of the metric. Should be set by descendant constructor.
- const bool m_rLogScale

  Are we using a logarithmic r coordinate?

6.38.1 Detailed Description

The abstract base class for all Metrics.

6.38.2 Constructor & Destructor Documentation

6.38.2.1 ~Metric()

```
virtual Metric::~Metric ()  [virtual], [default]
```

6.38.2.2 Metric()

```
Metric::Metric (
            bool rlogscale = false)
```

Construct a new Metric object.

Parameters

rlogscale	Whether we are using a logarithmic radial scale

6.38.3 Member Function Documentation

6.38.3.1 getChristoffel_udd()

```
ThreeIndex Metric::getChristoffel_udd (
            const Point & p) const  [virtual]
```

Return the Christoffel symbols of the metric (indices up, down, down)

Parameters

p	Point at which to evaluate the Christoffel symbols

Returns

ThreeIndex

6.38.3.2 getFullDescriptionStr()

```
std::string Metric::getFullDescriptionStr () const  [virtual]
```

Base class description string getter.

Returns

std::string

Reimplemented in BosonStarMetric, FlatSpaceMetric, JohannsenMetric, KerrMetric, KerrSchildMetric, MankoNovikovMetric, RasheedLarsenMetric, RotatingBosonStarMetric, and ST3CrMetric.

6.38.3.3 getKretschmann()

```
real Metric::getKretschmann (
            const Point & p) const  [virtual]
```

Get the Kretschmann scalar at a point.

The Kretschmann scalar is defined as R_{abcd} R^{abcd}

Note

This needs to be implemented still.

Parameters

p	Point at which to evaluate the Kretschmann scalar

Returns

real

6.38.3.4 getMetric_dd()

```
virtual TwoIndex Metric::getMetric_dd (
            const Point & p) const  [pure virtual]
```

Implemented in BosonStarMetric, FlatSpaceMetric, JohannsenMetric, KerrMetric, KerrSchildMetric, MankoNovikovMetric, RasheedLarsenMetric, RotatingBosonStarMetric, and ST3CrMetric.

6.38.3.5 getMetric_uu()

```
virtual TwoIndex Metric::getMetric_uu (
            const Point & p) const  [pure virtual]
```

Implemented in BosonStarMetric, FlatSpaceMetric, JohannsenMetric, KerrMetric, KerrSchildMetric, MankoNovikovMetric, RasheedLarsenMetric, RotatingBosonStarMetric, and ST3CrMetric.

6.38.3.6 getRiemann_uddd()

```
FourIndex Metric::getRiemann_uddd (
            const Point & p) const  [virtual]
```

Riemann tensor of the metric (indices up, down, down, down)

Parameters

p	Point at which to evaluate the Riemann tensor

Note

This needs to be implemented still.

Returns

FourIndex

6.38.3.7 getrLogScale()

```
bool Metric::getrLogScale () const
```

Get whether we are using a logarithmic radial scale.

Returns

true

false

6.38.4 Member Data Documentation

6.38.4.1 m_rLogScale

```
const bool Metric::m_rLogScale  [protected]
```

Are we using a logarithmic r coordinate?

6.38.4.2 m_Symmetries

```
std::vector<int> Metric::m_Symmetries {}  [protected]
```

The symmetries (coordinate Killing vectors) of the metric. Should be set by descendant constructor.

The documentation for this class was generated from the following files:

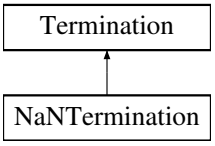
- FOORT/src/Metric.h
- FOORT/src/Metric.cpp

6.39 NaNTermination Class Reference

NaNTermination: Terminate geodesics if they contain a NaN in position or velocity.

```
#include <Terminations.h>
```

Inheritance diagram for NaNTermination:



Public Member Functions

- NaNTermination (Geodesic *const theGeodesic)
- Term CheckTermination () final

  Check if the geodesic contains a NaN.
- std::string getFullDescriptionStr () const final

  Description string getter for the NaN Termination.

Public Member Functions inherited from Termination

- Termination ()=delete
- Termination (Geodesic *const theGeodesic)
- virtual void Reset ()

  Reset to 0 to start integrating a new geodesic.
- virtual ~Termination ()=default

Static Public Attributes

- static std::unique_ptr< NaNTermOptions > TermOptions

  Options (contains horizon radius, if we are using logarithmic r coordinate, and distance allowed from the horizon)

Additional Inherited Members

Protected Member Functions inherited from Termination

- bool DecideUpdate (largecounter UpdateNSteps)

  Returns true if the Termination should update its internal status.

Protected Attributes inherited from Termination

- Geodesic ∗const m_OwnerGeodesic

  The geodesic that owns the Termination (a const pointer to the Geodesic)
- largecounter m_StepsSinceUpdated {}

  The termination is itself in charge of keeping track of how many steps it has been since it has been updated. The Termination's TerminationOptions struct tells it how many steps it needs to wait between updates.

6.39.1 Detailed Description

NaNTermination: Terminate geodesics if they contain a NaN in position or velocity.

6.39.2 Constructor & Destructor Documentation

6.39.2.1 NaNTermination()

```
NaNTermination::NaNTermination (
            Geodesic *const theGeodesic)  [inline]
```

6.39.3 Member Function Documentation

6.39.3.1 CheckTermination()

```
Term NaNTermination::CheckTermination ()  [final], [virtual]
```

Check if the geodesic contains a NaN.

Returns

Term

Implements Termination.

6.39.3.2 getFullDescriptionStr()

```
std::string NaNTermination::getFullDescriptionStr () const  [final], [virtual]
```

Description string getter for the NaN Termination.

Returns

std::string

Implements Termination.

6.39.4 Member Data Documentation

6.39.4.1 TermOptions

```
std::unique_ptr< NaNTermOptions > NaNTermination::TermOptions  [static]
```

Options (contains horizon radius, if we are using logarithmic r coordinate, and distance allowed from the horizon)

The documentation for this class was generated from the following files:

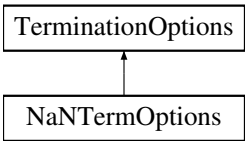
- FOORT/src/Terminations.h
- FOORT/src/Config.cpp
- FOORT/src/Terminations.cpp

6.40 NaNTermOptions Struct Reference

Options for NaNTermination.

```
#include <Terminations.h>
```

Inheritance diagram for NaNTermOptions:



Public Member Functions

- NaNTermOptions (bool consoleoutputon, largecounter Nsteps)

Public Member Functions inherited from TerminationOptions

- TerminationOptions (largecounter Nsteps)
- virtual ~TerminationOptions ()=default

Public Attributes

- const bool OutputToConsole

  Whether to output to the console when a NaN is encountered.

Public Attributes inherited from TerminationOptions

- const largecounter UpdateEveryNSteps

  Number of steps between updates.

6.40.1 Detailed Description

Options for NaNTermination.

6.40.2 Constructor & Destructor Documentation

6.40.2.1 NaNTermOptions()

```
NaNTermOptions::NaNTermOptions (
            bool consoleoutputon,
            largecounter Nsteps)  [inline]
```

6.40.3 Member Data Documentation

6.40.3.1 OutputToConsole

```
const bool NaNTermOptions::OutputToConsole
```

Whether to output to the console when a NaN is encountered.

The documentation for this struct was generated from the following file:

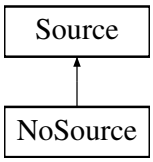
- FOORT/src/Terminations.h

6.41 NoSource Class Reference

NoSource class.

```
#include <Geodesic.h>
```

Inheritance diagram for NoSource:



Public Member Functions

- NoSource (const Metric *const theMetric)
- OneIndex getSource (Point pos, OneIndex vel) const final

  Returns zero source for the geodesic equation: no force felt by geodesic.
- std::string getFullDescriptionStr () const final

  Full description string getter for NoSource class.

Public Member Functions inherited from Source

- Source (const Metric *const theMetric)
- virtual ~Source ()=default

Additional Inherited Members

Protected Attributes inherited from Source

- const Metric *const m_theMetric

  A const pointer to a const metric.

6.41.1 Detailed Description

NoSource class.

There is no source, i.e. the geodesic is indeed a geodesic (and feels no force)

6.41.2 Constructor & Destructor Documentation

6.41.2.1 NoSource()

```
NoSource::NoSource (
            const Metric *const theMetric)  [inline]
```

6.41.3 Member Function Documentation

6.41.3.1 getFullDescriptionStr()

```
std::string NoSource::getFullDescriptionStr () const  [final], [virtual]
```

Full description string getter for NoSource class.

Returns

std::string

Reimplemented from Source.

6.41.3.2 getSource()

```
OneIndex NoSource::getSource (
            Point pos,
            OneIndex vel) const  [final], [virtual]
```

Returns zero source for the geodesic equation: no force felt by geodesic.

Parameters

pos	Position of the geodesic
vel	Velocity of the geodesic

Returns

OneIndex Source for the geodesic

Implements Source.

The documentation for this class was generated from the following files:

- FOORT/src/Geodesic.h
- FOORT/src/Geodesic.cpp

6.42 SquareSubdivisionMesh::PixelInfo Struct Reference

A struct the Mesh uses to keep all information about a given pixel.

Public Member Functions

- PixelInfo (ScreenIndex ind, int subdiv)

Public Attributes

- ScreenIndex Index {}

  The pixel's screenindex.
- int SubdivideLevel {}

  The level at which the pixel has been subdivided. Note: initial grid pixels are at 1; pixels at 0 are pixels that cannot be subdivided (for example, at the right or lower edges)
- real Weight {-1}

  Weight of the pixel: if negative, this signifies that it needs to be updated/calculated! The weight is determined as the max of the distance (as calculated by the value Diagnostic) between its values and those of its right, lower, and right-lower neighbors.
- std::vector< real > DiagValue {}

  The values associated to this pixel (as calculated by the value Diagnostic)
- largecounter LowerNbrIndex {0}

  Where its lower neighbor is located in m_AllPixels. Note: the pixel with index 0 is (0,0) and can never be the lower or right neighbor of any other pixel!
- largecounter RightNbrIndex {0}

  Where its right neighbor is located in m_AllPixels. Note: the pixel with index 0 is (0,0) and can never be the lower or right neighbor of any other pixel!

6.42.1 Detailed Description

A struct the Mesh uses to keep all information about a given pixel.

6.42.2 Constructor & Destructor Documentation

6.42.2.1 PixelInfo()

```
SquareSubdivisionMesh::PixelInfo::PixelInfo (
            ScreenIndex ind,
            int subdiv)  [inline]
```

6.42.3 Member Data Documentation

6.42.3.1 DiagValue

```
std::vector<real> SquareSubdivisionMesh::PixelInfo::DiagValue {}
```

The values associated to this pixel (as calculated by the value Diagnostic)

6.42.3.2 Index

```
ScreenIndex SquareSubdivisionMesh::PixelInfo::Index {}
```

The pixel's screenindex.

6.42.3.3 LowerNbrIndex

```
largecounter SquareSubdivisionMesh::PixelInfo::LowerNbrIndex {0}
```

Where its lower neighbor is located in m_AllPixels. Note: the pixel with index 0 is (0,0) and can never be the lower or right neighbor of any other pixel!

6.42.3.4 RightNbrIndex

```
largecounter SquareSubdivisionMesh::PixelInfo::RightNbrIndex {0}
```

Where its right neighbor is located in m_AllPixels. Note: the pixel with index 0 is (0,0) and can never be the lower or right neighbor of any other pixel!

6.42.3.5 SubdivideLevel

```
int SquareSubdivisionMesh::PixelInfo::SubdivideLevel {}
```

The level at which the pixel has been subdivided. Note: initial grid pixels are at 1; pixels at 0 are pixels that cannot be subdivided (for example, at the right or lower edges)

6.42.3.6 Weight

```
real SquareSubdivisionMesh::PixelInfo::Weight {-1}
```

Weight of the pixel: if negative, this signifies that it needs to be updated/calculated! The weight is determined as the max of the distance (as calculated by the value Diagnostic) between its values and those of its right, lower, and right-lower neighbors.

The documentation for this struct was generated from the following file:

- FOORT/src/Mesh.h

6.43 SquareSubdivisionMeshV2::PixelInfo Struct Reference

A struct the Mesh uses to keep all information about a given pixel.

Public Member Functions

- PixelInfo (ScreenIndex ind, int subdiv)

Public Attributes

- const ScreenIndex Index {}

  The pixel's screenindex: this gets set by the constructor and cannot change anymore.
- int SubdivideLevel {}

  The level at which the pixel has been subdivided. Note: initial grid pixels are at 1; pixels at 0 are pixels that cannot be subdivided (for example, at the right or lower edges)
- real Weight {-1}

  Weight of the pixel: if negative, this signifies that it needs to be updated/calculated! The weight is determined as the max of the distance (as calculated by the value Diagnostic) between its values and those of its right, lower, and right-lower neighbors.
- std::vector< real > DiagValue {}

  The values associated to this pixel (as calculated by the value Diagnostic)
- PixelInfo * LeftNbr {nullptr}

  Pointer to its left neighbor.
- PixelInfo * RightNbr {nullptr}

  Pointer to its right neighbor.
- PixelInfo * UpNbr {nullptr}

  Pointer to its upper neighbor.
- PixelInfo * DownNbr {nullptr}

  Pointer to its lower neighbor.
- PixelInfo * SEdiagNbr {nullptr}

  Pointer to its lower-rightt neighbor.

6.43.1 Detailed Description

A struct the Mesh uses to keep all information about a given pixel.

6.43.2 Constructor & Destructor Documentation

6.43.2.1 PixelInfo()

```
SquareSubdivisionMeshV2::PixelInfo::PixelInfo (
            ScreenIndex ind,
            int subdiv)  [inline]
```

6.43.3 Member Data Documentation

6.43.3.1 DiagValue

```
std::vector<real> SquareSubdivisionMeshV2::PixelInfo::DiagValue {}
```

The values associated to this pixel (as calculated by the value Diagnostic)

6.43.3.2 DownNbr

```
PixelInfo* SquareSubdivisionMeshV2::PixelInfo::DownNbr {nullptr}
```

Pointer to its lower neighbor.

6.43.3.3 Index

```
const ScreenIndex SquareSubdivisionMeshV2::PixelInfo::Index {}
```

The pixel's screenindex: this gets set by the constructor and cannot change anymore.

6.43.3.4 LeftNbr

```
PixelInfo* SquareSubdivisionMeshV2::PixelInfo::LeftNbr {nullptr}
```

Pointer to its left neighbor.

6.43.3.5 RightNbr

```
PixelInfo* SquareSubdivisionMeshV2::PixelInfo::RightNbr {nullptr}
```

Pointer to its right neighbor.

6.43.3.6 SEdiagNbr

```
PixelInfo* SquareSubdivisionMeshV2::PixelInfo::SEdiagNbr {nullptr}
```

Pointer to its lower-rightt neighbor.

6.43.3.7 SubdivideLevel

```
int SquareSubdivisionMeshV2::PixelInfo::SubdivideLevel {}
```

The level at which the pixel has been subdivided. Note: initial grid pixels are at 1; pixels at 0 are pixels that cannot be subdivided (for example, at the right or lower edges)

6.43.3.8 UpNbr

```
PixelInfo* SquareSubdivisionMeshV2::PixelInfo::UpNbr {nullptr}
```

Pointer to its upper neighbor.

6.43.3.9 Weight

```
real SquareSubdivisionMeshV2::PixelInfo::Weight {-1}
```

Weight of the pixel: if negative, this signifies that it needs to be updated/calculated! The weight is determined as the max of the distance (as calculated by the value Diagnostic) between its values and those of its right, lower, and right-lower neighbors.

The documentation for this struct was generated from the following file:

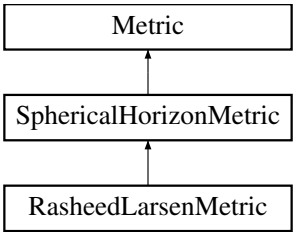
- FOORT/src/Mesh.h

6.44 RasheedLarsenMetric Class Reference

Rasheed-Larsen black hole.

```
#include <Metric.h>
```

Inheritance diagram for RasheedLarsenMetric:



Public Member Functions

- RasheedLarsenMetric ()=delete
- RasheedLarsenMetric (real mParam, real aParam, real pParam, real qParam, bool rLogScale=false)

  Construct a new Rasheed Larsen Metric object. All parameters are rescaled by the mass to end up with a M = 1 BH.
- TwoIndex getMetric_dd (const Point &p) const final

  Rasheed-Larsen metric getter, indices down.
- TwoIndex getMetric_uu (const Point &p) const final

  Rasheed-Larsen metric getter, indices up.
- std::string getFullDescriptionStr () const final

  Rasheed-Larsen metric description string getter.

Public Member Functions inherited from SphericalHorizonMetric

- SphericalHorizonMetric ()=delete
- SphericalHorizonMetric (real HorizonRadius, bool rLogScale)

  Construct a new Spherical Horizon Metric object.
- real getHorizonRadius () const

  Get the radius of the horizon.

Public Member Functions inherited from Metric

- virtual ~Metric ()=default
- Metric (bool rlogscale=false)

  Construct a new Metric object.
- virtual ThreeIndex getChristoffel_udd (const Point &p) const

  Return the Christoffel symbols of the metric (indices up, down, down)
- virtual FourIndex getRiemann_uddd (const Point &p) const

  Riemann tensor of the metric (indices up, down, down, down)
- virtual real getKretschmann (const Point &p) const

  Get the Kretschmann scalar at a point.
- bool getrLogScale () const

  Get whether we are using a logarithmic radial scale.

Private Attributes

- const real m_aParam

  a parameter for the Rasheed-Larsen metric
- const real m_mParam

  m parameter for the Rasheed-Larsen metric
- const real m_pParam

  p parameter for the Rasheed-Larsen metric
- const real m_qParam

  q parameter for the Rasheed-Larsen metric

Additional Inherited Members

Protected Attributes inherited from SphericalHorizonMetric

- const real m_HorizonRadius

  Radius of the horizon.

Protected Attributes inherited from Metric

- std::vector< int > m_Symmetries {}

  The symmetries (coordinate Killing vectors) of the metric. Should be set by descendant constructor.
- const bool m_rLogScale

  Are we using a logarithmic r coordinate?

6.44.1 Detailed Description

Rasheed-Larsen black hole.

6.44.2 Constructor & Destructor Documentation

6.44.2.1 RasheedLarsenMetric() [1/2]

```
RasheedLarsenMetric::RasheedLarsenMetric ()  [delete]
```

6.44.2.2 RasheedLarsenMetric() [2/2]

```
RasheedLarsenMetric::RasheedLarsenMetric (
            real mParam,
            real aParam,
            real pParam,
            real qParam,
            bool rLogScale = false)
```

Construct a new Rasheed Larsen Metric object. All parameters are rescaled by the mass to end up with a M = 1 BH.

Parameters

mParam	m parameter for the Rasheed-Larsen metric
aParam	a parameter for the Rasheed-Larsen metric
pParam	p parameter for the Rasheed-Larsen metric
qParam	q parameter for the Rasheed-Larsen metric
rLogScale	whether we are using a logarithmic radial scale

6.44.3 Member Function Documentation

6.44.3.1 getFullDescriptionStr()

```
std::string RasheedLarsenMetric::getFullDescriptionStr () const  [final], [virtual]
```

Rasheed-Larsen metric description string getter.

Returns

std::string

Reimplemented from Metric.

6.44.3.2 getMetric_dd()

```
TwoIndex RasheedLarsenMetric::getMetric_dd (
            const Point & p) const  [final], [virtual]
```

Rasheed-Larsen metric getter, indices down.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.44.3.3 getMetric_uu()

```
TwoIndex RasheedLarsenMetric::getMetric_uu (
            const Point & p) const  [final], [virtual]
```

Rasheed-Larsen metric getter, indices up.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.44.4 Member Data Documentation

6.44.4.1 m_aParam

```
const real RasheedLarsenMetric::m_aParam  [private]
```

a parameter for the Rasheed-Larsen metric

6.44.4.2 m_mParam

```
const real RasheedLarsenMetric::m_mParam  [private]
```

m parameter for the Rasheed-Larsen metric

6.44.4.3 m_pParam

```
const real RasheedLarsenMetric::m_pParam  [private]
```

p parameter for the Rasheed-Larsen metric

6.44.4.4 m_qParam

```
const real RasheedLarsenMetric::m_qParam  [private]
```

q parameter for the Rasheed-Larsen metric

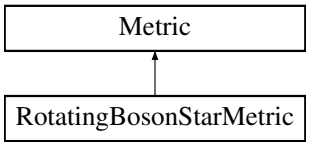
The documentation for this class was generated from the following files:

- FOORT/src/Metric.h
- FOORT/src/Metric.cpp

6.45 RotatingBosonStarMetric Class Reference

```
#include <Metric.h>
```

Inheritance diagram for RotatingBosonStarMetric:



Public Member Functions

- RotatingBosonStarMetric (bool rLogScale=false, std::string MetricFolder="RotatingBosonStar/data_Will/", int num_x=500, int num_th=399, bool FlipAngularMomentum=false)

  Construct a new Rotating Boson Star Metric object.
- ~RotatingBosonStarMetric ()=default
- TwoIndex getMetric_dd (const Point &p) const final

  RotatingBosonStar metric getter, indices down.
- TwoIndex getMetric_uu (const Point &p) const final

  RotatingBosonStar metric getter, indices up.
- std::string getFullDescriptionStr () const final

  RotatingBosonStar metric description string getter.

Public Member Functions inherited from Metric

- virtual ~Metric ()=default
- Metric (bool rlogscale=false)

  Construct a new Metric object.
- virtual ThreeIndex getChristoffel_udd (const Point &p) const

  Return the Christoffel symbols of the metric (indices up, down, down)
- virtual FourIndex getRiemann_uddd (const Point &p) const

  Riemann tensor of the metric (indices up, down, down, down)
- virtual real getKretschmann (const Point &p) const

  Get the Kretschmann scalar at a point.
- bool getrLogScale () const

  Get whether we are using a logarithmic radial scale.

Public Attributes

- std::unique_ptr< BicubicSplineInterpolator > m_fInterpolator

  The grid interpolators.
- std::unique_ptr< BicubicSplineInterpolator > m_lInterpolator
- std::unique_ptr< BicubicSplineInterpolator > m_gInterpolator
- std::unique_ptr< BicubicSplineInterpolator > m_OmegaInterpolator
- const int m_OmegaSign

Private Attributes

- std::unique_ptr< Grid > m_grid_f

  The grids with the metric functions.
- std::unique_ptr< Grid > m_grid_l
- std::unique_ptr< Grid > m_grid_g
- std::unique_ptr< Grid > m_grid_Omega

Additional Inherited Members

Protected Attributes inherited from Metric

- std::vector< int > m_Symmetries {}

  The symmetries (coordinate Killing vectors) of the metric. Should be set by descendant constructor.
- const bool m_rLogScale

  Are we using a logarithmic r coordinate?

6.45.1 Constructor & Destructor Documentation

6.45.1.1 RotatingBosonStarMetric()

```
RotatingBosonStarMetric::RotatingBosonStarMetric (
            bool rLogScale = false,
            std::string MetricFolder = "RotatingBosonStar/data_Will/",
            int num_x = 500,
            int num_th = 399,
            bool FlipAngularMomentum = false)
```

Construct a new Rotating Boson Star Metric object.

Parameters

rLogScale	whether we are using a logarithmic radial scale
FlipAngularMomentum	whether to flip the sign of Omega (reverse rotation direction)

6.45.1.2 ~RotatingBosonStarMetric()

```
RotatingBosonStarMetric::~RotatingBosonStarMetric () [default]
```

6.45.2 Member Function Documentation

6.45.2.1 getFullDescriptionStr()

```
std::string RotatingBosonStarMetric::getFullDescriptionStr () const  [final], [virtual]
```

RotatingBosonStar metric description string getter.

Returns

std::string

Reimplemented from Metric.

6.45.2.2 getMetric_dd()

```
TwoIndex RotatingBosonStarMetric::getMetric_dd (
            const Point & p) const  [final], [virtual]
```

RotatingBosonStar metric getter, indices down.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.45.2.3 getMetric_uu()

```
TwoIndex RotatingBosonStarMetric::getMetric_uu (
            const Point & p) const  [final], [virtual]
```

RotatingBosonStar metric getter, indices up.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.45.3 Member Data Documentation

6.45.3.1 m_fInterpolator

```
std::unique_ptr<BicubicSplineInterpolator> RotatingBosonStarMetric::m_fInterpolator
```

The grid interpolators.

6.45.3.2 m_gInterpolator

```
std::unique_ptr<BicubicSplineInterpolator> RotatingBosonStarMetric::m_gInterpolator
```

6.45.3.3 m_grid_f

```
std::unique_ptr<Grid> RotatingBosonStarMetric::m_grid_f  [private]
```

The grids with the metric functions.

6.45.3.4 m_grid_g

```
std::unique_ptr<Grid> RotatingBosonStarMetric::m_grid_g  [private]
```

6.45.3.5 m_grid_l

```
std::unique_ptr<Grid> RotatingBosonStarMetric::m_grid_l  [private]
```

6.45.3.6 m_grid_Omega

```
std::unique_ptr<Grid> RotatingBosonStarMetric::m_grid_Omega  [private]
```

6.45.3.7 m_lInterpolator

```
std::unique_ptr<BicubicSplineInterpolator> RotatingBosonStarMetric::m_lInterpolator
```

6.45.3.8 m_OmegaInterpolator

```
std::unique_ptr<BicubicSplineInterpolator> RotatingBosonStarMetric::m_OmegaInterpolator
```

6.45.3.9 m_OmegaSign

```
const int RotatingBosonStarMetric::m_OmegaSign
```

The documentation for this class was generated from the following files:

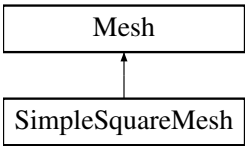
- FOORT/src/Metric.h
- FOORT/src/Metric.cpp

6.46 SimpleSquareMesh Class Reference

A simple square mesh that will integrate a square of evenly spaced pixels.

`#include <Mesh.h>`

Inheritance diagram for SimpleSquareMesh:



Public Member Functions

- SimpleSquareMesh ()=delete
- SimpleSquareMesh (largecounter totalPixels, DiagBitflag valdiag)
- largecounter getCurNrGeodesics () const final

  Get the current number of geodesics to be integrated.
- void getNewInitConds (largecounter index, ScreenPoint &newunitpoint, ScreenIndex &newscreenindex) const final

  Get the initial conditions for the next pixel to be integrated.
- void GeodesicFinished (largecounter index, std::vector< real > finalValues) final

  Function that is called when a geodesic is finished integrating.
- void EndCurrentLoop () final

  End the current loop of integrating pixels.
- bool IsFinished () const final

  Check whether the Mesh is finished integrating.
- std::string getFullDescriptionStr () const final

  Description string getter.

Public Member Functions inherited from Mesh

- Mesh (DiagBitflag valdiag)
- virtual ~Mesh ()=default

Private Attributes

- const largecounter m_TotalPixels

  Total amount of pixels in grid (is the square of m_RowColumnSize)
- const pixelcoord m_RowColumnSize

  Amount of pixels per row or column (square grid)
- bool m_Finished {false}

  Are we done integrating or not?

Additional Inherited Members

Protected Attributes inherited from Mesh

- const std::unique_ptr< const Diagnostic > m_DistanceDiagnostic

  The Diagnostic (a const pointer to a const Diagnostic object) that is used to calculate distances (using FinalData↩ ValDistance()) between the "values" that are assigned to Geodesics.

6.46.1 Detailed Description

A simple square mesh that will integrate a square of evenly spaced pixels.

6.46.2 Constructor & Destructor Documentation

6.46.2.1 SimpleSquareMesh() [1/2]

```
SimpleSquareMesh::SimpleSquareMesh ()  [delete]
```

6.46.2.2 SimpleSquareMesh() [2/2]

```
SimpleSquareMesh::SimpleSquareMesh (
            largecounter totalPixels,
            DiagBitflag valdiag)  [inline]
```

6.46.3 Member Function Documentation

6.46.3.1 EndCurrentLoop()

```
void SimpleSquareMesh::EndCurrentLoop ()  [final], [virtual]
```

End the current loop of integrating pixels.

We are done integrating after one loop

Implements Mesh.

6.46.3.2 GeodesicFinished()

```
void SimpleSquareMesh::GeodesicFinished (
            largecounter index,
            std::vector< real > finalValues)  [final], [virtual]
```

Function that is called when a geodesic is finished integrating.

This Mesh doesn't actually have to do anything with the geodesic (values) when it is done!

Parameters

index	index of the geodesic that is finished
finalValues	final values of the geodesic

Implements Mesh.

6.46.3.3 getCurNrGeodesics()

```
largecounter SimpleSquareMesh::getCurNrGeodesics () const  [final], [virtual]
```

Get the current number of geodesics to be integrated.

This is the total number of pixels in the grid

Returns

largecounter

Implements Mesh.

6.46.3.4 getFullDescriptionStr()

```
std::string SimpleSquareMesh::getFullDescriptionStr () const  [final], [virtual]
```

Description string getter.

Returns

std::string

Reimplemented from Mesh.

6.46.3.5 getNewInitConds()

```
void SimpleSquareMesh::getNewInitConds (
            largecounter index,
            ScreenPoint & newunitpoint,
            ScreenIndex & newscreenindex) const  [final], [virtual]
```

Get the initial conditions for the next pixel to be integrated.

The next pixel is the one with index m_CurrentPixel

Parameters

index	index of the pixel to be initialized
newunitpoint	new unit point to be initialized
newscreenindex	new screen index to be initialized

Implements Mesh.

6.46.3.6 IsFinished()

```
bool SimpleSquareMesh::IsFinished () const  [final], [virtual]
```

Check whether the Mesh is finished integrating.

We are done if we have sent all pixels to be integrated (there is only one iteration of pixels)

Returns

true

false

Implements Mesh.

6.46.4 Member Data Documentation

6.46.4.1 m_Finished

```
bool SimpleSquareMesh::m_Finished {false}  [private]
```

Are we done integrating or not?

6.46.4.2 m_RowColumnSize

```
const pixelcoord SimpleSquareMesh::m_RowColumnSize  [private]
```

Amount of pixels per row or column (square grid)

6.46.4.3 m_TotalPixels

```
const largecounter SimpleSquareMesh::m_TotalPixels  [private]
```

Total amount of pixels in grid (is the square of m_RowColumnSize)

The documentation for this class was generated from the following files:

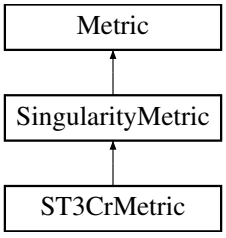
- FOORT/src/Mesh.h
- FOORT/src/Mesh.cpp

6.47 SingularityMetric Class Reference

Abstract base class for a metric with an arbitrary number of singularities (of arbitrary codimension)

`#include <Metric.h>`

Inheritance diagram for SingularityMetric:



Public Member Functions

- SingularityMetric (std::vector< Singularity > thesings, bool rLogScale)

  Construct a new Singularity Metric object.
- std::vector< Singularity > getSingularities () const

  Getters for the singularities.

Public Member Functions inherited from Metric

- virtual ~Metric ()=default
- Metric (bool rlogscale=false)

  Construct a new Metric object.
- virtual TwoIndex getMetric_dd (const Point &p) const =0
- virtual TwoIndex getMetric_uu (const Point &p) const =0
- virtual ThreeIndex getChristoffel_udd (const Point &p) const

  Return the Christoffel symbols of the metric (indices up, down, down)
- virtual FourIndex getRiemann_uddd (const Point &p) const

  Riemann tensor of the metric (indices up, down, down, down)
- virtual real getKretschmann (const Point &p) const

  Get the Kretschmann scalar at a point.
- virtual std::string getFullDescriptionStr () const

  Base class description string getter.
- bool getrLogScale () const

  Get whether we are using a logarithmic radial scale.

Protected Attributes

- const std::vector< Singularity > m_AllSingularities

  All singularities of the metric.

Protected Attributes inherited from Metric

- std::vector< int > m_Symmetries {}

  The symmetries (coordinate Killing vectors) of the metric. Should be set by descendant constructor.
- const bool m_rLogScale

  Are we using a logarithmic r coordinate?

6.47.1 Detailed Description

Abstract base class for a metric with an arbitrary number of singularities (of arbitrary codimension)

6.47.2 Constructor & Destructor Documentation

6.47.2.1 SingularityMetric()

```
SingularityMetric::SingularityMetric (
            std::vector< Singularity > thesings,
            bool rLogScale)
```

Construct a new Singularity Metric object.

Parameters

thesings	The singularities to include in the metric
rLogScale	whether we are using a logarithmic radial scale

6.47.3 Member Function Documentation

6.47.3.1 getSingularities()

```
std::vector< Singularity > SingularityMetric::getSingularities () const
```

Getters for the singularities.

Returns

std::vector<Singularity>

6.47.4 Member Data Documentation

6.47.4.1 m_AllSingularities

```
const std::vector<Singularity> SingularityMetric::m_AllSingularities  [protected]
```

All singularities of the metric.

The documentation for this class was generated from the following files:

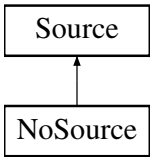
- FOORT/src/Metric.h
- FOORT/src/Metric.cpp

6.48 Source Class Reference

Abstract Source base class.

```
#include <Geodesic.h>
```

Inheritance diagram for Source:



Public Member Functions

- Source (const Metric ∗const theMetric)
- virtual ∼Source ()=default
- virtual OneIndex getSource (Point pos, OneIndex vel) const =0
- virtual std::string getFullDescriptionStr () const

  Basic full description string getter for Source base class.

Protected Attributes

- const Metric ∗const m_theMetric

  A const pointer to a const metric.

6.48.1 Detailed Description

Abstract Source base class.

6.48.2 Constructor & Destructor Documentation

6.48.2.1 Source()

```
Source::Source (
            const Metric *const theMetric)  [inline]
```

6.48.2.2 ∼Source()

```
virtual Source::~Source ()  [virtual], [default]
```

6.48.3 Member Function Documentation

6.48.3.1 getFullDescriptionStr()

```
std::string Source::getFullDescriptionStr () const  [virtual]
```

Basic full description string getter for Source base class.

Returns

std::string

Reimplemented in NoSource.

6.48.3.2 getSource()

```
virtual OneIndex Source::getSource (
            Point pos,
            OneIndex vel) const  [pure virtual]
```

Implemented in NoSource.

6.48.4 Member Data Documentation

6.48.4.1 m_theMetric

```
const Metric* const Source::m_theMetric  [protected]
```

A const pointer to a const metric.

The documentation for this class was generated from the following files:

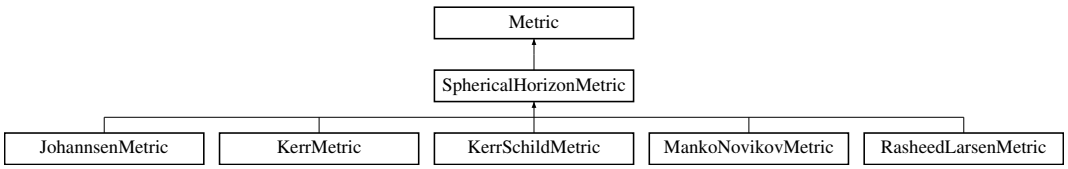
- FOORT/src/Geodesic.h
- FOORT/src/Geodesic.cpp

6.49 SphericalHorizonMetric Class Reference

Abstract base class for a metric with a spherical horizon (i.e. horizon at constant radius r)

```
#include <Metric.h>
```

Inheritance diagram for SphericalHorizonMetric:



Public Member Functions

- SphericalHorizonMetric ()=delete
- SphericalHorizonMetric (real HorizonRadius, bool rLogScale)

  Construct a new Spherical Horizon Metric object.
- real getHorizonRadius () const

  Get the radius of the horizon.

Public Member Functions inherited from Metric

- virtual ~Metric ()=default
- Metric (bool rlogscale=false)

  Construct a new Metric object.
- virtual TwoIndex getMetric_dd (const Point &p) const =0
- virtual TwoIndex getMetric_uu (const Point &p) const =0
- virtual ThreeIndex getChristoffel_udd (const Point &p) const

  Return the Christoffel symbols of the metric (indices up, down, down)
- virtual FourIndex getRiemann_uddd (const Point &p) const

  Riemann tensor of the metric (indices up, down, down, down)
- virtual real getKretschmann (const Point &p) const

  Get the Kretschmann scalar at a point.
- virtual std::string getFullDescriptionStr () const

  Base class description string getter.
- bool getrLogScale () const

  Get whether we are using a logarithmic radial scale.

Protected Attributes

- const real m_HorizonRadius

  Radius of the horizon.

Protected Attributes inherited from Metric

- std::vector< int > m_Symmetries {}

  The symmetries (coordinate Killing vectors) of the metric. Should be set by descendant constructor.
- const bool m_rLogScale

  Are we using a logarithmic r coordinate?

6.49.1 Detailed Description

Abstract base class for a metric with a spherical horizon (i.e. horizon at constant radius r)

6.49.2 Constructor & Destructor Documentation

6.49.2.1 SphericalHorizonMetric() [1/2]

```
SphericalHorizonMetric::SphericalHorizonMetric ()  [delete]
```

6.49.2.2 SphericalHorizonMetric() [2/2]

```
SphericalHorizonMetric::SphericalHorizonMetric (
            real HorizonRadius,
            bool rLogScale)
```

Construct a new Spherical Horizon Metric object.

Parameters

HorizonRadius	Radius of the horizon
rLogScale	Whether we are using a logarithmic radial scale

6.49.3 Member Function Documentation

6.49.3.1 getHorizonRadius()

```
real SphericalHorizonMetric::getHorizonRadius () const
```

Get the radius of the horizon.

Returns

real

6.49.4 Member Data Documentation

6.49.4.1 m_HorizonRadius

```
const real SphericalHorizonMetric::m_HorizonRadius  [protected]
```

Radius of the horizon.

The documentation for this class was generated from the following files:

- FOORT/src/Metric.h
- FOORT/src/Metric.cpp

6.50 tk::spline Class Reference

```
#include <Spline.h>
```

Public Types

- enum spline_type { linear = 10 , cspline = 30 , cspline_hermite = 31 }
- enum bd_type { first_deriv = 1 , second_deriv = 2 }

Public Member Functions

- spline ()
- spline (const std::vector< double > &X, const std::vector< double > &Y, spline_type type=cspline, bool make_monotonic=false, bd_type left=second_deriv, double left_value=0.0, bd_type right=second_deriv, double right_value=0.0)
- void set_boundary (bd_type left, double left_value, bd_type right, double right_value)
- void set_points (const std::vector< double > &x, const std::vector< double > &y, spline_type type=cspline)
- bool make_monotonic ()
- double operator() (double x) const
- double deriv (int order, double x) const
- std::vector< double > get_x () const
- std::vector< double > get_y () const
- double get_x_min () const
- double get_x_max () const

Protected Member Functions

- void set_coeffs_from_b ()
- size_t find_closest (double x) const

Protected Attributes

- std::vector< double > m_x
- std::vector< double > m_y
- std::vector< double > m_b
- std::vector< double > m_c
- std::vector< double > m_d
- double m_c0
- spline_type m_type
- bd_type m_left
- bd_type m_right
- double m_left_value
- double m_right_value
- bool m_made_monotonic

6.50.1 Member Enumeration Documentation

6.50.1.1 bd_type

enum tk::spline::bd_type

Enumerator

first_deriv	
second_deriv	

6.50.1.2 spline_type

enum tk::spline::spline_type

Enumerator

linear	
cspline	
cspline_hermite	

6.50.2 Constructor & Destructor Documentation

6.50.2.1 spline() [1/2]

```
tk::spline::spline ()
```

6.50.2.2 spline() [2/2]

```
tk::spline::spline (
            const std::vector< double > & X,
            const std::vector< double > & Y,
            spline_type type = cspline,
            bool make_monotonic = false,
            bd_type left = second_deriv,
            double left_value = 0.0,
            bd_type right = second_deriv,
            double right_value = 0.0)
```

6.50.3 Member Function Documentation

6.50.3.1 deriv()

```
double tk::spline::deriv (
            int order,
            double x) const
```

6.50.3.2 find_closest()

```
size_t tk::spline::find_closest (
            double x) const  [protected]
```

6.50.3.3 get_x()

```
std::vector< double > tk::spline::get_x () const
```

6.50.3.4 get_x_max()

```
double tk::spline::get_x_max () const
```

6.50.3.5 get_x_min()

```
double tk::spline::get_x_min () const
```

6.50.3.6 get_y()

```
std::vector< double > tk::spline::get_y () const
```

6.50.3.7 make_monotonic()

```
bool tk::spline::make_monotonic ()
```

6.50.3.8 operator()()

```
double tk::spline::operator() (
            double x) const
```

6.50.3.9 set_boundary()

```
void tk::spline::set_boundary (
            spline::bd_type left,
            double left_value,
            spline::bd_type right,
            double right_value)
```

6.50.3.10 set_coeffs_from_b()

```
void tk::spline::set_coeffs_from_b ()  [protected]
```

6.50.3.11 set_points()

```
void tk::spline::set_points (
            const std::vector< double > & x,
            const std::vector< double > & y,
            spline_type type = cspline)
```

6.50.4 Member Data Documentation

6.50.4.1 m_b

```
std::vector<double> tk::spline::m_b  [protected]
```

6.50.4.2 m_c

```
std::vector<double> tk::spline::m_c  [protected]
```

6.50.4.3 m_c0

```
double tk::spline::m_c0  [protected]
```

6.50.4.4 m_d

```
std::vector<double> tk::spline::m_d  [protected]
```

6.50.4.5 m_left

```
bd_type tk::spline::m_left  [protected]
```

6.50.4.6 m_left_value

```
double tk::spline::m_left_value  [protected]
```

6.50.4.7 m_made_monotonic

```
bool tk::spline::m_made_monotonic  [protected]
```

6.50.4.8 m_right

```
bd_type tk::spline::m_right  [protected]
```

6.50.4.9 m_right_value

```
double tk::spline::m_right_value  [protected]
```

6.50.4.10 m_type

```
spline_type tk::spline::m_type  [protected]
```

6.50.4.11 m_x

```
std::vector<double> tk::spline::m_x  [protected]
```

6.50.4.12 m_y

```
std::vector<double> tk::spline::m_y  [protected]
```

The documentation for this class was generated from the following files:

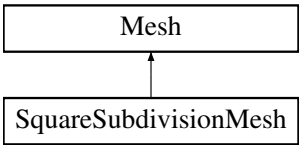
- FOORT/src/Spline.h
- FOORT/src/Spline.cpp

6.51 SquareSubdivisionMesh Class Reference

Adaptive subdivision Mesh.

```
#include <Mesh.h>
```

Inheritance diagram for SquareSubdivisionMesh:



Classes

- struct PixelInfo

  A struct the Mesh uses to keep all information about a given pixel.

Public Member Functions

- SquareSubdivisionMesh ()=delete
- SquareSubdivisionMesh (largecounter maxPixels, largecounter initialPixels, int maxSubdivide, largecounter iterationPixels, bool initialSubToFinal, DiagBitflag valdiag)
- largecounter getCurNrGeodesics () const final

  Return the current number of geodesics to be integrated.
- void getNewInitConds (largecounter index, ScreenPoint &newunitpoint, ScreenIndex &newscreenindex) const final

  Get the initial conditions for the next pixel to be integrated.
- void GeodesicFinished (largecounter index, std::vector< real > finalValues) final

  Called when a geodesic is finished integrating: the final values are stored.
- void EndCurrentLoop () final

  End the current loop of integrating pixels.
- bool IsFinished () const final

  Check whether the Mesh is finished integrating.
- std::string getFullDescriptionStr () const final

  Description string getter.

Public Member Functions inherited from Mesh

- Mesh (DiagBitflag valdiag)
- virtual ~Mesh ()=default

Private Member Functions

- void InitializeFirstGrid ()

  Helper function to initialize the first grid of pixels to integrate.
- void UpdateAllNeighbors ()

  Helper function to update all pixel neighbors (if needed)
- void UpdateAllWeights ()

  Helper function to update all pixel weights.
- void SubdivideAndQueue (largecounter ind)

  Helper function to subdivide a pixel and add up to 5 new pixels to the queue.
- pixelcoord ExpInt (int base, int exp)

  Helper function to exponentiate ints.

Private Attributes

- const largecounter m_InitialPixels

  How many initial pixels (spread uniformly over the grid) do we integrate?
- const int m_MaxSubdivide

  How many times are we allowed to subdivide a square? Note: the initial grid is already at 1.
- const pixelcoord m_RowColumnSize

  The total size in pixels of a row or column (square grid)
- const largecounter m_IterationPixels

  How many pixels per iteration can we subdivide?
- const largecounter m_MaxPixels

  How many pixels can we integrate in total over all iterations?
- const bool m_InitialSubDividideToFinal

  If we decide to subdivide a square, do we automatically subdivide it further to the max level?
- const bool m_InfinitePixels

  Are we allowed to integrate as many pixels as we want? (m_MaxPixels == 0)
- largecounter m_PixelsLeft

  How many pixels are we still allowed to integrate (if !m_InfinitePixels)?
- std::vector< PixelInfo > m_CurrentPixelQueue {}

  The current queue of pixels to be integrated.
- std::vector< bool > m_CurrentPixelQueueDone {}

  A bool for every pixel in the current queue: gets set to true when the pixel is done integrating and gets its values returned.
- std::vector< PixelInfo > m_AllPixels {}

  All pixels that have been integrated already (so does not include the pixels in the current queue)

Additional Inherited Members

Protected Attributes inherited from Mesh

- const std::unique_ptr< const Diagnostic > m_DistanceDiagnostic

  The Diagnostic (a const pointer to a const Diagnostic object) that is used to calculate distances (using FinalData↩ ValDistance()) between the "values" that are assigned to Geodesics.

6.51.1 Detailed Description

Adaptive subdivision Mesh.

starts with evenly spaced, square Mesh, then decides to subdivide certain squares of pixels into smaller squares, based on which pixels have a bigger "weight", which is defined as the maximum ""distance" (using the Diagnostic value distance) between the upper-left vertex of the square with the other three vertices of the square.

6.51.2 Constructor & Destructor Documentation

6.51.2.1 SquareSubdivisionMesh() [1/2]

```
SquareSubdivisionMesh::SquareSubdivisionMesh ()  [delete]
```

6.51.2.2 SquareSubdivisionMesh() [2/2]

```
SquareSubdivisionMesh::SquareSubdivisionMesh (
            largecounter maxPixels,
            largecounter initialPixels,
            int maxSubdivide,
            largecounter iterationPixels,
            bool initialSubToFinal,
            DiagBitflag valdiag)  [inline]
```

6.51.3 Member Function Documentation

6.51.3.1 EndCurrentLoop()

```
void SquareSubdivisionMesh::EndCurrentLoop ()  [final], [virtual]
```

End the current loop of integrating pixels.

This function is called at the end of each integration iteration loop. We must wrap up the current iteration and initialize the next one.

Implements Mesh.

6.51.3.2 ExpInt()

```
pixelcoord SquareSubdivisionMesh::ExpInt (
            int base,
            int exp)  [private]
```

Helper function to exponentiate ints.

Note that the result may be larger than fits in an int, but this is only called with int arguments

Parameters

base	base
exp	exponent

Returns

pixelcoord

6.51.3.3 GeodesicFinished()

```
void SquareSubdivisionMesh::GeodesicFinished (
            largecounter index,
            std::vector< real > finalValues)  [final], [virtual]
```

Called when a geodesic is finished integrating: the final values are stored.

Parameters

index	index of the geodesic that is finished
finalValues	final values of the diagnostics

Implements Mesh.

6.51.3.4 getCurNrGeodesics()

```
largecounter SquareSubdivisionMesh::getCurNrGeodesics () const  [final], [virtual]
```

Return the current number of geodesics to be integrated.

Returns

largecounter

Implements Mesh.

6.51.3.5 getFullDescriptionStr()

```
std::string SquareSubdivisionMesh::getFullDescriptionStr () const  [final], [virtual]
```

Description string getter.

Returns

std::string

Reimplemented from Mesh.

6.51.3.6 getNewInitConds()

```
void SquareSubdivisionMesh::getNewInitConds (
            largecounter index,
            ScreenPoint & newunitpoint,
            ScreenIndex & newscreenindex) const  [final], [virtual]
```

Get the initial conditions for the next pixel to be integrated.

The next pixel is the one with index m_CurrentPixel

Parameters

index	index of the pixel to be initialized
newunitpoint	new unit point to be initialized
newscreenindex	new screen index to be initialized

Implements Mesh.

6.51.3.7 InitializeFirstGrid()

```
void SquareSubdivisionMesh::InitializeFirstGrid ()  [private]
```

Helper function to initialize the first grid of pixels to integrate.

6.51.3.8 IsFinished()

```
bool SquareSubdivisionMesh::IsFinished () const  [final], [virtual]
```

Check whether the Mesh is finished integrating.

Returns

true

false

Implements Mesh.

6.51.3.9 SubdivideAndQueue()

```
void SquareSubdivisionMesh::SubdivideAndQueue (
            largecounter ind)  [private]
```

Helper function to subdivide a pixel and add up to 5 new pixels to the queue.

This will take the pixel m_AllPixels[ind] and subdivide it, adding up to <=5 pixels to the CurrentPixelQueue

Parameters

ind	index of the pixel to subdivide

6.51.3.10 UpdateAllNeighbors()

```
void SquareSubdivisionMesh::UpdateAllNeighbors ()  [private]
```

Helper function to update all pixel neighbors (if needed)

6.51.3.11 UpdateAllWeights()

```
void SquareSubdivisionMesh::UpdateAllWeights ()  [private]
```

Helper function to update all pixel weights.

Updates all weights of the pixels in m_AllPixels with weight $<$ 0 and subdiv $>$ 0 and subdiv $<$ m_MaxSubdivide

Note

Assumes all squares have neighbors assigned correctly!

6.51.4 Member Data Documentation

6.51.4.1 m_AllPixels

```
std::vector<PixelInfo> SquareSubdivisionMesh::m_AllPixels {}  [private]
```

All pixels that have been integrated already (so does not include the pixels in the current queue)

6.51.4.2 m_CurrentPixelQueue

```
std::vector<PixelInfo> SquareSubdivisionMesh::m_CurrentPixelQueue {}  [private]
```

The current queue of pixels to be integrated.

6.51.4.3 m_CurrentPixelQueueDone

```
std::vector<bool> SquareSubdivisionMesh::m_CurrentPixelQueueDone {}  [private]
```

A bool for every pixel in the current queue: gets set to true when the pixel is done integrating and gets its values returned.

6.51.4.4 m_InfinitePixels

```
const bool SquareSubdivisionMesh::m_InfinitePixels  [private]
```

Are we allowed to integrate as many pixels as we want? (m_MaxPixels == 0)

6.51.4.5 m_InitialPixels

```
const largecounter SquareSubdivisionMesh::m_InitialPixels  [private]
```

How many initial pixels (spread uniformly over the grid) do we integrate?

6.51.4.6 m_InitialSubDividideToFinal

```
const bool SquareSubdivisionMesh::m_InitialSubDividideToFinal  [private]
```

If we decide to subdivide a square, do we automatically subdivide it further to the max level?

6.51.4.7 m_IterationPixels

```
const largecounter SquareSubdivisionMesh::m_IterationPixels  [private]
```

How many pixels per iteration can we subdivide?

6.51.4.8 m_MaxPixels

```
const largecounter SquareSubdivisionMesh::m_MaxPixels  [private]
```

How many pixels can we integrate in total over all iterations?

6.51.4.9 m_MaxSubdivide

```
const int SquareSubdivisionMesh::m_MaxSubdivide  [private]
```

How many times are we allowed to subdivide a square? Note: the initial grid is already at 1.

6.51.4.10 m_PixelsLeft

```
largecounter SquareSubdivisionMesh::m_PixelsLeft  [private]
```

How many pixels are we still allowed to integrate (if !m_InfinitePixels)?

6.51.4.11 m_RowColumnSize

```
const pixelcoord SquareSubdivisionMesh::m_RowColumnSize  [private]
```

The total size in pixels of a row or column (square grid)

The documentation for this class was generated from the following files:

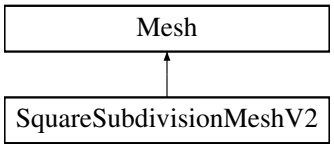
- FOORT/src/Mesh.h
- FOORT/src/Mesh.cpp

6.52 SquareSubdivisionMeshV2 Class Reference

Adaptive subdivision Mesh.

`#include <Mesh.h>`

Inheritance diagram for SquareSubdivisionMeshV2:



Classes

- struct PixelInfo

  A struct the Mesh uses to keep all information about a given pixel.

Public Member Functions

- SquareSubdivisionMeshV2 ()=delete
- SquareSubdivisionMeshV2 (largecounter maxPixels, largecounter initialPixels, int maxSubdivide, largecounter iterationPixels, bool initialSubToFinal, DiagBitflag valdiag)
- largecounter getCurNrGeodesics () const final

  Return the current number of geodesics to be integrated.
- void getNewInitConds (largecounter index, ScreenPoint &newunitpoint, ScreenIndex &newscreenindex) const final

  Get the initial conditions for the next pixel to be integrated.
- void GeodesicFinished (largecounter index, std::vector< real > finalValues) final

  Called when a geodesic is finished integrating: the final values are stored.
- void EndCurrentLoop () final

  Called at the end of each integration iteration loop. Wraps up the current iteration and initializes the next one.
- bool IsFinished () const final

  Check whether the Mesh is finished integrating.
- std::string getFullDescriptionStr () const final

  Description string getter.

Public Member Functions inherited from Mesh

- Mesh (DiagBitflag valdiag)
- virtual ~Mesh ()=default

Private Member Functions

- void InitializeFirstGrid ()

  Helper function to initialize the first grid of pixels to integrate.
- void UpdateAllWeights ()

  Helper function to update all pixel weights in m_CurrentPixelUpdating.
- PixelInfo * GetUp (PixelInfo *p, int subdiv) const

  Get the upper neighbor of a pixel at a certain subdivision level.
- PixelInfo * GetDown (PixelInfo *p, int subdiv) const

  Get the lower neighbor of a pixel at a certain subdivision level.
- PixelInfo * GetRight (PixelInfo *p, int subdiv) const

  Get the right neighbor of a pixel at a certain subdivision level.
- PixelInfo * GetLeft (PixelInfo *p, int subdiv) const

  Get the left neighbor of a pixel at a certain subdivision level.
- void SubdivideAndQueue (largecounter ind)

  Subdivide the pixel at index ind and add up to 5 new pixels to the integration queue.
- pixelcoord ExpInt (int base, int exp) const

  Helper function to exponentiate ints.

Private Attributes

- const largecounter m_InitialPixels

  How many initial pixels (spread uniformly over the grid) do we integrate?
- const int m_MaxSubdivide

  How many times are we allowed to subdivide a square? Note: the initial grid is already at 1.
- const pixelcoord m_RowColumnSize

  The total size in pixels of a row or column (square grid)
- const largecounter m_IterationPixels

  How many pixels per iteration can we subdivide?
- const largecounter m_MaxPixels

  How many pixels can we integrate in total over all iterations?
- const bool m_InitialSubDivideToFinal

  If we decide to subdivide a square, do we automatically subdivide it further to the max level?
- const bool m_InfinitePixels

  Are we allowed to integrate as many pixels as we want? (m_MaxPixels == 0)
- largecounter m_PixelsLeft

  How many pixels are we still allowed to integrate (if !m_InfinitePixels)?
- largecounter m_PixelsIntegrated {0}

  How many pixels we have integrated so far.
- std::forward_list< std::unique_ptr< PixelInfo > > m_AllPixels {}

  Master list of all pixels. std::forward_list is more space-efficient than std::list, bidirectional iteration is not needed, no random access supported. This list is only used to store the owner pointers (and thus the objects) of all pixels. The other pixel vectors are used to iterate through (and need random access)
- std::vector< PixelInfo * > m_ActivePixels {}

  List of active pixels, i.e. those that can be subdivided and have non-zero weight.
- std::vector< PixelInfo * > m_CurrentPixelQueue {}

  List of current queue of pixels to be sent to be integrated.
- std::vector< bool > m_CurrentPixelQueueDone {}

  A bool for every pixel in the current queue: gets set to true when the pixel is done integrating and gets its values returned.
- std::vector< PixelInfo * > m_CurrentPixelUpdating {}

  List of pixels that are already integrated but need updating weights after current queue is all integrated.

Additional Inherited Members

Protected Attributes inherited from Mesh

- const std::unique_ptr< const Diagnostic > m_DistanceDiagnostic

  The Diagnostic (a const pointer to a const Diagnostic object) that is used to calculate distances (using FinalData↩ ValDistance()) between the "values" that are assigned to Geodesics.

6.52.1 Detailed Description

Adaptive subdivision Mesh.

starts with evenly spaced, square Mesh, then decides to subdivide certain squares of pixels into smaller squares, based on which pixels have a bigger "weight", which is defined as the maximum "distance" (using the Diagnostic value distance) between the upper-left vertex of the square with the other three vertices of the square. V2: new way of dealing with neighbors and looping over pixels

6.52.2 Constructor & Destructor Documentation

6.52.2.1 SquareSubdivisionMeshV2() [1/2]

```
SquareSubdivisionMeshV2::SquareSubdivisionMeshV2 ()  [delete]
```

6.52.2.2 SquareSubdivisionMeshV2() [2/2]

```
SquareSubdivisionMeshV2::SquareSubdivisionMeshV2 (
            largecounter maxPixels,
            largecounter initialPixels,
            int maxSubdivide,
            largecounter iterationPixels,
            bool initialSubToFinal,
            DiagBitflag valdiag)  [inline]
```

6.52.3 Member Function Documentation

6.52.3.1 EndCurrentLoop()

```
void SquareSubdivisionMeshV2::EndCurrentLoop ()  [final], [virtual]
```

Called at the end of each integration iteration loop. Wraps up the current iteration and initializes the next one.

Implements Mesh.

6.52.3.2 ExpInt()

```
pixelcoord SquareSubdivisionMeshV2::ExpInt (
            int base,
            int exp) const  [private]
```

Helper function to exponentiate ints.

Note

The result may be larger than fits in an int, but this is only called with int arguments

Parameters

base	base
exp	exponent

Returns

pixelcoord

6.52.3.3 GeodesicFinished()

```
void SquareSubdivisionMeshV2::GeodesicFinished (
            largecounter index,
            std::vector< real > finalValues)  [final], [virtual]
```

Called when a geodesic is finished integrating: the final values are stored.

Parameters

index	index of the geodesic that is finished
finalValues	final values of the diagnostics

Implements Mesh.

6.52.3.4 getCurNrGeodesics()

```
largecounter SquareSubdivisionMeshV2::getCurNrGeodesics () const  [final], [virtual]
```

Return the current number of geodesics to be integrated.

Returns

largecounter

Implements Mesh.

6.52.3.5 GetDown()

```
SquareSubdivisionMeshV2::PixelInfo * SquareSubdivisionMeshV2::GetDown (
            PixelInfo * p,
            int subdiv) const  [private]
```

Get the lower neighbor of a pixel at a certain subdivision level.

Parameters

p	PixelInfo pointer to the pixel
subdiv	subdivision level of the neighbor

Returns

SquareSubdivisionMeshV2::PixelInfo*

6.52.3.6 getFullDescriptionStr()

```
std::string SquareSubdivisionMeshV2::getFullDescriptionStr () const  [final], [virtual]
```

Description string getter.

Returns

std::string

Reimplemented from Mesh.

6.52.3.7 GetLeft()

```
SquareSubdivisionMeshV2::PixelInfo * SquareSubdivisionMeshV2::GetLeft (
            PixelInfo * p,
            int subdiv) const  [private]
```

Get the left neighbor of a pixel at a certain subdivision level.

Parameters

p	PixelInfo pointer to the pixel
subdiv	subdivision level of the neighbor

Returns

SquareSubdivisionMeshV2::PixelInfo*

6.52.3.8 getNewInitConds()

```
void SquareSubdivisionMeshV2::getNewInitConds (
            largecounter index,
            ScreenPoint & newunitpoint,
            ScreenIndex & newscreenindex) const  [final], [virtual]
```

Get the initial conditions for the next pixel to be integrated.

The next pixel is the one with index m_CurrentPixel

Parameters

index	index of the pixel to be initialized
newunitpoint	new unit point to be initialized
newscreenindex	new screen index to be initialized

Implements Mesh.

6.52.3.9 GetRight()

```
SquareSubdivisionMeshV2::PixelInfo * SquareSubdivisionMeshV2::GetRight (
            PixelInfo * p,
            int subdiv) const  [private]
```

Get the right neighbor of a pixel at a certain subdivision level.

Parameters

p	PixelInfo pointer to the pixel
subdiv	subdivision level of the neighbor

Returns

SquareSubdivisionMeshV2::PixelInfo*

6.52.3.10 GetUp()

```
SquareSubdivisionMeshV2::PixelInfo * SquareSubdivisionMeshV2::GetUp (
            PixelInfo * p,
            int subdiv) const  [private]
```

Get the upper neighbor of a pixel at a certain subdivision level.

Parameters

p	PixelInfo pointer to the pixel
subdiv	subdivision level of the neighbor

Returns

SquareSubdivisionMeshV2::PixelInfo*

6.52.3.11 InitializeFirstGrid()

```
void SquareSubdivisionMeshV2::InitializeFirstGrid ()  [private]
```

Helper function to initialize the first grid of pixels to integrate.

6.52.3.12 IsFinished()

```
bool SquareSubdivisionMeshV2::IsFinished () const  [final], [virtual]
```

Check whether the Mesh is finished integrating.

Returns

true

false

Implements Mesh.

6.52.3.13 SubdivideAndQueue()

```
void SquareSubdivisionMeshV2::SubdivideAndQueue (
            largecounter ind)  [private]
```

Subdivide the pixel at index ind and add up to 5 new pixels to the integration queue.

The pixels are numbered as 1 2 3 4 5 6 7 8 9 i.e. the initial square is given by (1, 3, 7, 9), and the new pixels are (2, 4, 5, 6, 8).

Parameters

ind	index of the pixel to subdivide

6.52.3.14 UpdateAllWeights()

```
void SquareSubdivisionMeshV2::UpdateAllWeights ()  [private]
```

Helper function to update all pixel weights in m_CurrentPixelUpdating.

All pixels with weight $>$ 0 will be added to m_ActivePixels

Note

Assumes all pixels have subdiv $>$ 0 and subdiv $<$ m_MaxSubdivide, and all their neighbors assigned correctly

Assumes all pixels have their (and their neighbor's) values assigned correctly

6.52.4 Member Data Documentation

6.52.4.1 m_ActivePixels

```
std::vector<PixelInfo *> SquareSubdivisionMeshV2::m_ActivePixels {}  [private]
```

List of active pixels, i.e. those that can be subdivided and have non-zero weight.

6.52.4.2 m_AllPixels

```
std::forward_list<std::unique_ptr<PixelInfo> > SquareSubdivisionMeshV2::m_AllPixels {}  [private]
```

Master list of all pixels. std::forward_list is more space-efficient than std::list, bidirectional iteration is not needed, no random access supported. This list is only used to store the owner pointers (and thus the objects) of all pixels. The other pixel vectors are used to iterate through (and need random access)

6.52.4.3 m_CurrentPixelQueue

```
std::vector<PixelInfo *> SquareSubdivisionMeshV2::m_CurrentPixelQueue {}  [private]
```

List of current queue of pixels to be sent to be integrated.

6.52.4.4 m_CurrentPixelQueueDone

```
std::vector<bool> SquareSubdivisionMeshV2::m_CurrentPixelQueueDone {}  [private]
```

A bool for every pixel in the current queue: gets set to true when the pixel is done integrating and gets its values returned.

6.52.4.5 m_CurrentPixelUpdating

```
std::vector<PixelInfo *> SquareSubdivisionMeshV2::m_CurrentPixelUpdating {}  [private]
```

List of pixels that are already integrated but need updating weights after current queue is all integrated.

6.52.4.6 m_InfinitePixels

```
const bool SquareSubdivisionMeshV2::m_InfinitePixels  [private]
```

Are we allowed to integrate as many pixels as we want? (m_MaxPixels == 0)

6.52.4.7 m_InitialPixels

```
const largecounter SquareSubdivisionMeshV2::m_InitialPixels  [private]
```

How many initial pixels (spread uniformly over the grid) do we integrate?

6.52.4.8 m_InitialSubDividideToFinal

```
const bool SquareSubdivisionMeshV2::m_InitialSubDividideToFinal  [private]
```

If we decide to subdivide a square, do we automatically subdivide it further to the max level?

6.52.4.9 m_IterationPixels

```
const largecounter SquareSubdivisionMeshV2::m_IterationPixels  [private]
```

How many pixels per iteration can we subdivide?

6.52.4.10 m_MaxPixels

```
const largecounter SquareSubdivisionMeshV2::m_MaxPixels  [private]
```

How many pixels can we integrate in total over all iterations?

6.52.4.11 m_MaxSubdivide

```
const int SquareSubdivisionMeshV2::m_MaxSubdivide  [private]
```

How many times are we allowed to subdivide a square? Note: the initial grid is already at 1.

6.52.4.12 m_PixelsIntegrated

```
largecounter SquareSubdivisionMeshV2::m_PixelsIntegrated {0}  [private]
```

How many pixels we have integrated so far.

6.52.4.13 m_PixelsLeft

```
largecounter SquareSubdivisionMeshV2::m_PixelsLeft  [private]
```

How many pixels are we still allowed to integrate (if !m_InfinitePixels)?

6.52.4.14 m_RowColumnSize

```
const pixelcoord SquareSubdivisionMeshV2::m_RowColumnSize  [private]
```

The total size in pixels of a row or column (square grid)

The documentation for this class was generated from the following files:

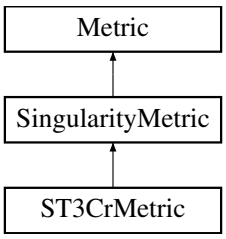
- FOORT/src/Mesh.h
- FOORT/src/Mesh.cpp

6.53 ST3CrMetric Class Reference

Ring fuzzball metric.

```
#include <Metric.h>
```

Inheritance diagram for ST3CrMetric:



Public Member Functions

- ST3CrMetric (real P, real q0, real lambda, bool rlogscale=false)
  Construct a new ST3Cr Metric object.
- TwoIndex getMetric_dd (const Point &p) const final
  ST3Cr metric getter, indices down.
- TwoIndex getMetric_uu (const Point &p) const final
  ST3Cr metric getter, indices up.
- std::string getFullDescriptionStr () const final
  ST3Cr metric description string getter.

Public Member Functions inherited from SingularityMetric

- SingularityMetric (std::vector< Singularity > thesings, bool rLogScale)
  Construct a new Singularity Metric object.
- std::vector< Singularity > getSingularities () const
  Getters for the singularities.

Public Member Functions inherited from Metric

- virtual ~Metric ()=default
- Metric (bool rlogscale=false)

  Construct a new Metric object.
- virtual ThreeIndex getChristoffel_udd (const Point &p) const

  Return the Christoffel symbols of the metric (indices up, down, down)
- virtual FourIndex getRiemann_uddd (const Point &p) const

  Riemann tensor of the metric (indices up, down, down, down)
- virtual real getKretschmann (const Point &p) const

  Get the Kretschmann scalar at a point.
- bool getrLogScale () const

  Get whether we are using a logarithmic radial scale.

Private Member Functions

- real get_omega (real r, real theta, real l) const

  Get omega for the ST3Cr metric.
- real f_phi (real phi, real r, real theta, real l, real R) const

  *function for prefactor * dphi' for ring*
- real f_om_phi (real phi, real r, real theta, real l, real R) const

  omega_phi ring functions

Private Attributes

- const real m_P
- const real m_q0
- const real m_lambda

Additional Inherited Members

Protected Attributes inherited from SingularityMetric

- const std::vector< Singularity > m_AllSingularities

  All singularities of the metric.

Protected Attributes inherited from Metric

- std::vector< int > m_Symmetries {}

  The symmetries (coordinate Killing vectors) of the metric. Should be set by descendant constructor.
- const bool m_rLogScale

  Are we using a logarithmic r coordinate?

6.53.1 Detailed Description

Ring fuzzball metric.

Note

Implementation by Lies Van Dael

6.53.2 Constructor & Destructor Documentation

6.53.2.1 ST3CrMetric()

```
ST3CrMetric::ST3CrMetric (
            real P,
            real q0,
            real lambda,
            bool rlogscale = false)
```

Construct a new ST3Cr Metric object.

Parameters

P	P parameter for the ST3Cr metric
q0	q0 parameter for the ST3Cr metric
lambda	lambda parameter for the ST3Cr metric
rlogscale	whether we are using a logarithmic radial scale

6.53.3 Member Function Documentation

6.53.3.1 f_om_phi()

```
real ST3CrMetric::f_om_phi (
            real phi,
            real r,
            real theta,
            real l,
            real R) const  [private]
```

omega_phi ring functions

Parameters

phi	
r	
theta	
l	
R	

Returns

real

6.53.3.2 f_phi()

```
real ST3CrMetric::f_phi (
            real phi,
            real r,
            real theta,
            real l,
            real R) const  [private]
```

function for prefactor $*$ dphi' for ring

Parameters

phi	
r	
theta	
l	
R	

Returns

real

6.53.3.3 get_omega()

```
real ST3CrMetric::get_omega (
            real r,
            real theta,
            real l) const  [private]
```

Get omega for the ST3Cr metric.

Parameters

r	r_ij
theta	theta↩ _ij
l	l_ij

Returns

real

6.53.3.4 getFullDescriptionStr()

```
std::string ST3CrMetric::getFullDescriptionStr () const  [final], [virtual]
```

ST3Cr metric description string getter.

Returns

std::string

Reimplemented from Metric.

6.53.3.5 getMetric_dd()

```
TwoIndex ST3CrMetric::getMetric_dd (
            const Point & p) const  [final], [virtual]
```

ST3Cr metric getter, indices down.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.53.3.6 getMetric_uu()

```
TwoIndex ST3CrMetric::getMetric_uu (
            const Point & p) const  [final], [virtual]
```

ST3Cr metric getter, indices up.

Parameters

p	Point at which to evaluate the metric

Returns

TwoIndex

Implements Metric.

6.53.4 Member Data Documentation

6.53.4.1 m_lambda

```
const real ST3CrMetric::m_lambda  [private]
```

6.53.4.2 m_P

```
const real ST3CrMetric::m_P  [private]
```

6.53.4.3 m_q0

```
const real ST3CrMetric::m_q0  [private]
```

The documentation for this class was generated from the following files:

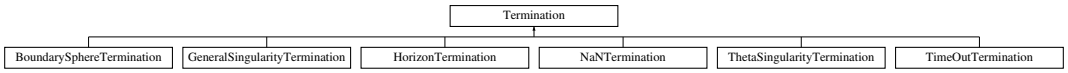
- FOORT/src/Metric.h
- FOORT/src/Metric.cpp

6.54 Termination Class Reference

Abstract base class for all Terminations.

`#include <Terminations.h>`

Inheritance diagram for Termination:



Public Member Functions

- Termination ()=delete
- Termination (Geodesic ∗const theGeodesic)
- virtual void Reset ()

  Reset to 0 to start integrating a new geodesic.
- virtual ∼Termination ()=default
- virtual Term CheckTermination ()=0
- virtual std::string getFullDescriptionStr () const =0

Protected Member Functions

- bool DecideUpdate (largecounter UpdateNSteps)

  Returns true if the Termination should update its internal status.

Protected Attributes

- Geodesic ∗const m_OwnerGeodesic

  The geodesic that owns the Termination (a const pointer to the Geodesic)
- largecounter m_StepsSinceUpdated {}

  The termination is itself in charge of keeping track of how many steps it has been since it has been updated. The Termination's TerminationOptions struct tells it how many steps it needs to wait between updates.

6.54.1 Detailed Description

Abstract base class for all Terminations.

6.54.2 Constructor & Destructor Documentation

6.54.2.1 Termination() [1/2]

`Termination::Termination () [delete]`

6.54.2.2 Termination() [2/2]

```
Termination::Termination (
            Geodesic *const theGeodesic)  [inline]
```

6.54.2.3 ~Termination()

```
virtual Termination::~Termination ()  [virtual], [default]
```

6.54.3 Member Function Documentation

6.54.3.1 CheckTermination()

```
virtual Term Termination::CheckTermination ()  [pure virtual]
```

Implemented in BoundarySphereTermination, GeneralSingularityTermination, HorizonTermination, NaNTermination, ThetaSingularityTermination, and TimeOutTermination.

6.54.3.2 DecideUpdate()

```
bool Termination::DecideUpdate (
            largecounter UpdateNSteps)  [protected]
```

Returns true if the Termination should update its internal status.

Should be called from within CheckTermination() with the appropriate TermOptions::UpdateEveryNSteps

Parameters

UpdateNSteps	the number of steps after which the terminations should be updated

Returns

true

false

6.54.3.3 getFullDescriptionStr()

```
virtual std::string Termination::getFullDescriptionStr () const  [pure virtual]
```

Implemented in BoundarySphereTermination, GeneralSingularityTermination, HorizonTermination, NaNTermination, ThetaSingularityTermination, and TimeOutTermination.

6.54.3.4 Reset()

```
void Termination::Reset ()  [virtual]
```

Reset to 0 to start integrating a new geodesic.

Reimplemented in TimeOutTermination.

6.54.4 Member Data Documentation

6.54.4.1 m_OwnerGeodesic

```
Geodesic* const Termination::m_OwnerGeodesic  [protected]
```

The geodesic that owns the Termination (a const pointer to the Geodesic)

6.54.4.2 m_StepsSinceUpdated

```
largecounter Termination::m_StepsSinceUpdated {}  [protected]
```

The termination is itself in charge of keeping track of how many steps it has been since it has been updated. The Termination's TerminationOptions struct tells it how many steps it needs to wait between updates.

The documentation for this class was generated from the following files:

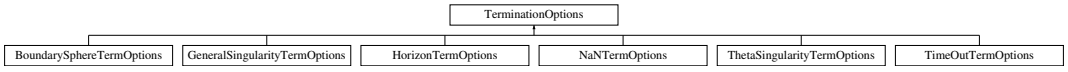
- FOORT/src/Terminations.h
- FOORT/src/Terminations.cpp

6.55 TerminationOptions Struct Reference

Base class for all TerminationOptions structs. Other TerminationOptions can inherit from here if they require more options.

```
#include <Terminations.h>
```

Inheritance diagram for TerminationOptions:



Public Member Functions

- TerminationOptions (largecounter Nsteps)
- virtual ~TerminationOptions ()=default

Public Attributes

- const largecounter UpdateEveryNSteps

  Number of steps between updates.

6.55.1 Detailed Description

Base class for all TerminationOptions structs. Other TerminationOptions can inherit from here if they require more options.

6.55.2 Constructor & Destructor Documentation

6.55.2.1 TerminationOptions()

```
TerminationOptions::TerminationOptions (
            largecounter Nsteps)  [inline]
```

6.55.2.2 ~TerminationOptions()

```
virtual TerminationOptions::~TerminationOptions ()  [virtual], [default]
```

6.55.3 Member Data Documentation

6.55.3.1 UpdateEveryNSteps

```
const largecounter TerminationOptions::UpdateEveryNSteps
```

Number of steps between updates.

The documentation for this struct was generated from the following file:

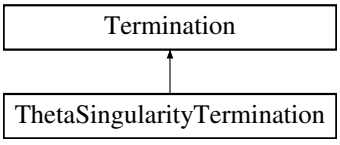
- FOORT/src/Terminations.h

6.56 ThetaSingularityTermination Class Reference

ThetaSingularityTermination: Terminate geodesics if they get too close to a polar coordinate singularity.

```
#include <Terminations.h>
```

Inheritance diagram for ThetaSingularityTermination:



Public Member Functions

- ThetaSingularityTermination (Geodesic ∗const theGeodesic)
- Term CheckTermination () final

  Check if the geodesic is too close to a pole.
- std::string getFullDescriptionStr () const final

  Description string getter for the Theta Singularity Termination.

Public Member Functions inherited from Termination

- Termination ()=delete
- Termination (Geodesic ∗const theGeodesic)
- virtual void Reset ()

  Reset to 0 to start integrating a new geodesic.
- virtual ∼Termination ()=default

Static Public Attributes

- static std::unique_ptr< ThetaSingularityTermOptions > TermOptions

  The options that the Termination keeps (will probably be a descendant struct instead, which specifies any additional options the Termination needs)

Additional Inherited Members

Protected Member Functions inherited from Termination

- bool DecideUpdate (largecounter UpdateNSteps)

  Returns true if the Termination should update its internal status.

Protected Attributes inherited from Termination

- Geodesic ∗const m_OwnerGeodesic

  The geodesic that owns the Termination (a const pointer to the Geodesic)
- largecounter m_StepsSinceUpdated {}

  The termination is itself in charge of keeping track of how many steps it has been since it has been updated. The Termination's TerminationOptions struct tells it how many steps it needs to wait between updates.

6.56.1 Detailed Description

ThetaSingularityTermination: Terminate geodesics if they get too close to a polar coordinate singularity.

6.56.2 Constructor & Destructor Documentation

6.56.2.1 ThetaSingularityTermination()

```
ThetaSingularityTermination::ThetaSingularityTermination (
            Geodesic *const theGeodesic)  [inline]
```

6.56.3 Member Function Documentation

6.56.3.1 CheckTermination()

```
Term ThetaSingularityTermination::CheckTermination ()  [final], [virtual]
```

Check if the geodesic is too close to a pole.

Returns

Term

Implements Termination.

6.56.3.2 getFullDescriptionStr()

```
std::string ThetaSingularityTermination::getFullDescriptionStr () const  [final], [virtual]
```

Description string getter for the Theta Singularity Termination.

Returns

std::string

Implements Termination.

6.56.4 Member Data Documentation

6.56.4.1 TermOptions

```
std::unique_ptr< ThetaSingularityTermOptions > ThetaSingularityTermination::TermOptions  [static]
```

The options that the Termination keeps (will probably be a descendant struct instead, which specifies any additional options the Termination needs)

The documentation for this class was generated from the following files:

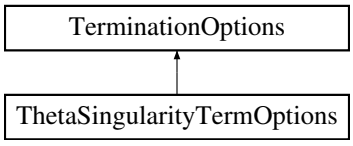
- FOORT/src/Terminations.h
- FOORT/src/Config.cpp
- FOORT/src/Terminations.cpp

6.57 ThetaSingularityTermOptions Struct Reference

Options for ThetaSingularityTermination.

```
#include <Terminations.h>
```

Inheritance diagram for ThetaSingularityTermOptions:



Public Member Functions

- ThetaSingularityTermOptions (real epsilon, largecounter Nsteps)

Public Member Functions inherited from TerminationOptions

- TerminationOptions (largecounter Nsteps)
- virtual ~TerminationOptions ()=default

Public Attributes

- const real ThetaSingEpsilon

  Minimum difference between theta and 0 or pi.

Public Attributes inherited from TerminationOptions

- const largecounter UpdateEveryNSteps

  Number of steps between updates.

6.57.1 Detailed Description

Options for ThetaSingularityTermination.

Keeps track of the minimum required distance from the singularity

6.57.2 Constructor & Destructor Documentation

6.57.2.1 ThetaSingularityTermOptions()

```
ThetaSingularityTermOptions::ThetaSingularityTermOptions (
            real epsilon,
            largecounter Nsteps)  [inline]
```

6.57.3 Member Data Documentation

6.57.3.1 ThetaSingEpsilon

```
const real ThetaSingularityTermOptions::ThetaSingEpsilon
```

Minimum difference between theta and 0 or pi.

The documentation for this struct was generated from the following file:

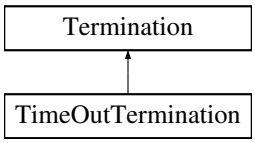
- FOORT/src/Terminations.h

6.58 TimeOutTermination Class Reference

TimeOutTermination: Terminate geodesics if they take too many steps.

`#include <Terminations.h>`

Inheritance diagram for TimeOutTermination:



Public Member Functions

- TimeOutTermination (Geodesic *const theGeodesic)
- void Reset () final

  Reset m_CurNrSteps to 0 for new geodesic, and call base class implementation to reset base class member variables.
- Term CheckTermination () final

  Check if the geodesic has taken too many steps.
- std::string getFullDescriptionStr () const final

  Description string getter for the Time Out Termination.

Public Member Functions inherited from Termination

- Termination ()=delete
- Termination (Geodesic *const theGeodesic)
- virtual ~Termination ()=default

Static Public Attributes

- static std::unique_ptr< TimeOutTermOptions > TermOptions

  The options that the TimeOutTermination keeps (contains max number of steps allowed)

Private Attributes

- largecounter m_CurNrSteps {0}

  Keep track of the number of steps that the geodesic has taken so far.

Additional Inherited Members

Protected Member Functions inherited from Termination

- bool DecideUpdate (largecounter UpdateNSteps)

  Returns true if the Termination should update its internal status.

Protected Attributes inherited from Termination

- Geodesic *const m_OwnerGeodesic

  The geodesic that owns the Termination (a const pointer to the Geodesic)
- largecounter m_StepsSinceUpdated {}

  The termination is itself in charge of keeping track of how many steps it has been since it has been updated. The Termination's TerminationOptions struct tells it how many steps it needs to wait between updates.

6.58.1 Detailed Description

TimeOutTermination: Terminate geodesics if they take too many steps.

6.58.2 Constructor & Destructor Documentation

6.58.2.1 TimeOutTermination()

```
TimeOutTermination::TimeOutTermination (
            Geodesic *const theGeodesic)  [inline]
```

6.58.3 Member Function Documentation

6.58.3.1 CheckTermination()

```
Term TimeOutTermination::CheckTermination ()  [final], [virtual]
```

Check if the geodesic has taken too many steps.

Returns

Term

Implements Termination.

6.58.3.2 getFullDescriptionStr()

```
std::string TimeOutTermination::getFullDescriptionStr () const  [final], [virtual]
```

Description string getter for the Time Out Termination.

Returns

std::string

Implements Termination.

6.58.3.3 Reset()

```
void TimeOutTermination::Reset ()  [final], [virtual]
```

Reset m_CurNrSteps to 0 for new geodesic, and call base class implementation to reset base class member variables.

Reimplemented from Termination.

6.58.4 Member Data Documentation

6.58.4.1 m_CurNrSteps

```
largecounter TimeOutTermination::m_CurNrSteps {0}  [private]
```

Keep track of the number of steps that the geodesic has taken so far.

6.58.4.2 TermOptions

```
std::unique_ptr< TimeOutTermOptions > TimeOutTermination::TermOptions  [static]
```

The options that the TimeOutTermination keeps (contains max number of steps allowed)

The documentation for this class was generated from the following files:

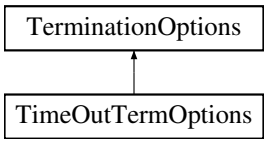
- FOORT/src/Terminations.h
- FOORT/src/Config.cpp
- FOORT/src/Terminations.cpp

6.59 TimeOutTermOptions Struct Reference

Options for TimeOutTermination.

```
#include <Terminations.h>
```

Inheritance diagram for TimeOutTermOptions:



Public Member Functions

- TimeOutTermOptions (largecounter MaxStepsAllowed, largecounter Nsteps)

Public Member Functions inherited from TerminationOptions

- TerminationOptions (largecounter Nsteps)
- virtual ~TerminationOptions ()=default

Public Attributes

- const largecounter MaxSteps

  Max. number of integration steps allowed.

Public Attributes inherited from TerminationOptions

- const largecounter UpdateEveryNSteps

  Number of steps between updates.

6.59.1 Detailed Description

Options for TimeOutTermination.

Keeps track of the max. number of integration steps allowed

6.59.2 Constructor & Destructor Documentation

6.59.2.1 TimeOutTermOptions()

```
TimeOutTermOptions::TimeOutTermOptions (
            largecounter MaxStepsAllowed,
            largecounter Nsteps)  [inline]
```

6.59.3 Member Data Documentation

6.59.3.1 MaxSteps

```
const largecounter TimeOutTermOptions::MaxSteps
```

Max. number of integration steps allowed.

The documentation for this struct was generated from the following file:

- FOORT/src/Terminations.h

6.60 Utilities::Timer Class Reference

Timer class to keep track of elapsed time.

```
#include <Utilities.h>
```

Public Member Functions

- void reset ()

  Reset begin time.
- double elapsed () const

  Returns time elapsed since begin time.

Private Types

- using Clock = std::chrono::steady_clock

  Type alias to make accessing nested type easier.
- using Second = std::chrono::duration<double, std::ratio<1>>

  Type alias to make accessing nested type easier.

Private Attributes

- std::chrono::time_point< Clock > m_beg {Clock::now()}

  begin time

6.60.1 Detailed Description

Timer class to keep track of elapsed time.

6.60.2 Member Typedef Documentation

6.60.2.1 Clock

```
using Utilities::Timer::Clock = std::chrono::steady_clock  [private]
```

Type alias to make accessing nested type easier.

6.60.2.2 Second

```
using Utilities::Timer::Second = std::chrono::duration<double, std::ratio<1>>  [private]
```

Type alias to make accessing nested type easier.

6.60.3 Member Function Documentation

6.60.3.1 elapsed()

```
double Utilities::Timer::elapsed () const
```

Returns time elapsed since begin time.

Returns

double

6.60.3.2 reset()

```
void Utilities::Timer::reset ()
```

Reset begin time.

6.60.4 Member Data Documentation

6.60.4.1 m_beg

```
std::chrono::time_point<Clock> Utilities::Timer::m_beg {Clock::now()}  [private]
```

begin time

The documentation for this class was generated from the following files:

- FOORT/src/Utilities.h
- FOORT/src/Utilities.cpp

6.61 UpdateFrequency Struct Reference

Struct to hold the information for a Diagnostic to update itself.

```
#include <Diagnostics.h>
```

Public Attributes

- largecounter UpdateNSteps {0}
- bool UpdateStart {false}
- bool UpdateFinish {false}

6.61.1 Detailed Description

Struct to hold the information for a Diagnostic to update itself.

If UpdateNsteps $>$ 0, then every so many steps. If UpdateNSteps == 0, then it only updates at the start and/or finish of integration if the appropriate bool is set.

6.61.2 Member Data Documentation

6.61.2.1 UpdateFinish

```
bool UpdateFrequency::UpdateFinish {false}
```

6.61.2.2 UpdateNSteps

```
largecounter UpdateFrequency::UpdateNSteps {0}
```

6.61.2.3 UpdateStart

```
bool UpdateFrequency::UpdateStart {false}
```

The documentation for this struct was generated from the following file:

- FOORT/src/Diagnostics.h

6.62 ViewScreen Class Reference

This class is in charge of converting a pixel on the screen into physical initial conditions for a geodesic.

```
#include <ViewScreen.h>
```

Public Member Functions

- ViewScreen ()=delete
- ViewScreen (Point pos, OneIndex dir, ScreenPoint screensize, ScreenPoint screencenter, std::unique_ptr< Mesh > theMesh, const Metric *const theMetric, GeodesicType thegeodtype=GeodesicType::Null)
- void SetNewInitialConditions (largecounter index, Point &pos, OneIndex &vel, ScreenIndex &scrIndex) const

  Set new initial conditions for geodesic nr index.
- bool IsFinished () const

  Check if the ViewScreen is finished.
- largecounter getCurNrGeodesics () const

  Get the current number of geodesics.
- void EndCurrentLoop ()

  End the current loop of geodesics.
- void GeodesicFinished (largecounter index, std::vector< real > finalValues)

  Pass on information that a geodesic has finished.
- std::string getFullDescriptionStr () const

  Get the full description string of the ViewScreen.

Private Member Functions

- void ConstructVielbein ()

  Construct the vielbein at the position of the camera.

Private Attributes

- TwoIndex m_Metric_dd {}

  The metric at the position of the viewscreen (we only need indices down)
- TwoIndex m_Vielbein {}

  The vielbein used to transform from the curved spacetime at the viewscreen to a locally flat frame.
- const Point m_Pos

  The position of the camera.
- const OneIndex m_Direction

  The looking direction of the camera (always pointing towards the origin)
- const ScreenPoint m_ScreenSize

  The screensize (in physical units of length)
- const ScreenPoint m_ScreenCenter

  The screen center.
- const bool m_rLogScale

  Whether the metric uses a logarithmic r coordinate or not.
- const Metric *const m_theMetric

  const pointer to const Metric
- const GeodesicType m_GeodType {GeodesicType::Null}

  The geodesic type to be integrated.
- const std::unique_ptr< Mesh > m_theMesh

  The const pointer to the Mesh we are using to determine pixels to be integrated.

6.62.1 Detailed Description

This class is in charge of converting a pixel on the screen into physical initial conditions for a geodesic.

The ViewScreen class is in charge of converting a pixel on the screen (which the Mesh wants to integrate) into physical initial conditions for the position and velocity of a geodesic. It owns a Mesh instance, which will tell it which pixels to integrate etc.

6.62.2 Constructor & Destructor Documentation

6.62.2.1 ViewScreen() [1/2]

```
ViewScreen::ViewScreen ()  [delete]
```

6.62.2.2 ViewScreen() [2/2]

```
ViewScreen::ViewScreen (
            Point pos,
            OneIndex dir,
            ScreenPoint screensize,
            ScreenPoint screencenter,
            std::unique_ptr< Mesh > theMesh,
            const Metric *const theMetric,
            GeodesicType thegeodtype = GeodesicType::Null)  [inline]
```

6.62.3 Member Function Documentation

6.62.3.1 ConstructVielbein()

```
void ViewScreen::ConstructVielbein ()  [private]
```

Construct the vielbein at the position of the camera.

6.62.3.2 EndCurrentLoop()

```
void ViewScreen::EndCurrentLoop ()
```

End the current loop of geodesics.

6.62.3.3 GeodesicFinished()

```
void ViewScreen::GeodesicFinished (
            largecounter index,
            std::vector< real > finalValues)
```

Pass on information that a geodesic has finished.

Parameters

index	index of the geodesic
finalValues	final values of the geodesic

6.62.3.4 getCurNrGeodesics()

```
largecounter ViewScreen::getCurNrGeodesics () const
```

Get the current number of geodesics.

Returns

largecounter

6.62.3.5 getFullDescriptionStr()

```
std::string ViewScreen::getFullDescriptionStr () const
```

Get the full description string of the ViewScreen.

Returns

std::string

6.62.3.6 IsFinished()

```
bool ViewScreen::IsFinished () const
```

Check if the ViewScreen is finished.

Returns

true if the ViewScreen is finished, false otherwise

6.62.3.7 SetNewInitialConditions()

```
void ViewScreen::SetNewInitialConditions (
            largecounter index,
            Point & pos,
            OneIndex & vel,
            ScreenIndex & scrIndex) const
```

Set new initial conditions for geodesic nr index.

Parameters

index	index of the geodesic
pos	pointer to the starting position of the geodesic - which is the camera position
vel	pointer to the starting velocity of the geodesic
scrIndex	pointer to the screen index of the geodesic

6.62.4 Member Data Documentation

6.62.4.1 m_Direction

```
const OneIndex ViewScreen::m_Direction  [private]
```

The looking direction of the camera (always pointing towards the origin)

6.62.4.2 m_GeodType

```
const GeodesicType ViewScreen::m_GeodType {GeodesicType::Null}  [private]
```

The geodesic type to be integrated.

6.62.4.3 m_Metric_dd

```
TwoIndex ViewScreen::m_Metric_dd {}  [private]
```

The metric at the position of the viewscreen (we only need indices down)

6.62.4.4 m_Pos

```
const Point ViewScreen::m_Pos  [private]
```

The position of the camera.

6.62.4.5 m_rLogScale

```
const bool ViewScreen::m_rLogScale  [private]
```

Whether the metric uses a logarithmic r coordinate or not.

6.62.4.6 m_ScreenCenter

```
const ScreenPoint ViewScreen::m_ScreenCenter  [private]
```

The screen center.

6.62.4.7 m_ScreenSize

```
const ScreenPoint ViewScreen::m_ScreenSize  [private]
```

The screensize (in physical units of length)

6.62.4.8 m_theMesh

```
const std::unique_ptr<Mesh> ViewScreen::m_theMesh  [private]
```

The const pointer to the Mesh we are using to determine pixels to be integrated.

6.62.4.9 m_theMetric

```
const Metric* const ViewScreen::m_theMetric  [private]
```

const pointer to const Metric

6.62.4.10 m_Vielbein

```
TwoIndex ViewScreen::m_Vielbein {}  [private]
```

The vielbein used to transform from the curved spacetime at the viewscreen to a locally flat frame.

The documentation for this class was generated from the following files:

- FOORT/src/ViewScreen.h
- FOORT/src/ViewScreen.cpp

Chapter 7

File Documentation

7.1 FOORT/src/Config.cpp File Reference

```
#include "Config.h"
#include "Utilities.h"
#include <algorithm>
#include <cctype>
#include <utility>
```

7.1.1 Detailed Description

Author

Daniel R. Mayerson

Version

1.0

Date

2024-12-12

Copyright

Copyright (c) 2024

7.2 FOORT/src/Config.h File Reference

Functions that read the configuration file and initialize all objects.

```
#include "Geometry.h"
#include "Metric.h"
#include "Diagnostics.h"
#include "Terminations.h"
#include "ViewScreen.h"
#include "Geodesic.h"
#include "Integrators.h"
#include <memory>
#include <string>
#include <exception>
#include "ConfigReader.h"
```

Namespaces

- namespace Config

  Namespace for all configuration functions that initialize objects based on configuration file.

Typedefs

- using Config::ConfigCollection = ConfigReader::ConfigCollection
- using Config::SettingError = std::invalid_argument

Functions

- void Config::InitializeScreenOutput (const ConfigCollection &theCfg)

  Initialize screen output options.
- std::unique_ptr< Metric > Config::GetMetric (const ConfigCollection &theCfg)

  Use configuration to create the correct Metric with specified parameters.
- std::unique_ptr< Source > Config::GetSource (const ConfigCollection &theCfg, const Metric *const the↩ Metric)

  Use configuration to create the correct Source with specified parameters.
- void Config::InitializeDiagnostics (const ConfigCollection &theCfg, DiagBitflag &alldiags, DiagBitflag &valdiag, const Metric *const theMetric)

  Use configuration to set the Diagnostics bitflag appropriately; initialize all DiagnosticOptions for all Diagnostics that are turned on; and set bitflag for diagnostic to be used for coarseness evaluating in Mesh.
- void Config::InitializeTerminations (const ConfigCollection &theCfg, TermBitflag &allterms, const Metric *const theMetric)

  Use configuration to set the Termination bitflag appropriately; and initialize all TerminationOptions for all Terminations that are turned on.
- std::unique_ptr< ViewScreen > Config::GetViewScreen (const ConfigCollection &theCfg, DiagBitflag valdiag, const Metric *const theMetric)

  Use configuration to create the ViewScreen object with specified parameters.
- std::unique_ptr< Mesh > Config::GetMesh (const ConfigCollection &theCfg, DiagBitflag valdiag)

  Use configuration to create the Mesh object with specified parameters.
- GeodesicIntegratorFunc Config::GetGeodesicIntegrator (const ConfigCollection &theCfg)

  Use configuration to set the GeodesicIntegratorFunc pointer to the correct integrator function.
- std::unique_ptr< GeodesicOutputHandler > Config::GetOutputHandler (const ConfigCollection &theCfg, DiagBitflag alldiags, DiagBitflag valdiag, std::string FirstLineInfo)

  Use configuration to create the GeodesicOutputHandler object with specified parameters.

Variables

- constexpr auto Config::Output_Important_Default = OutputLevel::Level_0_WARNING
- constexpr auto Config::Output_Other_Default = OutputLevel::Level_1_PROC

7.2.1 Detailed Description

Functions that read the configuration file and initialize all objects.

Author

Daniel R. Mayerson

Version

1.0

Date

2024-12-12

Copyright

Copyright (c) 2024

7.3 Config.h

Go to the documentation of this file.

```
00001 #ifndef _FOORT_CONFIG_H
00002 #define _FOORT_CONFIG_H
00003
00012
00013 #include "Geometry.h"
00014 #include "Metric.h"
00015 #include "Diagnostics.h"
00016 #include "Terminations.h"
00017 #include "ViewScreen.h"
00018 #include "Geodesic.h"
00019 #include "Integrators.h"
00020
00021 #include <memory> // std::unique_ptr
00022 #include <string> // std::string
00023
00024 #include <exception> // needed to define our own configuration error
00025
00026 #include "ConfigReader.h"
00027
00031 namespace Config
00032 {
00033     // Output level for important missing information that will default
00034     // (e.g. no metric selected, no diagnostics selected, ...)
00035     constexpr auto Output_Important_Default = OutputLevel::Level_0_WARNING;
00036     // Output level for less important information that will default
00037     // (e.g. Kerr metric a parameter not specified)
00038     constexpr auto Output_Other_Default = OutputLevel::Level_1_PROC;
00039
00040     // Total configuration object, as loaded from configuration file
00041     using ConfigCollection = ConfigReader::ConfigCollection;
00042
00043     // An exception to throw whenever an important setting is not found
00044     // Always should be caught and then reverted to default settings
00045     using SettingError = std::invalid_argument;
00046
00049
00050     // Use configuration to initialize the screen output level
00051     void InitializeScreenOutput(const ConfigCollection &theCfg);
00052
00053     // Use configuration to create the correct Metric with specified parameters
00054     std::unique_ptr<Metric> GetMetric(const ConfigCollection &theCfg);
00055
00056     // Use configuration to create the correct Source with specified parameters
00057     std::unique_ptr<Source> GetSource(const ConfigCollection &theCfg, const Metric *const theMetric);
00058
```

```
00059     // Use configuration to set the Diagnostics bitflag appropriately;
00060     // initialize all DiagnosticOptions for all Diagnostics that are turned on;
00061     // and set bitflag for diagnostic to be used for coarseness evaluating in Mesh
00062     void InitializeDiagnostics(const ConfigCollection &theCfg, DiagBitflag &alldiags, DiagBitflag
      &valdiag, const Metric *const theMetric);
00063
00064     // Use configuration to set the Terminations bitflag appropriately;
00065     // initialize all TerminationOptions for all Terminations that are turned on;
00066     void InitializeTerminations(const ConfigCollection &theCfg, TermBitflag &allterms, const Metric
      *const theMetric);
00067
00068     // Use configuration to create ViewScreen appropriately
00069     std::unique_ptr<ViewScreen> GetViewScreen(const ConfigCollection &theCfg, DiagBitflag valdiag,
      const Metric *const theMetric);
00070     // GetViewScreen calls GetMesh to create the correct Mesh
00071     std::unique_ptr<Mesh> GetMesh(const ConfigCollection &theCfg, DiagBitflag valdiag);
00072
00073     // Use configuration to return a pointer to the correct integration function to use
00074     GeodesicIntegratorFunc GetGeodesicIntegrator(const ConfigCollection &theCfg);
00075
00076     // Use configuration to initialize the output handler
00077     std::unique_ptr<GeodesicOutputHandler> GetOutputHandler(const ConfigCollection &theCfg,
00078                                                             DiagBitflag alldiags, DiagBitflag valdiag,
      std::string FirstLineInfo);
00079
00080 } // end namespace Config
00081
00082 #endif
```

7.4 FOORT/src/ConfigReader.cpp File Reference

```
#include "ConfigReader.h"
#include <ios>
#include <type_traits>
```

7.4.1 Detailed Description

Author

Daniel R. Mayerson

Version

1.0

Date

2024-12-12

Copyright

Copyright (c) 2024

7.5 FOORT/src/ConfigReader.h File Reference

Classes used to read in a configuration file and store the settings.

```
#include <string_view>
#include <string>
#include <iostream>
#include <fstream>
#include <vector>
#include <variant>
#include <stdexcept>
#include <limits>
#include <memory>
#include <type_traits>
```

Classes

- class ConfigReader::ConfigReaderException
- class ConfigReader::ConfigCollection
- struct ConfigReader::ConfigCollection::ConfigSetting

Namespaces

- namespace ConfigReader

  Namespace to put all ConfigReader objects.

Variables

- const long long ConfigReader::MaxStreamSize {std::numeric_limits<std::streamsize>::max()}

7.5.1 Detailed Description

Classes used to read in a configuration file and store the settings.

Author

Daniel R. Mayerson

Version

1.0

Date

2024-12-12

Copyright

Copyright (c) 2024

7.6 ConfigReader.h

Go to the documentation of this file.

```
00001 #ifndef _CONFIGREADER_H
00002 #define _CONFIGREADER_H
00003
00012
00013 #include <string_view> // std::string_view
00014 #include <string>      // std::string
00015 #include <iostream>    // needed for file and console output
00016 #include <fstream>     // needed for file input
00017 #include <vector>      // needed to create vectors of Settings
00018 #include <variant>     // needed for std::variant
00019 #include <stdexcept>   // needed for exceptions
00020 #include <limits>      // for std::numeric_limits
00021 #include <memory>      // for std::unique_ptr
00022 #include <type_traits> // for std::is_floating_point_v
00023
00027 namespace ConfigReader
00028 {
00029     // Shorthand for the maximum number of characters we can ignore in a stream
00030     const long long MaxStreamSize{std::numeric_limits<std::streamsize>::max()};
00031
00032     // Exception that is thrown if anything bad happens when reading in configuration file
00033     class ConfigReaderException : public std::runtime_error
00034     {
00035     public:
00036         ConfigReaderException(const std::string &error, std::vector<int> settingtrace = {})
00037             : std::runtime_error{error.c_str()}, m_settingtrace{settingtrace}
00038         {
00039         }
00040         // no need to override what() since we can just use std::runtime_error::what()
00041
00042         std::vector<int> trace() const
00043         {
00044             return m_settingtrace;
00045         };
00046
00047     private:
00048         std::vector<int> m_settingtrace;
00049     };
00050
00051     // A collection of configuration settings
00052     class ConfigCollection
00053     {
00054     public:
00056         // Returns true if the collection contains a setting with the given name
00057         bool Exists(std::string_view SettingName) const;
00058         // Returns total number of settings in this collection
00059         int NrSettings() const;
00060
00061         // Returns true if the setting with the given name (resp. index) is a collection of
    subsettings
00062         // (Returns false if this setting name does not exist, or index is out of bounds)
00063         bool IsCollection(std::string_view SettingName) const;
00064         bool IsCollection(int SettingIndex) const;
00065         // If CollectionName (resp. CollectionIndex) is the name (resp. index)
00066         // of a setting that is a subcollection, then this returns
00067         // a const reference to this subcollection; otherwise throws ConfigReaderException
00068         const ConfigCollection &operator[](std::string_view CollectionName) const;
00069         const ConfigCollection &operator[](int CollectionIndex) const;
00070
00072         // Generic functions that look up the value of a given setting
00073         // and put it in the output variable
00074         // Returns true if successful, false if unsuccesful
00075         // (either due to type mismatch of output and setting value, or if setting doesn't exist)
00076         // Templated functions implemented below in .h file!
00077         template <class OutputType>
00078         bool LookupValue(std::string_view SettingName, OutputType &theOutput) const;
00079         template <class OutputType>
00080         bool LookupValue(int SettingIndex, OutputType &theOutput) const;
00081
00082         // Specific functions for looking up signed integral types
00083         // These will first see if the setting value is of the given type, then
00084         // they will try smaller types and then recast to the larger type
00085         // (so e.g. for long long, first long long is tried, then long, then int)
00086         bool LookupValueInteger(std::string_view SettingName, int &theOutput) const;
00087         bool LookupValueInteger(std::string_view SettingName, long &theOutput) const;
00088         bool LookupValueInteger(std::string_view SettingName, long long &theOutput) const;
00089         // Specific functions for looking up unsigned integral types
00090         // These will first test if the setting value is of the signed case, if it is and it is >=0
    then it always fits in
00091         // the unsigned version; then it will test larger (signed) types and see if the result still
    fits in the unsigned version
00092         bool LookupValueInteger(std::string_view SettingName, unsigned int &theOutput) const;
```

```
00093         bool LookupValueInteger(std::string_view SettingName, unsigned long &theOutput) const;
00094         bool LookupValueInteger(std::string_view SettingName, unsigned long long &theOutput) const;
00095         // Templated functions implemented below in .h file!
00096         template <class OutputType>
00097         bool LookupValueInteger(int SettingIndex, OutputType &theOutput) const;
00098 
00100         void DisplayCollection(std::ostream &OutputStream, int Indent = 0) const;
00101 
00103         // this overwrites the current collection with the contents of the config file
00104         // Will throw ConfigReaderException if parse/syntax error in configuration file
00105         // Returns true if successful
00106         bool ReadFile(const std::string &FileName);
00107 
00108     private:
00109         // A configuration setting value can be an integral type, bool, double, string, or a
      (sub)collection of settings
00110         // Note: when reading in a configuration file, an integral value will always be stored in the
      smallest possible type!
00111         using ConfigSettingValue = std::variant<bool,
00112                                                 int, long, long long,
00113                                                 double,
00114                                                 std::string,
00115                                                 std::unique_ptr<ConfigCollection»;
00116         // A setting is a name and setting value
00117         struct ConfigSetting
00118         {
00119             std::string SettingName;
00120             ConfigSettingValue SettingValue;
00121         };
00122         // The vector of all settings in this collection
00123         std::vector<ConfigSetting> m_Settings{};
00124 
00125         // Helper function that returns the index of the setting with the given name
00126         // returns -1 if setting not found
00127         int GetSettingIndex(std::string_view SettingName) const;
00128 
00129         // Helper functions for DisplayCollection
00130         // DisplaySetting displays one setting; if it's a subcollection then it calls
00131         // DisplayCollection on the subcollection
00132         void DisplaySetting(std::ostream &OutputStream, int SettingIndex, int Indent) const;
00133         // Output given number of tabs
00134         void DisplayTabs(std::ostream &OutputStream, int NrTabs) const;
00135 
00136         // Helper functions for ReadFile()
00137         // These all take an ifstream at a given location and will read in one specific object
00138         //
00139         // Read in entire collection (including subcollections)
00140         // Pre: stream is positioned right after { (or at beginning of file)
00141         // Post: stream is positioned right after } (or at EOF)
00142         void ReadCollection(std::ifstream &InputFile);
00143         // Read in a setting name
00144         // Pre: next non-whitespace character in stream is beginning of name
00145         // Post: stream has just read in all characters of name (but no more);
00146         // returns "" if there is no more setting to read in the current collection, in this case
00147         // '}' has been read in (for subcollections)
00148         void ReadSettingName(std::ifstream &InputFile, std::string &theName);
00149         // Read in one specific character only (not '/'!)
00150         // Pre: next non-whitespace character in stream is this character
00151         // Post: stream is just after character
00152         void ReadSettingSpecificChar(std::ifstream &InputFile, char theChar) const;
00153         // Read in setting value (could be subcollection)
00154         // Pre: next non-whitespace character in stream is beginning of value
00155         // (can be '{' for subcollection, digit for number, '"' for string, or 't'/'f' for boolean)
00156         // Post: Value has entirely been read in (i.e. next character should be ';')
00157         void ReadSettingValue(std::ifstream &InputFile, ConfigSettingValue &theValue);
00158     };
00159 
00160 } // end namespace declarations
00161 
00164 
00174 template <class OutputType>
00175 bool ConfigReader::ConfigCollection::LookupValue(int SettingIndex, OutputType &theOutput) const
00176 {
00177     // Get the setting value, if it is indeed of this type
00178     const ConfigSettingValue &theValue{m_Settings[SettingIndex].SettingValue};
00179     const OutputType *OutputPointer{std::get_if<OutputType>(&theValue)};
00180     if (OutputPointer)
00181     {
00182         // Success, the setting is indeed of this type; set the output accordingly
00183         theOutput = *OutputPointer;
00184         return true;
00185     }
00186 
00187     // A real-valued setting may have been written without a decimal point (e.g. "a = 0;"),
00188     // in which case it was stored as an integral type: accept it and convert
00189     if constexpr (std::is_floating_point_v<OutputType>)
00190     {
```

```
00191         if (const int *p{std::get_if<int>(&theValue)})             { theOutput =
      static_cast<OutputType>(*p); return true; }
00192         if (const long *p{std::get_if<long>(&theValue)})           { theOutput =
      static_cast<OutputType>(*p); return true; }
00193         if (const long long *p{std::get_if<long long>(&theValue)}) { theOutput =
      static_cast<OutputType>(*p); return true; }
00194     }
00195     return false;
00196 }
00197
00207 template <class OutputType>
00208 bool ConfigReader::ConfigCollection::LookupValue(std::string_view SettingName, OutputType &theOutput)
      const
00209 {
00210     // Look up setting index and then defer to index-based implementation, if setting exists
00211     int SettingIndex{GetSettingIndex(SettingName)};
00212     if (SettingIndex >= 0)
00213     {
00214         return LookupValue(SettingIndex, theOutput);
00215     }
00216     return false;
00217 }
00218
00229 template <class OutputType>
00230 bool ConfigReader::ConfigCollection::LookupValueInteger(int SettingIndex, OutputType &theOutput) const
00231 {
00232     return LookupValueInteger(m_Settings[SettingIndex].SettingName, theOutput);
00233 }
00234
00235 #endif
```

7.7 FOORT/src/Diagnostics.cpp File Reference

```
#include "Diagnostics.h"
#include "DiagnosticsEmission.h"
#include "Geodesic.h"
#include "InputOutput.h"
#include "Integrators.h"
#include <algorithm>
#include <cmath>
```

Functions

- DiagnosticUniqueVector CreateDiagnosticVector (DiagBitflag diagflags, DiagBitflag valdiag, Geodesic *const theGeodesic)

  Diagnostic helper function.

7.7.1 Detailed Description

Author

Daniel R. Mayerson

Version

1.0

Date

2024-12-16

Copyright

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7.7.2 Function Documentation

7.7.2.1 CreateDiagnosticVector()

```
DiagnosticUniqueVector CreateDiagnosticVector (
            DiagBitflag diagflags,
            DiagBitflag valdiag,
            Geodesic *const theGeodesic)
```

Diagnostic helper function.

Create a Diagnostic Vector object

Parameters

diagflags	Diagnostic bitflags
valdiag	Value Diagnostic bitflag
theGeodesic	Pointer to the Geodesic object

Returns

DiagnosticUniqueVector

7.8 FOORT/src/Diagnostics.h File Reference

Declarations of abstract base Diagnostic class and all its descendants. All Definitions in Diagnostics.cpp.

```
#include "Geometry.h"
#include "Metric.h"
#include "InputOutput.h"
#include "DiagnosticsEmission.h"
#include <cstdint>
#include <string>
#include <memory>
#include <vector>
```

Classes

- struct UpdateFrequency

  Struct to hold the information for a Diagnostic to update itself.
- class Diagnostic

  Abstract base class for all diagnostics.
- class FourColorScreenDiagnostic

  The four color screen diagnostic.
- class GeodesicPositionDiagnostic

  Geodesic position tracker.
- class EquatorialPassesDiagnostic

  Diagnostic that counts the number of passes through the equatorial plane.
- class ClosestRadiusDiagnostic

Diagnostic that keeps track of the closest (true, not log) radius that the geodesic passes through.

- class EquatorialEmissionDiagnostic

  Diagnostic that calculates brightness intensity for the geodesic, based on a specified equatorial disc emission model.
- struct DiagnosticOptions

  Base class for DiagnosticOptions. Other Diagnostics can inherit from here if they require more options.
- struct GeodesicPositionOptions

  Options for GeodesicPositionDiagnostic.
- struct EquatorialPassesOptions

  Options for EquatorialPassesDiagnostic.
- struct ClosestRadiusOptions

  Options for ClosestRadiusDiagnostic.
- struct EquatorialEmissionOptions

  Options for EquatorialEmissionDiagnostic.

Typedefs

- using DiagBitflag = std::uint16_t

  Diagnostic bitflags.
- using DiagnosticUniqueVector = std::vector<std::unique_ptr<Diagnostic>>

  Owner vector of derived Diagnostics classes.

Functions

- DiagnosticUniqueVector CreateDiagnosticVector (DiagBitflag diagflags, DiagBitflag valdiag, Geodesic *const theGeodesic)

  Diagnostic helper function.

Variables

- constexpr DiagBitflag Diag_None {0b0000'0000'0000'0000}
- constexpr DiagBitflag Diag_GeodesicPosition {0b0000'0000'0000'0001}
- constexpr DiagBitflag Diag_FourColorScreen {0b0000'0000'0000'0010}
- constexpr DiagBitflag Diag_EquatorialPasses {0b0000'0000'0000'0100}
- constexpr DiagBitflag Diag_ClosestRadius {0b0000'0000'0000'1000}
- constexpr DiagBitflag Diag_EquatorialEmission {0b0000'0000'0001'0000}

7.8.1 Detailed Description

Declarations of abstract base Diagnostic class and all its descendants. All Definitions in Diagnostics.cpp.

Author

Daniel R. Mayerson

Version

1.0

Date

2024-12-16

Copyright

Copyright (c) 2024

7.8.2 Typedef Documentation

7.8.2.1 DiagBitflag

```
using DiagBitflag = std::uint16_t
```

Diagnostic bitflags.

Used for constructing vector of Diagnostics ...

7.8.2.2 DiagnosticUniqueVector

```
using DiagnosticUniqueVector = std::vector<std::unique_ptr<Diagnostic>>
```

Owner vector of derived Diagnostics classes.

7.8.3 Function Documentation

7.8.3.1 CreateDiagnosticVector()

```
DiagnosticUniqueVector CreateDiagnosticVector (
            DiagBitflag diagflags,
            DiagBitflag valdiag,
            Geodesic *const theGeodesic)
```

Diagnostic helper function.

Helper to create a new vector of Diagnostic options, based on the bitflag The first diagnostic is the value diagnostic

Create a Diagnostic Vector object

Parameters

diagflags	Diagnostic bitflags
valdiag	Value Diagnostic bitflag
theGeodesic	Pointer to the Geodesic object

Returns

DiagnosticUniqueVector

7.8.4 Variable Documentation

7.8.4.1 Diag_ClosestRadius

```
DiagBitflag Diag_ClosestRadius {0b0000'0000'0000'1000}  [constexpr]
```

7.8.4.2 Diag_EquatorialEmission

```
DiagBitflag Diag_EquatorialEmission {0b0000'0000'0001'0000}  [constexpr]
```

7.8.4.3 Diag_EquatorialPasses

```
DiagBitflag Diag_EquatorialPasses {0b0000'0000'0000'0100}  [constexpr]
```

7.8.4.4 Diag_FourColorScreen

```
DiagBitflag Diag_FourColorScreen {0b0000'0000'0000'0010}  [constexpr]
```

7.8.4.5 Diag_GeodesicPosition

```
DiagBitflag Diag_GeodesicPosition {0b0000'0000'0000'0001}  [constexpr]
```

7.8.4.6 Diag_None

```
DiagBitflag Diag_None {0b0000'0000'0000'0000}  [constexpr]
```

7.9 Diagnostics.h

Go to the documentation of this file.

```
00001 #ifndef _FOORT_DIAGNOSTICS_H
00002 #define _FOORT_DIAGNOSTICS_H
00003
00012
00013 #include "Geometry.h"  // for tensors
00014 #include "Metric.h"      // for metric
00015 #include "InputOutput.h" // for ScreenOutput
00016
00017 #include "DiagnosticsEmission.h" // Emission models
00018
00019 #include <cstdint> // for std::uint16_t
00020 #include <string>  // for strings
00021 #include <memory>  // for std::unique_ptr
00022 #include <vector>  // for std::vector
00023
00024 // Forward declaration of Geodesic class needed here, since Diagnostics are passed a pointer to their
      owner Geodesic
00025 // (note "Geodesic.h" is NOT included to avoid header loop, and we do not need Geodesic member
      functions here!)
00026 class Geodesic;
00027
00030
00031 // Diagnostic bitflags
00032 // Used for constructing vector of Diagnostics
00033 // Note that this means every diagnostic is either "on" or "off";
00034 // it is not possible to have a Diagnostic "on" more than once
00035 // Note: why not std::bitset<size>? Because in this way we can use expressions with DiagBitflag in
      conditional expressions,
00036 // e.g. if ( mydiagbitflag & Diag_FourColorScreen)
00041 using DiagBitflag = std::uint16_t;
00042
00043 // Define a bitflag per existing diagnostic
00044 constexpr DiagBitflag Diag_None{0b0000'0000'0000'0000};
00045 constexpr DiagBitflag Diag_GeodesicPosition{0b0000'0000'0000'0001};
00046 constexpr DiagBitflag Diag_FourColorScreen{0b0000'0000'0000'0010};
00047 constexpr DiagBitflag Diag_EquatorialPasses{0b0000'0000'0000'0100};
00048 constexpr DiagBitflag Diag_ClosestRadius{0b0000'0000'0000'1000};
```

```
00049 constexpr DiagBitflag Diag_EquatorialEmission{0b0000'0000'0001'0000};
00050
00052 // Add a DiagBitflag for your new diagnostic. Make sure you use a bitflag that has not been used
      before!
00053 // Sample code:
00054 /*
00055 constexpr DiagBitflag Diag_MyDiag               { 0b0000'0000'0000'1000 };
00056 */
00058
00063 struct UpdateFrequency
00064 {
00065     largecounter UpdateNSteps{0};
00066     bool UpdateStart{false};
00067     bool UpdateFinish{false};
00068 };
00069
00074 class Diagnostic
00075 {
00076 public:
00077     // Constructor must initialize the pointer to its owner Geodesic
00078     Diagnostic() = delete;
00079     Diagnostic(Geodesic *const theGeodesic) : m_OwnerGeodesic{theGeodesic}
00080     {
00081     }
00082
00083     // Resets Diagnostic object. This is called when the owner Geodesic is reset in order to start
      integrating
00084     // a new geodesic.
00085     // The base class implementation only resets m_StepsSinceUpdated
00086     // Descendants can override this if they need to reset additional internal variables
00087     virtual void Reset();
00088
00089     // virtual destructor to ensure correct destruction of descendants
00090     virtual ~Diagnostic() = default;
00091
00092     // This is the heart of the Diagnostic. It calls the helper function DecideUpdate(), and if this
00093     // returns true, will update its internal status based on the current (new) state of its owner
      geodesic.
00094     virtual void UpdateData() = 0;
00095
00096     // These functions are for use at the end of integration of a geodesic.
00097     // getFullData() returns all the data stored in the Diagnostic as a string (for output to file)
00098     virtual std::string getFullDataStr() const = 0;
00099     // getFinalDataVal() associates a value to the geodesic corresponding to the final value of this
      diagnostic
00100     // (used for determining coarseness of nearby geodesics)
00101     virtual std::vector<real> getFinalDataVal() const = 0;
00102
00103     // Function used to determine distance between two values obtained from getFinalDataVal()
00104     // (for determining coarseness of nearby geodesics)
00105     // This should return a number >=0
00106     virtual real FinalDataValDistance(const std::vector<real> &val1, const std::vector<real> &val2)
      const = 0;
00107
00108     // Getters for descriptions
00109     // This returns the name (only) of the Diagnostic, as a string without spaces that will be
      appended
00110     // to an output file (e.g. prefix_DiagName.ext). Must be implemented!
00111     virtual std::string getNameStr() const = 0;
00112     // This returns the full description of the Diagnostic. Default implementation
00113     // returns GetDiagNameStr()
00114     virtual std::string getFullDescriptionStr() const;
00115
00116 protected:
00117     // The geodesic that owns the Diagnostic (a const pointer to the Geodesic)
00118     Geodesic *const m_OwnerGeodesic;
00119
00120     // Helper function to decide if the Diagnostic should indeed update its status, based on
00121     // its UpdateFrequency struct information (which will come from the Diagnostics's
      DiagnosticOptions)
00122     bool DecideUpdate(const UpdateFrequency &myUpdateFrequency);
00123
00124     // The diagnostic is itself in charge of keeping track of how many steps it has been since it has
      been updated
00125     // The Diagnostic's DiagnosticOptions struct tells it how many steps it needs to wait between
      updates
00126     largecounter m_StepsSinceUpdated{};
00127 };
00128
00130 using DiagnosticUniqueVector = std::vector<std::unique_ptr<Diagnostic>>;
00131
00134 DiagnosticUniqueVector CreateDiagnosticVector(DiagBitflag diagflags, DiagBitflag valdiag, Geodesic
      *const theGeodesic);
00135
00138
00144 class FourColorScreenDiagnostic final : public Diagnostic
00145 {
```

```
00146 public:
00147     // Basic constructor only passes on Geodesic pointer to base class constructor
00148     FourColorScreenDiagnostic(Geodesic *const theGeodesic) : Diagnostic(theGeodesic) {}
00149
00150     // Reset needs to be overridden in order to also reset m_quadrant
00151     void Reset() final;
00152
00153     // Sets the quadrant to the appropriate value, IF the boundary sphere has been reached
00154     void UpdateData() override;
00155
00156     // Both of these output functions simply returns the quadrant number associated with the
      geodesic's end position
00157     std::string getFullDataStr() const final;
00158     std::vector<real> getFinalDataVal() const final;
00159
00160     // Discrete metric for distance: returns 0 if the quadrants are the same, 1 if they are not
00161     real FinalDataValDistance(const std::vector<real> &val1, const std::vector<real> &val2) const
      final;
00162
00163     // Description string getters
00164     std::string getNameStr() const final;
00165     std::string getFullDescriptionStr() const final;
00166
00167     // FourColorScreen does not need any (static) options!
00168
00169 private:
00171     int m_quadrant{0};
00172 };
00173
00175 struct GeodesicPositionOptions;
00179 class GeodesicPositionDiagnostic final : public Diagnostic
00180 {
00181 public:
00182     // Basic constructor only passes on Geodesic pointer to base class constructor
00183     GeodesicPositionDiagnostic(Geodesic *const theGeodesic) : Diagnostic(theGeodesic) {}
00184
00185     // Override Reset to also clear m_AllSavedPoints
00186     void Reset() final;
00187
00188     // Stores the current position of the geodesic
00189     void UpdateData() final;
00190
00191     // This returns as many stored positions as is specified in the options struct
00192     std::string getFullDataStr() const final;
00193     // This returns the final (theta,phi) value of the geodesic
00194     std::vector<real> getFinalDataVal() const final;
00195
00196     // This determines the angular distance between two geodesic
00197     // (based on their final angles on the boundary sphere (theta, phi))
00198     real FinalDataValDistance(const std::vector<real> &val1, const std::vector<real> &val2) const
      final;
00199
00200     // Description string getters
00201     std::string getNameStr() const final;
00202     std::string getFullDescriptionStr() const final;
00203
00204     // The options: specifies the update frequency but also how many points we want to output in the
      end
00205     static std::unique_ptr<GeodesicPositionOptions> DiagOptions;
00206
00207 private:
00208     // Keeps track of the points that are saved
00209     std::vector<Point> m_AllSavedPoints{};
00210 };
00211
00213 struct EquatorialPassesOptions;
00217 class EquatorialPassesDiagnostic : public Diagnostic
00218 {
00219 public:
00220     // Basic constructor only passes on Geodesic pointer to base class constructor
00221     EquatorialPassesDiagnostic(Geodesic *const theGeodesic) : Diagnostic(theGeodesic) {}
00222
00223     // Override Reset to also reset m_EquatPasses and m_PrevTheta
00224     void Reset() override;
00225
00226     // Checks to see if we have a new cross over the equatorial plane
00227     void UpdateData() override;
00228
00229     // Returns the number of passes over the equatorial plane
00230     std::string getFullDataStr() const override;
00231     std::vector<real> getFinalDataVal() const override;
00232
00233     // Simple absolute value of difference of passes
00234     real FinalDataValDistance(const std::vector<real> &val1, const std::vector<real> &val2) const
      override;
00235
00236     // Description string getters
```

```
00237     std::string getNameStr() const override;
00238     std::string getFullDescriptionStr() const override;
00239
00241     static std::unique_ptr<EquatorialPassesOptions> DiagOptions;
00242
00243 protected:
00245     int m_EquatPasses{0};
00246
00247 private:
00249     real m_PrevTheta{-1};
00250 };
00251
00252 // Forward declaration needed before Diagnostic
00253 struct ClosestRadiusOptions;
00257 class ClosestRadiusDiagnostic final : public Diagnostic
00258 {
00259 public:
00260     // Basic constructor only passes on Geodesic pointer to base class constructor
00261     ClosestRadiusDiagnostic(Geodesic *const theGeodesic) : Diagnostic(theGeodesic) {}
00262
00263     // re-initialize closest radius
00264     void Reset() final;
00265
00266     // Check to see if we have travelled closer
00267     void UpdateData() final;
00268
00269     // Return closest radius travelled
00270     std::string getFullDataStr() const final;
00271     std::vector<real> getFinalDataVal() const final;
00272
00273     // Returns the absolute value of the difference between closest radii
00274     real FinalDataValDistance(const std::vector<real> &val1, const std::vector<real> &val2) const
      final;
00275
00276     // Description string getters
00277     std::string getNameStr() const final;
00278     std::string getFullDescriptionStr() const final;
00279
00280     // Option struct to keep track of RLogScale of Metric
00281     static std::unique_ptr<ClosestRadiusOptions> DiagOptions;
00282
00283 private:
00284     // Keeps track of closest radius travelled
00285     real m_ClosestRadius{-1};
00286 };
00287
00288 // Forward declaration needed before Diagnostic
00289 struct EquatorialEmissionOptions;
00296 class EquatorialEmissionDiagnostic final : public EquatorialPassesDiagnostic
00297 {
00298 public:
00299     // Basic constructor only passes on Geodesic pointer to parent class constructor
00300     EquatorialEmissionDiagnostic(Geodesic *const theGeodesic) :
      EquatorialPassesDiagnostic(theGeodesic) {}
00301
00302     // Reset intensity back to 0.0, and call parent class Reset()
00303     void Reset() final;
00304
00305     // Use parent class UpdateData() to decide whether the update the intensity
00306     // using the specified emission model
00307     void UpdateData() final;
00308
00309     // Returns total intensity and total number of equatorial passes (i.e. normal output from
      EquatorialPassesDiagnostic)
00310     std::string getFullDataStr() const final;
00311     std::vector<real> getFinalDataVal() const final;
00312
00313     // Distance is difference in intensities multiplied by a factor magnifying difference in
      equatorial passes
00314     real FinalDataValDistance(const std::vector<real> &val1, const std::vector<real> &val2) const
      final;
00315
00316     // Description string getters
00317     std::string getNameStr() const final;
00318     std::string getFullDescriptionStr() const final;
00319
00320     // Option struct to keep track emission model specifics
00321     static std::unique_ptr<EquatorialEmissionOptions> DiagOptions;
00322
00323 private:
00324     // Brightness intensity of the geodesic
00325     real m_Intensity{0.0};
00326 };
00327
00329 // Declare your Diagnostic class here, inheriting from Diagnostic.
00330 // Sample code:
00331 /*
```

```
00332 // Forward declaration needed before Diagnostic (if using base class DiagnosticOptions, this is
      strictly speaking not necessary)
00333 struct DiagnosticOptions;
00334 // class definition
00335 class MyDiagnostic final : public Diagnostic // good practice to make the class final unless
      descendant classes are possible
00336 {
00337 public:
00338     // constructor must at least take and pass along the const pointer to the owner Geodesic
00339     MyDiagnostic(Geodesic* const theGeodesic) : Diagnostic(theGeodesic) {}
00340
00341     // If you have MyDiagnostic-specific member variables, then they probably need to be reset at the
      beginning
00342     // of integration for each new geodesic; in that case, you need to override Reset() to do so.
00343     // Note: make sure to call the base class implementation Diagnostic::Reset()
00344     // from within your implementation of MyDiagnostic::Reset(), so that the base class internal
      variable is also reset!
00345     void Reset() final;
00346
00347     // This is the heart of the Diagnostic: here the Diagnostic updates its internal state according
      to
00348     // the owner Geodesic's current state
00349     void UpdateData() final;
00350
00351     // This should return the string that is to be outputted to the file as the final output of this
      Diagnostic
00352     // for its owner Geodesic
00353     std::string getFullDataStr() const final;
00354     // This should return a vector of real numbers that indicates the final "value" that should be
      associated to
00355     // the owner Geodesic --- this is then used in FinalDataValDistance() to find "distances" between
      geodesics.
00356     std::vector<real> getFinalDataVal() const final;
00357
00358     // This should return a (positive) distance of two values returned by getFinalDataVal(), indicated
      the
00359     // "distance" of two geodesics (this is used for Mesh refinement)
00360     real FinalDataValDistance(const std::vector<real>& val1, const std::vector<real>& val2) const
      final;
00361
00362     // Must implement getNameStr() (and recommended also to implement getFullDescriptionStr)
00363     // getNameStr() is a simple, short string (without spaces) that will be appended to the file name
00364     // where this Diagnostic's output is written. getFullDescriptionStr() is a descriptive string that
      should list
00365     // all relevant options set; this is outputted to e.g. the screen at runtime.
00366     std::string getNameStr() const final;
00367     std::string getFullDescriptionStr() const final;
00368
00369     // Diagnostic options: basic DiagnosticOptions only contains UpdateFrequency information,
00370     // if needed, can use a descendant class with more options (see e.g. GeodesicPositionDiagnostic)
00371     static std::unique_ptr<DiagnosticOptions> DiagOptions;
00372
00373 private:
00374     // will probably want some private member variable(s) to keep track of whatever geodesic property
      is desired
00375 };
00376 */
00378
00381
00385 struct DiagnosticOptions
00386 {
00387 public:
00388     // Basic constructor only sets the number of steps between updates
00389     DiagnosticOptions(UpdateFrequency thefrequency) : theUpdateFrequency{thefrequency}
00390     {
00391     }
00392
00393     virtual ~DiagnosticOptions() = default;
00394
00395     const UpdateFrequency theUpdateFrequency;
00396 };
00402 struct GeodesicPositionOptions : public DiagnosticOptions
00403 {
00404 public:
00405     GeodesicPositionOptions(largecounter outputsteps, UpdateFrequency thefrequency) :
      OutputNrSteps{outputsteps},
00406
      DiagnosticOptions(thefrequency)
00407     {
00408     }
00409
00410     const largecounter OutputNrSteps;
00411 };
00412
00418 struct EquatorialPassesOptions : public DiagnosticOptions
00419 {
00420 public:
```

```
00421     EquatorialPassesOptions(real thethreshold, UpdateFrequency thefrequency) :
      Threshold{thethreshold},
00422
      DiagnosticOptions(thefrequency)
00423     {
00424     }
00425
00426     const real Threshold;
00427 };
00428
00434 struct ClosestRadiusOptions : public DiagnosticOptions
00435 {
00436 public:
00437     ClosestRadiusOptions(bool rlog, UpdateFrequency thefrequency) : RLogScale{rlog},
00438                                                                     DiagnosticOptions(thefrequency)
00439     {
00440     }
00441
00442     const bool RLogScale;
00443 };
00444
00445 // Forward declarations of abstract EmissionModel and FluidVelocityModel classes.
00446 // See Diagnostics_Emission.h and .cpp for declarations and definitions of these and their descendant
      classes!
00447 struct EmissionModel;
00448 struct FluidVelocityModel;
00454 struct EquatorialEmissionOptions : public EquatorialPassesOptions
00455 {
00456 public:
00457     EquatorialEmissionOptions(real thefudgefactor, int equatupper,
00458                               std::unique_ptr<EmissionModel> theemission,
      std::unique_ptr<FluidVelocityModel> thefluidmodel,
00459                               bool rlog, int theredshiftpower, real thethreshold, UpdateFrequency
      thefrequency) : RedShiftPower{theredshiftpower}, RLogScale{rlog},
      GeometricFudgeFactor{thefudgefactor},
00460
      EquatPassUpperBound{equatupper},
00461
      TheEmissionModel{std::move(theemission)}, TheFluidVelocityModel{std::move(thefluidmodel)},
00462
      EquatorialPassesOptions(thethreshold, thefrequency) // constructor for parent class
00463     {
00464     }
00465
00466     // Geometric fudge factor for n>0 passes
00467     const real GeometricFudgeFactor;
00468     // Upper bound allowed for contribution to intensity
00469     const int EquatPassUpperBound;
00470
00471     // Logarithmic radius scale used or not
00472     const bool RLogScale;
00473
00474     // Power of redshift in intensity contribution (should be 3 or 4)
00475     const int RedShiftPower;
00476
00477     // The emission model used (see Diagnostics_Emission.h and .cpp for specific models)
00478     const std::unique_ptr<EmissionModel> TheEmissionModel;
00479     // The fluid velocity model used (see Diagnostics_Emission.h and .cpp for specific models)
00480     const std::unique_ptr<FluidVelocityModel> TheFluidVelocityModel;
00481 };
00482
00484 // if necessary, define your new DiagnosticOptions class here
00485 // Sample code:
00486 /*
00487 struct MyDiagnosticOptions : public DiagnosticOptions
00488 {
00489 public:
00490     // Constructor should pass along UpdateFrequency information to base class
00491     MyDiagnosticOptions(UpdateFrequency thefrequency) : DiagnosticOptions(thefrequency) //, (...)
      other initializations
00492     {}
00493
00494     // other member variables here - make them const!
00495     // ...
00496 };
00497 */
00499
00500 #endif
```

7.10 FOORT/src/DiagnosticsEmission.cpp File Reference

```
#include "DiagnosticsEmission.h"
#include "Integrators.h"
#include <cmath>
```

7.10.1 Detailed Description

Author

Daniel R. Mayerson

Version

1.0

Date

2024-12-16

Copyright

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7.11 FOORT/src/DiagnosticsEmission.h File Reference

Declarations of emission and fluid velocity models used for equatorial disc emission.

```
#include "Geometry.h"
#include "Metric.h"
#include "InputOutput.h"
#include <string>
#include <cmath>
#include <stdexcept>
```

Classes

- struct EmissionModel

  Emmision model abstract base class.
- struct GLMJohnsonSUEmission

  The Johnson SU emission model used in GLM 2020, PRD 102(12):124004.
- struct FluidVelocityModel

  Fluid velocity model abstract base class.
- struct GeneralCircularRadialFluid

  General circular radial fluid velocity model.

7.11.1 Detailed Description

Declarations of emission and fluid velocity models used for equatorial disc emission.

Author

Daniel R. Mayerson

Version

1.0

Date

2024-12-16

Copyright

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7.12 DiagnosticsEmission.h

Go to the documentation of this file.

```
00001 #ifndef _FOORT_DIAGNOSTICS_EMISSION_H
00002 #define _FOORT_DIAGNOSTICS_EMISSION_H
00003
00004 #include "Geometry.h"  // Tensor objects
00005 #include "Metric.h"      // for Metric functions
00006 #include "InputOutput.h" // for ScreenOutput
00007
00008 #include <string>    // for strings
00009 #include <cmath>     // for fmax, fmin
00010 #include <stdexcept> // for std::runtime_error
00011
00020
00025 struct EmissionModel
00026 {
00027 public:
00028     // Virtual destructor to ensure correct destruction
00029     virtual ~EmissionModel() = default;
00030
00031     // Function which returns the emitted brightness intensity at Point p
00032     // Note: always pass the true radius (not log(r)) to EmissionModel!
00033     virtual real GetEmission(const Point &p) const = 0;
00034
00035     // Description string getter
00036     virtual std::string getFullDescriptionStr() const;
00037 };
00038
00042 struct GLMJohnsonSUEmission final : public EmissionModel
00043 {
00044 public:
00045     // Constructor
00046     GLMJohnsonSUEmission(real mu, real gamma, real sigma) : m_mu{mu}, m_gamma{gamma}, m_sigma{sigma}
00047     {
00048     }
00049
00050     // Emitted brightness (only depends on radius)
00051     real GetEmission(const Point &p) const final;
00052
00053     // Description string getter
00054     std::string getFullDescriptionStr() const final;
00055
00056 private:
00058     const real m_mu;
00060     const real m_gamma;
00062     const real m_sigma;
00063 };
```

```
00064
00066
00071 struct FluidVelocityModel
00072 {
00073 public:
00074     // Constructor is passed Metric pointer
00075     FluidVelocityModel(const Metric *const theMetric) : m_theMetric{theMetric} {}
00076
00077     // Virtual destructor to ensure correct destruction
00078     virtual ~FluidVelocityModel() = default;
00079
00080     // Get the local four-velocity (with index down!) of the fluid at Point p
00081     virtual OneIndex GetFourVelocityd(const Point &p) const = 0;
00082
00083     // Description string getter
00084     virtual std::string getFullDescriptionStr() const;
00085
00086 protected:
00087     // Metric pointer, used for e.g. calculating geodesic orbits
00088     const Metric *const m_theMetric;
00089 };
00090
00096 struct GeneralCircularRadialFluid final : public FluidVelocityModel
00097 {
00098     // Constructor with three parameters and Metric pointer (which is passed to base class
      constructor)
00099     GeneralCircularRadialFluid(real subKeplerParam, real betar, real betaphi, const Metric *const
      theMetric,
00100                                real ISCO_lowerbound, real ISCO_upperbound) :
      m_subKeplerParam{fmin(fmax(subKeplerParam, 0.0), 1.0)}, m_betaR{fmin(fmax(betar, 0.0), 1.0)},
00101
      m_betaPhi{fmin(fmax(betaphi, 0.0), 1.0)}, FluidVelocityModel(theMetric),
00102
      m_ISCOlowerbound{ISCO_lowerbound}, m_ISCOupperbound{ISCO_upperbound}
00103     {
00104         // Do some checks on three params, which must lie between 0.0 and 1.0 (note that they are
      adjusted as such in
00105         // initializer above)
00106         if (subKeplerParam < 0.0)
00107             ScreenOutput("Sub-Keplerian parameter must be between 0 and 1; adjusting to 0",
      OutputLevel::Level_0_WARNING);
00108         if (subKeplerParam > 1.0)
00109             ScreenOutput("Sub-Keplerian parameter must be between 0 and 1; adjusting to 1",
      OutputLevel::Level_0_WARNING);
00110
00111         if (betar < 0.0)
00112             ScreenOutput("beta_r parameter must be between 0 and 1; adjusting to 0",
      OutputLevel::Level_0_WARNING);
00113         if (betar > 1.0)
00114             ScreenOutput("beta_r parameter must be between 0 and 1; adjusting to 1",
      OutputLevel::Level_0_WARNING);
00115
00116         if (betaphi < 0.0)
00117             ScreenOutput("beta_phi parameter must be between 0 and 1; adjusting to 0",
      OutputLevel::Level_0_WARNING);
00118         if (betaphi > 1.0)
00119             ScreenOutput("beta_phi parameter must be between 0 and 1; adjusting to 1",
      OutputLevel::Level_0_WARNING);
00120
00121         // ISCO search bounds must be positive and correctly ordered, or the binary search in
00122         // FindISCO() silently produces a wrong or unconverged ISCO (NaN bounds under log(r)
00123         // coordinates, or an empty search range)
00124         if (ISCO_lowerbound <= 0.0 || ISCO_upperbound <= 0.0 || ISCO_lowerbound >= ISCO_upperbound)
00125             throw std::runtime_error("ISCOLowerBound and ISCOUpperBound must be positive with
      ISCOLowerBound < ISCOUpperBound (got lower=" +
00126                                      std::to_string(ISCO_lowerbound) + ", upper=" +
      std::to_string(ISCO_upperbound) + ")");
00127
00128         // Find the (equatorial) ISCO for this Metric
00129         FindISCO();
00130     }
00131
00132     // Get the local four-velocity of the fluid according to this model
00133     // Note: will always calculate with Point p exactly on the equator theta=pi/2, despite what p[2]
      may be passed
00134     OneIndex GetFourVelocityd(const Point &p) const final;
00135
00136     // Description string getter
00137     std::string getFullDescriptionStr() const final;
00138
00139 private:
00140     // Three parameters determining the flow
00142     const real m_subKeplerParam;
00144     const real m_betaR;
00146     const real m_betaPhi;
00147
00148     // Helper function to get circular (sub)Keplerian velocity outside the ISCO
```

```
00149     OneIndex GetCircularVelocityd(const Point &p, bool subKeplerianOn = true) const;
00150     // Helper function to get circular and infalling (sub)Keplerian velocity inside ISCO
00151     // (according to prescription of Cunningham)
00152     OneIndex GetInsideISCOCircularVelocityd(const Point &p) const;
00153
00154     // Helper function to get radial infalling velocity
00155     OneIndex GetRadialVelocityd(const Point &p) const;
00156
00157     // Helper function to find the ISCO (called in Constructor)
00158     void FindISCO();
00160     bool m_ISCOexists{false};
00162     real m_ISCOr{-1.0};
00164     real m_ISCOlowerbound;
00166     real m_ISCOupperbound;
00168     real m_ISCOpt{};
00170     real m_ISCOpphi{};
00171
00172     // Helper function which returns \partial_r(g^{ab}\Gamma^r_{bc}g^{cd}),
00173     // whose sign (after contracted with p_a p_d of the corresponding circular orbit)
00174     // tells us if the circular orbit considered is stable or not
00175     TwoIndex GetChristrRaisedDer(real r) const;
00176 };
00177
00178 #endif
```

7.13 FOORT/src/Geodesic.cpp File Reference

```
#include "Geodesic.h"
#include "InputOutput.h"
```

7.13.1 Detailed Description

Author

Daniel R. Mayerson

Version

1.0

Date

2024-12-16

Copyright

Copyright (c) 2024

7.14 FOORT/src/Geodesic.h File Reference

Declarations of abstract base Source class and all its descendants. Declaration of Geodesic class.

```
#include "Geometry.h"
#include "Metric.h"
#include "Diagnostics.h"
#include "Terminations.h"
#include "Integrators.h"
#include <string>
#include <vector>
```

Classes

- class Source

  Abstract Source base class.
- class NoSource

  NoSource class.
- class Geodesic

  Geodesic class: an instance of this class is created for each Geodesic that is integrated.

7.14.1 Detailed Description

Declarations of abstract base Source class and all its descendants. Declaration of Geodesic class.

Author

Daniel R. Mayerson

Version

1.0

Date

2024-12-16

Copyright

Copyright (c) 2024

7.15 Geodesic.h

Go to the documentation of this file.

```
00001 #ifndef _FOORT_GEODESIC_H
00002 #define _FOORT_GEODESIC_H
00003
00004 #include "Geometry.h"   // for tensor objects
00005 #include "Metric.h"       // for the metric
00006 #include "Diagnostics.h"  // Geodesics own Diagnostics
00007 #include "Terminations.h" // Geodesics own Terminations
00008 #include "Integrators.h"  // Geodesics use an GeodesicIntegratorFunc to integrate itself
00009
00010 #include <string> // for strings
00011 #include <vector> // for std::vector
00012
00021
00024
00029 class Source
00030 {
00031 public:
00032     // Constructor initializes Metric
00033     Source(const Metric *const theMetric) : m_theMetric{theMetric} {}
00034
00035     // Virtual destructor to ensure correct descendant destruction
00036     virtual ~Source() = default;
00037
00038     // Get the source for the current geodesic position and velocity
00039     virtual OneIndex getSource(Point pos, OneIndex vel) const = 0;
00040
00041     // Full description string (space allowed), to be outputted to file
00042     virtual std::string getFullDescriptionStr() const;
```

```
00043
00044 protected:
00046     const Metric *const m_theMetric;
00047 };
00048
00054 class NoSource final : public Source
00055 {
00056 public:
00057     // Simple constructor passes on the Metric pointer to the base constructor
00058     NoSource(const Metric *const theMetric) : Source(theMetric) {}
00059
00060     // Returns zero source
00061     OneIndex getSource(Point pos, OneIndex vel) const final;
00062
00063     // Description string getter
00064     std::string getFullDescriptionStr() const final;
00065 };
00066
00069
00076 class Geodesic
00077 {
00078 public:
00079     // Default constructor not allowed
00080     Geodesic() = delete;
00081     // Copy constructor or copy assignment not allowed
00082     Geodesic(const Geodesic &) = delete;
00083     Geodesic &operator=(const Geodesic &) = delete;
00084
00095     Geodesic(const Metric *const theMetric, const Source *const theSource,
00096              DiagBitflag diagbit, DiagBitflag valdiagbit,
00097              TermBitflag termbit, GeodesicIntegratorFunc theIntegrator) : m_theMetric{theMetric},
     m_theSource{theSource},
00098
     m_AllDiagnostics{CreateDiagnosticVector(diagbit, valdiagbit, this)},
00099
     m_AllTerminations{CreateTerminationVector(termbit, this)},
00100
     m_theIntegrator{theIntegrator}
00101     {
00102     }
00103
00104     // This initializes/resets the geodesic with a given ScreenIndex, initial position, and initial
     velocity
00105     // Also resets all Diagnostics and Terminations, resets the TermCondition to Term::Continue,
00106     // and puts the Geodesic back to lambda = 0.0.
00107     void Reset(ScreenIndex scrindex, Point initpos, OneIndex initvel);
00108
00109     // This makes the Geodesic integrate itself one step; then the Geodesic loops through all
     Terminations and Diagnostics to update
00110     Term Update();
00111
00112     // Getters for properties of its internal state
00113     Term getTermCondition() const;      // Current termination condition (Term::Continue if not done
     integrating)
00114     Point getCurrentPos() const;       // Current position
00115     OneIndex getCurrentVel() const;     // Current velocity
00116     real getCurrentLambda() const;      // Current value of affine parameter
00117     ScreenIndex getScreenIndex() const; // screen index
00118
00119     // Output getters, to be called after the Geodesic terminates
00120     // This gets the complete output that should be written to the output files;
00121     // there is one string more than the count of Diagnostics: one string per Diagnostic,
00122     // PLUS the first string is the screen index.
00123     std::vector<std::string> getAllOutputStr() const;
00124     // This returns the "value" (from the Diagnostic that was set to the value Diagnostic) that is
     associated
00125     // to the Geodesic. Will be used to determine "distance" between Geodesics which is used in Mesh
     refinement.
00126     std::vector<real> getDiagnosticFinalValue() const;
00127
00128 private:
00129     // These variables define its internal state
00131     Term m_TermCond{Term::Uninitialized};
00133     Point m_CurrentPos{};
00135     OneIndex m_CurrentVel{};
00137     real m_curLambda{0.0};
00138
00141     ScreenIndex m_ScreenIndex{};
00142
00143     // These are const pointers (or const vectors of pointers) that contain all the information the
     Geodesic needs
00145     const Metric *const m_theMetric;
00147     const Source *const m_theSource;
00149     const DiagnosticUniqueVector m_AllDiagnostics;
00151     const TerminationUniqueVector m_AllTerminations;
00153     const GeodesicIntegratorFunc m_theIntegrator;
00154 };
```

```
00155
00156 #endif
```

7.16 FOORT/src/Geometry.h File Reference

Definitions and some operations with geometric objects.alignas.

```
#include <limits>
#include <string>
#include <array>
#include <utility>
#include <vector>
```

Macros

- #define LARGECOUNTER_MAX std::numeric_limits<largecounter>::max()

  Macro definition of maximum value that can be held in this large counter.
- #define PIXEL_MAX std::numeric_limits<pixelcoord>::max()

  Macro definition of maximum value that can be held in this large counter.

Typedefs

- using real = double

  A real number (could be changed to use arbitrary precision in the future).
- using largecounter = unsigned long

  This type is used to count geodesics integrated.
- using pixelcoord = largecounter

  A pixel coordinate: always >=0 and integer; we use the largecounter type.
- using Point = std::array<real, dimension>

  A point in spacetime. Note that coordinates are always assumed to be (t, r, theta, phi)
- using ScreenPoint = std::array<real, dimension - 2>

  A point on the ViewScreen; this does not have a time or radial extent.
- using ScreenIndex = std::array<pixelcoord, dimension - 2>

  An index on the ViewScreen (row, column)
- using OneIndex = Point

  Object with one index has the same structure as a Point.
- using TwoIndex = std::array<OneIndex, dimension>

  Object with two indices is an array of OneIndex objects.
- using ThreeIndex = std::array<TwoIndex, dimension>

  Object with three indices is an array of TwoIndex objects.
- using FourIndex = std::array<ThreeIndex, dimension>

  Object with four indices is an array of ThreeIndex objects.
- using SingularityCoord = std::pair<int, real>

  SingularityCoord: pair of (coordinate number, coordinate value)
- using Singularity = std::vector<SingularityCoord>

  Singularity: a number of SingularityCoords together that define a arbitrary codimension surface/line/point.

Functions

- template<size_t TensorDim>
  std::string toString (const std::array< largecounter, TensorDim > &theTensor)
  Base case for single index tensor of unsigned integers (ScreenIndex).
- template<size_t TensorDim>
  std::string toString (const std::array< real, TensorDim > &theTensor)
  Base case for single index tensor of reals (Point, OneIndex, ScreenPoint).
- template<typename Tensor, size_t TensorDim>
  std::string toString (const std::array< Tensor, TensorDim > &theTensor)
  General case for tensors with more than one index (TwoIndex, ThreeIndex, FourIndex).
- template<typename t, size_t TensorDim>
  std::array< t, TensorDim > operator+ (const std::array< t, TensorDim > &a1, const std::array< t, TensorDim > &a2)
  Function to recursively call + on the lower rank tensor (OR the underlying reals/ints, if the tensor is rank 1)
- template<typename t, size_t TensorDim>
  std::array< t, TensorDim > operator- (const std::array< t, TensorDim > &a1, const std::array< t, TensorDim > &a2)
  Function to recursively call - on the lower rank tensor (OR the underlying reals/ints, if the tensor is rank 1)
- template<typename t, size_t TensorDim>
  std::array< t, TensorDim > operator* (const std::array< t, TensorDim > &t1, real lambda)
  Function to recursively scalar multiply the lower-rank tensors (OR the underlying reals/ints for the rank-1 tensor)
- template<typename t, size_t TensorDim>
  std::array< t, TensorDim > operator* (real lambda, const std::array< t, TensorDim > &t1)
  Function to recursively scalar multiply the lower-rank tensors (OR the underlying reals/ints for the rank-1 tensor)
- template<typename t, size_t TensorDim>
  std::array< t, TensorDim > operator/ (const std::array< t, TensorDim > &t1, real lambda)
  Function to recursively scalar divide the lower-rank tensors (OR the underlying reals/ints for the rank-1 tensor)

Variables

- constexpr real pi {3.141592653589793}
  The value of pi.
- constexpr int dimension {4}
  The spacetime dimension.

7.16.1 Detailed Description

Definitions and some operations with geometric objects.alignas.

Author

Daniel R. Mayerson

Defines Point, tensors with 1-4 indices, and the operator toString for them. Also defines basic tensor arithmetic (+, -, *, /). No .cpp with implementations; all functions are inline.

Version

0.1

Date

2025-01-14

Copyright

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7.16.2 Macro Definition Documentation

7.16.2.1 LARGECOUNTER_MAX

```
#define LARGECOUNTER_MAX std::numeric_limits<largecounter>::max()
```

Macro definition of maximum value that can be held in this large counter.

7.16.2.2 PIXEL_MAX

```
#define PIXEL_MAX std::numeric_limits<pixelcoord>::max()
```

Macro definition of maximum value that can be held in this large counter.

7.16.3 Typedef Documentation

7.16.3.1 FourIndex

```
using FourIndex = std::array<ThreeIndex, dimension>
```

Object with four indices is an array of ThreeIndex objects.

7.16.3.2 largecounter

```
using largecounter = unsigned long
```

This type is used to count geodesics integrated.

7.16.3.3 OneIndex

```
using OneIndex = Point
```

Object with one index has the same structure as a Point.

7.16.3.4 pixelcoord

```
using pixelcoord = largecounter
```

A pixel coordinate: always >=0 and integer; we use the largecounter type.

7.16.3.5 Point

```
using Point = std::array<real, dimension>
```

A point in spacetime. Note that coordinates are always assumed to be (t, r, theta, phi)

7.16.3.6 real

```
using real = double
```

A real number (could be changed to use arbitrary precision in the future).

7.16.3.7 ScreenIndex

```
using ScreenIndex = std::array<pixelcoord, dimension - 2>
```

An index on the ViewScreen (row, column)

7.16.3.8 ScreenPoint

```
using ScreenPoint = std::array<real, dimension - 2>
```

A point on the ViewScreen; this does not have a time or radial extent.

7.16.3.9 Singularity

```
using Singularity = std::vector<SingularityCoord>
```

Singularity: a number of SingularityCoords together that define a arbitrary codimension surface/line/point.

7.16.3.10 SingularityCoord

```
using SingularityCoord = std::pair<int, real>
```

SingularityCoord: pair of (coordinate number, coordinate value)

7.16.3.11 ThreeIndex

```
using ThreeIndex = std::array<TwoIndex, dimension>
```

Object with three indices is an array of TwoIndex objects.

7.16.3.12 TwoIndex

```
using TwoIndex = std::array<OneIndex, dimension>
```

Object with two indices is an array of OneIndex objects.

7.16.4 Function Documentation

7.16.4.1 operator∗() [1/2]

```
template<typename t, size_t TensorDim>
std::array< t, TensorDim > operator* (
            const std::array< t, TensorDim > & t1,
            real lambda)
```

Function to recursively scalar multiply the lower-rank tensors (OR the underlying reals/ints for the rank-1 tensor)

Template Parameters

t	Type of the tensor
TensorDim	Dimension of the tensor

Parameters

t1	Tensor
lambda	Scalar

Returns

std::array<t, TensorDim> Scalar product of t1 and lambda

7.16.4.2 operator∗() [2/2]

```
template<typename t, size_t TensorDim>
std::array< t, TensorDim > operator* (
            real lambda,
            const std::array< t, TensorDim > & t1)
```

Function to recursively scalar multiply the lower-rank tensors (OR the underlying reals/ints for the rank-1 tensor)

This function calls the right multiplication operator.

Template Parameters

t	Type of the tensor
TensorDim	Dimension of the tensor

Parameters

lambda	Scalar
t1	Tensor

Returns

std::array<t, TensorDim> Scalar product of lambda and t1

7.16.4.3 operator+()

```
template<typename t, size_t TensorDim>
std::array< t, TensorDim > operator+ (
            const std::array< t, TensorDim > & a1,
            const std::array< t, TensorDim > & a2)
```

Function to recursively call + on the lower rank tensor (OR the underlying reals/ints, if the tensor is rank 1)

Template Parameters

t	data type of the tensor
TensorDim	Dimension of the tensor

Parameters

a1	Tensor 1
a2	Tensor 2

Returns

std::array<t, TensorDim> Tensor sum of a1 and a2

7.16.4.4 operator-()

```
template<typename t, size_t TensorDim>
std::array< t, TensorDim > operator- (
            const std::array< t, TensorDim > & a1,
            const std::array< t, TensorDim > & a2)
```

Function to recursively call - on the lower rank tensor (OR the underlying reals/ints, if the tensor is rank 1)

Template Parameters

t	data type of the tensor
TensorDim	Dimension of the tensor

Parameters

a1	Tensor 1
a2	Tensor 2

Returns

std::array<t, TensorDim> Tensor difference of a1 and a2

7.16.4.5 operator/()

```
template<typename t, size_t TensorDim>
std::array< t, TensorDim > operator/ (
            const std::array< t, TensorDim > & t1,
            real lambda)
```

Function to recursively scalar divide the lower-rank tensors (OR the underlying reals/ints for the rank-1 tensor)

This function calls the right multiplication operator.

Template Parameters

t	Type of the tensor
TensorDim	Dimension of the tensor

Parameters

t1	Tensor
lambda	Scalar

Returns

std::array<t, TensorDim> Scalar division of t1 by lambda

7.16.4.6 toString() [1/3]

```
template<size_t TensorDim>
std::string toString (
            const std::array< largecounter, TensorDim > & theTensor)
```

Base case for single index tensor of unsigned integers (ScreenIndex).

We do not want toString(ScreenIndex) to convert its entries to reals and use the implementation for a single index tensor of reals, because we do not want decimal points in our string for the ints!

Template Parameters

TensorDim	Dimension of the tensor

Parameters

theTensor	Tensor object

Returns

std::string

7.16.4.7 toString() [2/3]

```
template<size_t TensorDim>
std::string toString (
            const std::array< real, TensorDim > & theTensor)
```

Base case for single index tensor of reals (Point, OneIndex, ScreenPoint).

Template Parameters

TensorDim	Dimension of the tensor

Parameters

theTensor	Tensor object

Returns

std::string

7.16.4.8 toString() [3/3]

```
template<typename Tensor, size_t TensorDim>
std::string toString (
            const std::array< Tensor, TensorDim > & theTensor)
```

General case for tensors with more than one index (TwoIndex, ThreeIndex, FourIndex).

This function recursively calls the lower rank tensor to print itself.

Template Parameters

Tensor	The type of the tensor
TensorDim	Dimension of the tensor

Parameters

theTensor	Tensor object

Returns

std::string

7.16.5 Variable Documentation

7.16.5.1 dimension

```
int dimension {4}  [inline], [constexpr]
```

The spacetime dimension.

7.16.5.2 pi

```
real pi {3.141592653589793}  [inline], [constexpr]
```

The value of pi.

7.17 Geometry.h

Go to the documentation of this file.

```
00001 #ifndef _FOORT_GEOMETRY_H
00002 #define _FOORT_GEOMETRY_H
00003
00014
00015 #include <limits>  // for std::numeric_limits
00016 #include <string>  // needed for toString(...) to convert tensors to strings
00017 #include <array>   // needed to define tensors as fixed-size arrays of real or pixelcoord
00018 #include <utility> // needed for std::pair
00019 #include <vector>  // for std::vector
00020
00022 using real = double;
00023
00024 // Note: An unsigned long is guaranteed to be able to hold at least 4 294 967 295 (4.10^10).
00025 // An unsigned int is only guaranteed to be able to hold 65 535, although
00026 // many modern-day implementations will actually make the int 32-bit and so much larger
00027
00029 using largecounter = unsigned long;
00031 using pixelcoord = largecounter;
00032
00034 #ifndef LARGECOUNTER_MAX
00035 #define LARGECOUNTER_MAX std::numeric_limits<largecounter>::max()
00036 #endif
00038 #ifndef PIXEL_MAX
00039 #define PIXEL_MAX std::numeric_limits<pixelcoord>::max()
00040 #endif
00041
00042 // CONSTANTS
00043
00045 inline constexpr real pi{3.141592653589793};
00046
00048 inline constexpr int dimension{4};
00049
00050 // TENSOR DEFINITIONS
00051
00053 using Point = std::array<real, dimension>;
00054
00056 using ScreenPoint = std::array<real, dimension - 2>;
00057
00059 using ScreenIndex = std::array<pixelcoord, dimension - 2>;
00060
00062 using OneIndex = Point;
00064 using TwoIndex = std::array<OneIndex, dimension>;
00066 using ThreeIndex = std::array<TwoIndex, dimension>;
00068 using FourIndex = std::array<ThreeIndex, dimension>;
00069
00070 // Definition used to define singularity of arbitrary codimension
00072 using SingularityCoord = std::pair<int, real>;
00074 using Singularity = std::vector<SingularityCoord>;
00075
00076 // PRINTING TENSORS TO STRING
00077
00087 template <size_t TensorDim>
00088 std::string toString(const std::array<largecounter, TensorDim> &theTensor)
00089 {
00090     std::string theStr{"("}; // no spaces for the innermost brackets
00091     for (int i = 0; i < TensorDim - 1; ++i)
00092     {
00093         // Here we use the std::to_string to convert the real to string
00094         theStr += std::to_string(theTensor[i]);
00095         theStr += ", ";
00096     }
00097     theStr += std::to_string(theTensor[TensorDim - 1]);
00098     theStr += ")"; // no spaces for the innermost brackets
00099
00100     return theStr;
00101 }
00102
00110 template <size_t TensorDim>
00111 std::string toString(const std::array<real, TensorDim> &theTensor)
00112 {
00113     std::string theStr{"("}; // no spaces for the innermost brackets
00114
00115     for (int i = 0; i < TensorDim - 1; ++i)
00116     {
00117         // Here we use the std::to_string to convert the real to string
00118         theStr += std::to_string(theTensor[i]);
00119         theStr += ", ";
00120     }
00121     theStr += std::to_string(theTensor[TensorDim - 1]);
00122     theStr += ")"; // no spaces for the innermost brackets
00123
00124     return theStr;
```

```
00125 }
00126
00135 template <typename Tensor, size_t TensorDim>
00136 std::string toString(const std::array<Tensor, TensorDim> &theTensor)
00137 {
00138     std::string theStr{"( "}; // All but the innermost brackets have an extra space padding the
      bracket
00139
00140     for (int i = 0; i < TensorDim - 1; ++i)
00141     {
00142         theStr += toString(theTensor[i]);
00143         theStr += ", ";
00144     }
00145     theStr += toString(theTensor[TensorDim - 1]); // the last element doesn't have a comma after it
00146
00147     theStr += " )"; // All but the innermost brackets have an extra space padding the bracket
00148
00149     return theStr;
00150 }
00151
00152 // TENSOR ARITHMETIC: addition/subtraction of tensors, scalar multiplication/division
00153
00163 template <typename t, size_t TensorDim>
00164 std::array<t, TensorDim> operator+(const std::array<t, TensorDim> &a1, const std::array<t, TensorDim>
      &a2)
00165 {
00166     std::array<t, TensorDim> temp{a1};
00167     for (int i = 0; i < TensorDim; ++i)
00168         temp[i] = temp[i] + a2[i];
00169
00170     return temp;
00171 }
00172
00182 template <typename t, size_t TensorDim>
00183 std::array<t, TensorDim> operator-(const std::array<t, TensorDim> &a1, const std::array<t, TensorDim>
      &a2)
00184 {
00185     std::array<t, TensorDim> temp{a1};
00186     for (int i = 0; i < TensorDim; ++i)
00187         temp[i] = temp[i] - a2[i];
00188
00189     return temp;
00190 }
00191
00201 template <typename t, size_t TensorDim>
00202 std::array<t, TensorDim> operator*(const std::array<t, TensorDim> &t1, real lambda)
00203 {
00204     std::array<t, TensorDim> temp{t1};
00205     for (int i = 0; i < TensorDim; ++i)
00206         temp[i] = static_cast<t>(temp[i] * lambda); // static_cast necessary if t is integral type!
00207     return temp;
00208 }
00209
00219 template <typename t, size_t TensorDim>
00220 std::array<t, TensorDim> operator*(real lambda, const std::array<t, TensorDim> &t1)
00221 {
00222     return t1 * lambda;
00223 }
00224
00234 template <typename t, size_t TensorDim>
00235 std::array<t, TensorDim> operator/(const std::array<t, TensorDim> &t1, real lambda)
00236 {
00237     return t1 * (1 / lambda);
00238 }
00239
00240 #endif
```

7.18 FOORT/src/Grid.cpp File Reference

```
#include <fstream>
#include <iostream>
#include <sstream>
#include <stdexcept>
#include "Grid.h"
```

7.19 FOORT/src/Grid.h File Reference

```
#include <fstream>
#include <iostream>
```

Classes

- class Grid

  A class for a 2D grid that contains the values of the necessary functions.

7.20 Grid.h

Go to the documentation of this file.

```
00001 #include <fstream>
00002 #include <iostream>
00003
00004 #ifndef GRID_H
00005 #define GRID_H
00006
00008 class Grid
00009 {
00010 public:
00012     int N_row;
00014     int N_col;
00016     double *data;
00018     int size;
00019     Grid(int N_row, int N_col);
00020
00022     ~Grid() { delete[] this->data; }
00023
00025     double &operator()(int i, int j) { return this->data[i * N_col + j]; }
00027     const double &operator()(int i, int j) const { return this->data[i * N_col + j]; }
00028
00029     void initialize_from_file(std::string file);
00030 };
00031
00032 #endif // GRID_H
```

7.21 FOORT/src/InputOutput.cpp File Reference

```
#include "InputOutput.h"
#include <algorithm>
#include <filesystem>
```

Functions

- void SetOutputLevel (OutputLevel theLvl)

  Set the Output Level.
- void SetLoopMessageFrequency (largecounter thefreq)

  Set the Loop Message Frequency.
- largecounter GetLoopMessageFrequency ()

  Get the Loop Message Frequency during integration loops.
- void ScreenOutput (std::string_view theOutput, OutputLevel lvl, bool newLine)

  Outputs line to screen console, contingent on it being allowed by the set outputlevel.

7.21.1 Detailed Description

Author

Daniel R. Mayerson

Version

0.1

Date

2025-01-14

Copyright

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7.21.2 Function Documentation

7.21.2.1 GetLoopMessageFrequency()

```
largecounter GetLoopMessageFrequency ()
```

Get the Loop Message Frequency during integration loops.

Returns

largecounter

7.21.2.2 ScreenOutput()

```
void ScreenOutput (
            std::string_view theOutput,
            OutputLevel lvl,
            bool newLine)
```

Outputs line to screen console, contingent on it being allowed by the set outputlevel.

Defaults are lvl = OutputLevel::Level_3_ALLDETAIL and newLine = true

Parameters

theOutput	output to be put on the screen
lvl	the output level
newLine	indicates whether we want a new line after the output

7.21.2.3 SetLoopMessageFrequency()

```
void SetLoopMessageFrequency (
            largecounter thefreq)
```

Set the Loop Message Frequency.

Parameters

thefreq	the desired frequency

7.21.2.4 SetOutputLevel()

```
void SetOutputLevel (
            OutputLevel theLvl)
```

Set the Output Level.

Parameters

theLvl	The desired level

7.22 FOORT/src/InputOutput.h File Reference

Declarations of functions & classes that deal with the output to files and to the screen.

```
#include "Geometry.h"
#include <string_view>
#include <iostream>
#include <fstream>
#include <string>
#include <vector>
```

Classes

- class GeodesicOutputHandler

  // GeodesicOutputHandler handles all of the output to file. It gets passed all of the output strings for every Geodesic, it then stores this data until it eventually writes all data to the appropriate files

Enumerations

- enum class OutputLevel {
  Level_0_WARNING = 0 , Level_1_PROC = 1 , Level_2_SUBPROC = 2 , Level_3_ALLDETAIL = 3 ,
  Level_4_DEBUG = 4 , MaxLevel }

  The output level determines at what priority level the output is generated to the console.

Functions

- void SetOutputLevel (OutputLevel theLvl)

  Set the Output Level.
- void ScreenOutput (std::string_view theOutput, OutputLevel lvl=OutputLevel::Level_3_ALLDETAIL, bool newLine=true)

  Outputs line to screen console, contingent on it being allowed by the set outputlevel.
- void SetLoopMessageFrequency (largecounter thefreq)

  Set the Loop Message Frequency.
- largecounter GetLoopMessageFrequency ()

  Get the Loop Message Frequency during integration loops.

7.22.1 Detailed Description

Declarations of functions & classes that deal with the output to files and to the screen.

Author

Daniel R. Mayerson

Version

0.1

Date

2025-01-14

Copyright

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7.22.2 Enumeration Type Documentation

7.22.2.1 OutputLevel

```
enum class OutputLevel  [strong]
```

The output level determines at what priority level the output is generated to the console.

Level_0_WARNING: Only warnings are outputted. Level_1_PROC: Coarsest level output; only the major procedures produce output. Level_2_SUBPROC: Subprocedures can also produce output. Level_3_ALLDETAIL: Finest level output; all details are shown. Level_4_DEBUG: Finest level output AND debug messages as well.

Enumerator

Level_0_WARNING	
Level_1_PROC	
Level_2_SUBPROC	
Level_3_ALLDETAIL	
Level_4_DEBUG	
MaxLevel	

7.22.3 Function Documentation

7.22.3.1 GetLoopMessageFrequency()

```
largecounter GetLoopMessageFrequency ()
```

Get the Loop Message Frequency during integration loops.

Returns

largecounter

7.22.3.2 ScreenOutput()

```
void ScreenOutput (
            std::string_view theOutput,
            OutputLevel lvl,
            bool newLine)
```

Outputs line to screen console, contingent on it being allowed by the set outputlevel.

Defaults are lvl = OutputLevel::Level_3_ALLDETAIL and newLine = true

Parameters

theOutput	output to be put on the screen
lvl	the output level
newLine	indicates whether we want a new line after the output

7.22.3.3 SetLoopMessageFrequency()

```
void SetLoopMessageFrequency (
            largecounter thefreq)
```

Set the Loop Message Frequency.

Parameters

thefreq	the desired frequency

7.22.3.4 SetOutputLevel()

```
void SetOutputLevel (
            OutputLevel theLvl)
```

Set the Output Level.

Parameters

theLvl	The desired level

7.23 InputOutput.h

Go to the documentation of this file.

```
00001 #ifndef _FOORT_INPUTOUTPUT_H
00002 #define _FOORT_INPUTOUTPUT_H
00003
00012
00013 #include "Geometry.h" // for basic tensor objects
00014
00015 #include <string_view> // std::string_view
00016 #include <iostream>    // needed for file and console output
00017 #include <fstream>     // needed for file ouput
00018 #include <string>      // std::string used in various places
00019 #include <vector>      // needed to create vectors of strings
00020
00021 // SCREENOUTPUT
00022
00031 enum class OutputLevel
00032 {
00033     Level_0_WARNING = 0,
00034     Level_1_PROC = 1,
00035     Level_2_SUBPROC = 2,
00036     Level_3_ALLDETAIL = 3,
00037     Level_4_DEBUG = 4,
00038
00039     MaxLevel // (unused)
00040 };
00041
00042 // Set the output level used; note: the output level itself is a static variable in InputOutput.cpp
00043 void SetOutputLevel(OutputLevel theLvl);
00044
00045 // Outputs line to screen console), contingent on it being allowed by the set outputlevel
00046 void ScreenOutput(std::string_view theOutput, OutputLevel lvl = OutputLevel::Level_3_ALLDETAIL, bool
      newLine = true);
00047
00048 // Set and Get for the loop message frequency (messages indicating progress during each integration
      loop)
00049 void SetLoopMessageFrequency(largecounter thefreq);
00050 largecounter GetLoopMessageFrequency();
00051
00052 // FILE OUTPUT
00053
00059 class GeodesicOutputHandler
00060 {
00061 public:
00062     // No default constructor possible
00063     GeodesicOutputHandler() = delete;
00064     // Constructor must pass the following strings that are used to construct the file names of the
      output file:
00065     // FilePrefix, TimeStamp, FileExtension, and a vector of strings DiagNames (the names of each of
      the Diagnostics that will
00066     // be outputting). It must also specify how many outputs to cache before outputting to a file, and
00067     // how many geodesics are allowed per file created
00068     GeodesicOutputHandler(std::string FilePrefix, std::string TimeStamp, std::string FileExtension,
00069                           std::vector<std::string> DiagNames,
00070                           largecounter nroutputstocache = LARGECOUNTER_MAX - 1, // note -1,
00071                                                                                 // since we will
      actually cache one more then this number before outputting everything
00072                           largecounter geodperfile = LARGECOUNTER_MAX,
00073                           std::string firstlineinfo = "");
00074
00075     // This tells the OutputHandler to prepare for this many geodesic outputs to arrive;
00076     // the internal state needs to be prepared such that they can come in without providing a data
      race
00077     void PrepareForOutput(largecounter nrOutputToCome);
00078
00079     // A new vector of output strings from a (single) Geodesic;
00080     // the length of the vector should be m_DiagNames.size()+1, since the first entry
00081     // is the screen index
00082     // NOTE: this procedure needs to be thread-safe!
00083     void NewGeodesicOutput(largecounter index, std::vector<std::string> theOutput);
00084
00085     // Calling this indicates that there is no further output to be expected;
00086     // this means we will write all remaining cached output to file
00087     void OutputFinished();
00088
00089     // Returns full description string of output handler
00090     std::string getFullDescriptionStr() const;
00091
00092 private:
00093     // Helper function: write everything that is cached to file now (clear the cache)
00094     void WriteCachedOutputToFile();
00095
00096     // Helper function: open the file with name filename for the first time, preparing it to write
00097     // This will effectively clear this file of any pre-existing content.
```

```
00098     void OpenForFirstTime(const std::string &filename);
00099
00100     // Helper function: return the full file name for the n-th output file
00101     // for the Diagnostic diagnr (this is an entry in m_DiagNames)
00102     std::string GetFileName(int diagnr, unsigned short filenr) const;
00103
00104     // The const strings that are used to construct the output file names
00106     const std::string m_FilePrefix;
00108     const std::string m_TimeStamp;
00110     const std::string m_FileExtension;
00112     const std::vector<std::string> m_DiagNames;
00113
00115     const bool m_PrintFirstLineInfo;
00117     const std::string m_FirstLineInfoString;
00118
00119     // Const setting the maximum number of outputs that can be cached before writing output to file.
00120     const largecounter m_nrOutputsToCache{};
00122     const largecounter m_nrGeodesicsPerFile{};
00123
00125     bool m_WriteToConsole{false};
00126
00128     largecounter m_PrevCached{0};
00129
00131     largecounter m_CurrentGeodesicsInFile{0};
00132
00134     unsigned short m_CurrentFullFiles{0};
00135
00137     std::vector<std::vector<std::string» m_AllCachedData{};
00138 };
00139
00140 #endif
```

7.24 FOORT/src/Integrators.cpp File Reference

```
#include "Integrators.h"
#include "Geodesic.h"
#include <algorithm>
#include <cmath>
#include <sstream>
#include <iostream>
```

7.24.1 Detailed Description

Author

Daniel R. Mayerson

Version

0.1

Date

2025-01-14

Copyright

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7.25 FOORT/src/Integrators.h File Reference

Declarations of integrator functions that integrate geodesic equation.

```
#include "Geometry.h"
#include "Metric.h"
#include <string>
```

Namespaces

- namespace Integrators

Typedefs

- using GeodesicIntegratorFunc = void (∗)(Point, OneIndex, Point &, OneIndex &, real &, const Metric ∗, const Source ∗)

Functions

- std::string Integrators::GetFullIntegratorDescription ()

  Description string getter for the Integrator.
- real Integrators::GetAdaptiveStep (Point curpos, OneIndex curvel)

  Determine the (affine parameter) step size to take.
- void Integrators::IntegrateGeodesicStep_RK4 (Point curpos, OneIndex curvel, Point &nextpos, OneIndex &nextvel, real &stepsize, const Metric ∗theMetric, const Source ∗theSource)

  Integrate the geodesic equation by one step using Runge-Kutta-4.
- void Integrators::IntegrateGeodesicStep_Verlet (Point curpos, OneIndex curvel, Point &nextpos, OneIndex &nextvel, real &stepsize, const Metric ∗theMetric, const Source ∗theSource)

  Integrate the geodesic equation by one step using the Verlet algorithm.

Variables

- constexpr real Integrators::delta_nodiv0 = 1e-20

  This is used to avoid dividing by zero.
- real Integrators::Derivative_hval {1e-7}

  The spacing used in stencils for the calculation of numerical derivatives (e.g. in Christoffel symbols)
- std::string Integrators::IntegratorDescription {"Verlet"}

  The name of the integrator selected.
- real Integrators::epsilon {0.03}

  This is the base step size to be taken (the integrator will adapt this if necessary)
- real Integrators::SmallestPossibleStepsize {1e-12}

  The affine parameter must always go forward by at least this amount.
- real Integrators::VerletVelocityTolerance {-1.0}

7.25.1 Detailed Description

Declarations of integrator functions that integrate geodesic equation.

Author

Daniel R. Mayerson

Version

0.1

Date

2025-01-14

Copyright

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7.25.2 Typedef Documentation

7.25.2.1 GeodesicIntegratorFunc

```
using GeodesicIntegratorFunc = void (*)(Point, OneIndex, Point &, OneIndex &, real &, const
Metric *, const Source *)
```

7.26 Integrators.h

Go to the documentation of this file.

```
00001 #ifndef _FOORT_INTEGRATORS_H
00002 #define _FOORT_INTEGRATORS_H
00003
00012
00013 #include "Geometry.h" // for basic tensor objects
00014 #include "Metric.h"   // for the metric
00015
00016 #include <string> // std::string
00017
00019 class Source;
00020
00021 // This is the structure of a function that integrates the geodesic equation one step
00022 // It takes as arguments:
00023 // - current position, current velocity
00024 // - references to: next position, next velocity, affine parameter step size (which the functions
      sets)
00025 // - pointers to: the Metric, the Source that are used to evaluate the geodesic equation
00026 using GeodesicIntegratorFunc = void (*)(Point, OneIndex, Point &, OneIndex &, real &, const Metric *,
      const Source *);
00027
00028 // Namespace for integrator constants and functions
00029 namespace Integrators
00030 {
00032     constexpr real delta_nodiv0 = 1e-20;
00033
00035     inline real Derivative_hval{1e-7};
00036
00038     inline std::string IntegratorDescription{"Verlet"};
00039
```

```
00040     // Full descriptive string of integrator and all integrator options
00041     std::string GetFullIntegratorDescription();
00042
00044     inline real epsilon{0.03};
00045
00047     inline real SmallestPossibleStepsize{1e-12};
00048
00049     // Function to get  (adaptive) step size
00050     real GetAdaptiveStep(Point curpos, OneIndex curvel);
00051
00052     // This is a GeodesicIntegratorFunc
00053     // Using the Runge-Kutta-4 algorithm to integrate the geodesic equation
00054     void IntegrateGeodesicStep_RK4(Point curpos, OneIndex curvel,
00055                                    Point &nextpos, OneIndex &nextvel, real &stepsize, const Metric
      *theMetric, const Source *theSource);
00056
00059     inline real VerletVelocityTolerance{-1.0};
00060
00061     // This is a GeodesicIntegratorFunc
00062     // Using the velocity Verlet algorithm to integrate the geodesic equation
00063     void IntegrateGeodesicStep_Verlet(Point curpos, OneIndex curvel,
00064                                       Point &nextpos, OneIndex &nextvel, real &stepsize, const Metric
      *theMetric, const Source *theSource);
00065 }
00066
00067 #endif
```

7.27 FOORT/src/Interpolator.cpp File Reference

```
#include <fstream>
#include <iomanip>
#include <iostream>
#include <sstream>
#include <string>
#include "Interpolator.h"
#include "Grid.h"
```

7.28 FOORT/src/Interpolator.h File Reference

```
#include <fstream>
#include <iostream>
#include <sstream>
#include <string>
#include <vector>
#include <array>
#include <stdexcept>
#include <algorithm>
#include <cmath>
#include "Grid.h"
```

Classes

- class BicubicSplineInterpolator

7.29 Interpolator.h

Go to the documentation of this file.

```
00001 #include <fstream>
00002 #include <iostream>
00003 #include <sstream>
00004 #include <string>
00005 #include <vector>
00006 #include <array>
00007 #include <stdexcept>
00008 #include <algorithm>
00009 #include <cmath>
00010 #include "Grid.h"
00011
00012 #ifndef INTERPOLATOR_H
00013 #define INTERPOLATOR_H
00014
00015 class BicubicSplineInterpolator
00016 {
00017 public:
00018     BicubicSplineInterpolator(const std::string x_file = "RotatingBosonStar/x.txt",
00019                               const std::string th_file = "RotatingBosonStar/theta.txt")
00020     {
00021         // Load x and theta coordinates from files
00022         std::ifstream x_input(x_file);
00023         std::ifstream th_input(th_file);
00024
00025         if (!th_input || !x_input)
00026         {
00027             std::cerr « "Error opening file!" « std::endl;
00028             throw std::runtime_error("File not found");
00029         }
00030
00031         double val;
00032         while (x_input » val)
00033         {
00034             m_x.push_back(val);
00035         }
00036         while (th_input » val)
00037         {
00038             m_theta.push_back(val);
00039         }
00040
00041         // Explicitly close files
00042         x_input.close();
00043         th_input.close();
00044     }
00045
00046     // BicubicSplineInterpolator(const std::vector<double> &x_coords,
00047     //                           const std::vector<double> &theta_coords,
00048     //                           const Grid *grid)
00049     //     : m_x(x_coords), m_theta(theta_coords)
00050     // {
00051
00052     //     compute_spline_coefficients(grid);
00053     // }
00054
00055     void set_grid(const Grid *grid)
00056     {
00057         // Precompute spline coefficients for better performance
00058         // This is more complex but provides C2 continuity
00059         compute_spline_coefficients(grid);
00060         std::cout « "Interpolation domain set: x in [" « m_x.front() « ", " « m_x.back() « "], theta
      in [" « m_theta.front() « ", " « m_theta.back() « "].\n";
00061     }
00062     double interpolate(double p_x, double p_theta,
00063                        bool allow_extrapolation = false) const;
00064
00065 private:
00066     std::vector<double> m_x;
00067     std::vector<double> m_theta;
00068     std::vector<std::array<double, 16» m_coeffs; // 16 coefficients per cell
00069
00070     void compute_spline_coefficients(const Grid *grid);
00071     double estimate_dx(const Grid *grid, size_t j, size_t i) const;
00072     double estimate_dy(const Grid *grid, size_t j, size_t i) const;
00073     double estimate_dxdy(const Grid *grid, size_t j, size_t i) const;
00074     void compute_cell_coefficients(std::array<double, 16> &coeffs,
00075                                    double f00, double f10, double f01, double f11,
00076                                    double fx00, double fx10, double fx01, double fx11,
00077                                    double fy00, double fy10, double fy01, double fy11,
00078                                    double fxy00, double fxy10, double fxy01, double fxy11) const;
00079 };
00080
00081 #endif // INTERPOLATOR_H
```

7.30 FOORT/src/Main.cpp File Reference

Main function for FOORT.

```
#include "Config.h"
#include "Diagnostics.h"
#include "Geodesic.h"
#include "Geometry.h"
#include "InputOutput.h"
#include "Integrators.h"
#include "Metric.h"
#include "Terminations.h"
#include "Utilities.h"
#include "ViewScreen.h"
#include <omp.h>
#include <iostream>
```

Functions

- int main (int argc, char ∗argv[ ])

7.30.1 Detailed Description

Main function for FOORT.

Author

Daniel R. Mayerson

Version

1.0

Date

2025-1-21

Copyright

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7.30.2 Function Documentation

7.30.2.1 main()

```
int main (
            int argc,
            char * argv[ ])
```

7.31 FOORT/src/Mesh.cpp File Reference

```
#include "Mesh.h"
#include "Utilities.h"
#include <algorithm>
#include <limits>
#include <iostream>
```

7.31.1 Detailed Description

Author

Daniel R. Mayerson

Version

0.1

Date

2025-01-14

Copyright

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7.32 FOORT/src/Mesh.h File Reference

Declarations of abstract base Mesh class and all its descendants.

```
#include "Geometry.h"
#include "Diagnostics.h"
#include "InputOutput.h"
#include <cmath>
#include <utility>
#include <forward_list>
#include <memory>
#include <vector>
#include <array>
#include <string>
```

Classes

- class Mesh

  Abstract base Mesh class.
- class SimpleSquareMesh

  A simple square mesh that will integrate a square of evenly spaced pixels.
- class InputCertainPixelsMesh

  Mesh which integrates only certain user-inputted pixels.
- class SquareSubdivisionMesh

  Adaptive subdivision Mesh.
- struct SquareSubdivisionMesh::PixelInfo

  A struct the Mesh uses to keep all information about a given pixel.
- class SquareSubdivisionMeshV2

  Adaptive subdivision Mesh.
- struct SquareSubdivisionMeshV2::PixelInfo

  A struct the Mesh uses to keep all information about a given pixel.

7.32.1 Detailed Description

Declarations of abstract base Mesh class and all its descendants.

Author

Daniel R. Mayerson

Version

0.1

Date

2025-01-14

Copyright

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7.33 Mesh.h

Go to the documentation of this file.

```
00001 #ifndef _FOORT_MESH_H
00002 #define _FOORT_MESH_H
00003
00012
00013 #include "Geometry.h"  // needed for basic tensor objects
00014 #include "Diagnostics.h" // needed for Diagnostic "value" and "distance" functions
00015 #include "InputOutput.h" // needed for ScreenOutput
00016
00017 #include <cmath>        // needed for sqrt (only on Linux)
00018 #include <utility>      // std::move
00019 #include <forward_list> // std::forward_list
00020 #include <memory>       // std::unique_ptr
00021 #include <vector>       // std::vector
00022 #include <array>        // std::array
```

```
00023 #include <string>       // for strings
00024
00028 class Mesh
00029 {
00030 public:
00031     // Basic constructor constructs Diagnostic that is used for determinines "values" and "distances"
      between values
00032     Mesh(DiagBitflag valdiag)
00033         // Calling CreateDiagnosticVector in this way will create a vector with exactly one element in
      it,
00034         // i.e. the Diagnostic we need!
00035         : m_DistanceDiagnostic{std::move((CreateDiagnosticVector(valdiag, valdiag, nullptr))[0])}
00036     {
00037     }
00038
00039     // virtual destructor to ensure correct destruction of descendants
00040     virtual ~Mesh() = default;
00041
00042     // Getter for how many geodesics the Mesh currently wants to integrate in this iteration
00043     virtual largecounter getCurNrGeodesics() const = 0;
00044
00045     // This sets a new initial conditions (in the form of a ScreenPoint and ScreenIndex)
00046     // for a next pixel to be integrated in the current iteration
00047     virtual void getNewInitConds(largecounter index, ScreenPoint &newunitpoint, ScreenIndex
      &newscreenindex) const = 0;
00048
00049     // When a geodesic is finished integrating, it tells the Mesh and passes on its final "value"
00050     // NOTE: despite being a non-const member function, this must be designed to be thread-safe!
00051     virtual void GeodesicFinished(largecounter index, std::vector<real> finalValues) = 0;
00052
00053     // This is called when the current iteration is finished. The Mesh can now evaluate whether to
      continue or not
00054     virtual void EndCurrentLoop() = 0;
00055
00056     // Returns false if the Mesh wants another iteration of pixels to integrate
00057     virtual bool IsFinished() const = 0;
00058
00059     // Returns a string description of the Mesh (spaces allowed), describing its options
00060     virtual std::string getFullDescriptionStr() const;
00061
00062 protected:
00064     const std::unique_ptr<const Diagnostic> m_DistanceDiagnostic;
00065 };
00066
00070 class SimpleSquareMesh final : public Mesh
00071 {
00072 public:
00073     // Default constructor not possible
00074     SimpleSquareMesh() = delete;
00075     // Constructor initializes total number of pixels and passes valdiag to base constructor
00076     // Note that we static_cast the sqrt() to round off the row/column size to an integer number
00077     SimpleSquareMesh(largecounter totalPixels, DiagBitflag valdiag)
00078         : m_TotalPixels{static_cast<pixelcoord>(sqrt(totalPixels)) *
      static_cast<pixelcoord>(sqrt(totalPixels))},
00079           m_RowColumnSize{static_cast<pixelcoord>(sqrt(totalPixels))},
00080           Mesh(valdiag)
00081     {
00082         if constexpr (dimension != 4)
00083             ScreenOutput("SimpleSquareMesh only defined in 4D!", OutputLevel::Level_0_WARNING);
00084     }
00085
00086     // Declarations of overriding virtual functions
00087
00088     largecounter getCurNrGeodesics() const final;
00089
00090     void getNewInitConds(largecounter index, ScreenPoint &newunitpoint, ScreenIndex &newscreenindex)
      const final;
00091
00092     void GeodesicFinished(largecounter index, std::vector<real> finalValues) final;
00093
00094     void EndCurrentLoop() final;
00095
00096     bool IsFinished() const final;
00097
00098     // Description string getter
00099     std::string getFullDescriptionStr() const final;
00100
00101 private:
00103     const largecounter m_TotalPixels;
00105     const pixelcoord m_RowColumnSize;
00107     bool m_Finished{false};
00108 };
00109
00113 class InputCertainPixelsMesh : public Mesh
00114 {
00115 public:
00116     // Default constructor not possible
```

```
00117     InputCertainPixelsMesh() = delete;
00118     // Copy constructor not possible
00119     InputCertainPixelsMesh(const InputCertainPixelsMesh &) = delete;
00120     // Constructor given in Mesh.cpp file; constructor asks for input of pixels
00121     InputCertainPixelsMesh(largecounter totalPixels, DiagBitflag valdiag);
00122
00123     // Declarations of overriding virtual functions
00124
00125     largecounter getCurNrGeodesics() const final;
00126
00127     void getNewInitConds(largecounter index, ScreenPoint &newunitpoint, ScreenIndex &newscreenindex)
      const final;
00128
00129     void GeodesicFinished(largecounter index, std::vector<real> finalValues) final;
00130
00131     void EndCurrentLoop() final;
00132
00133     bool IsFinished() const final;
00134
00135     // Description string getter
00136     std::string getFullDescriptionStr() const final;
00137
00138 private:
00140     const pixelcoord m_RowColumnSize;
00141
00143     largecounter m_TotalPixels{0};
00145     std::vector<ScreenIndex> m_PixelsToIntegrate{};
00147     bool m_Finished{false};
00148 };
00149
00157 class SquareSubdivisionMesh : public Mesh
00158 {
00159 public:
00160     // default constructor not possible
00161     SquareSubdivisionMesh() = delete;
00162     // Constructor must be called with arguments:
00163     // - maxPixels: max. nr of pixels that can be integrated in TOTAL, over all iterations (if 0, then
      this is infinite,
00164     // i.e. we keep integrating until all squares are maximally subdivided or have weight 0)
00165     // - initialPixels: initial number of pixels to integrate (spaced equally over the screen)
00166     // - maxSubdivide: maximum number of times that we can subdivide squares (note: 1 denotes the
00167     // initial grid, so 2 would be subdividing the squares once)
00168     // - iterationPixels: maximum number of pixels to subdivide in each integration iteration
00169     // (max number of pixels that will be integrates is then 5*iterationPixels)
00170     // - initialSubToFinal: once we decide to subdivide a square, do we automatically keep subdividing
      it
00171     // until we reach maxSubdivision?
00172     // - valdiag: the "value" and "distance" Diagnostic to use
00173     SquareSubdivisionMesh(largecounter maxPixels, largecounter initialPixels, int maxSubdivide,
      largecounter iterationPixels, bool initialSubToFinal,
00174                           DiagBitflag valdiag)
00175         : m_InitialPixels{static_cast<pixelcoord>(sqrt(initialPixels)) *
      static_cast<pixelcoord>(sqrt(initialPixels))},
00176           m_MaxSubdivide{maxSubdivide},
00177           m_RowColumnSize{static_cast<pixelcoord>((sqrt(initialPixels) - 1) * ExpInt(2, maxSubdivide -
      1) + 1)},
00178           m_PixelsLeft{maxPixels}, m_MaxPixels{maxPixels}, m_InfinitePixels{maxPixels == 0},
      m_IterationPixels{iterationPixels},
00179           m_InitialSubDividideToFinal{initialSubToFinal}, Mesh(valdiag)
00180     {
00181         if constexpr (dimension != 4)
00182             ScreenOutput("SquareSubdivisionMesh only defined in 4D!", OutputLevel::Level_0_WARNING);
00183
00184         // DEBUG message for constructor (can delete)
00185         ScreenOutput("SquareSubdivisionMesh constructed: maxPixels: " + (m_InfinitePixels ? "infinite"
      : std::to_string(maxPixels)) + "; m_InitialPixels: " + std::to_string(m_InitialPixels) + ";
      m_RowColumnSize: " + std::to_string(m_RowColumnSize), OutputLevel::Level_4_DEBUG);
00186
00187         // Initialize the initial square, equally spaced grid
00188         InitializeFirstGrid();
00189     }
00190
00191     // Declarations of overriding virtual functions
00192
00193     largecounter getCurNrGeodesics() const final;
00194
00195     void getNewInitConds(largecounter index, ScreenPoint &newunitpoint, ScreenIndex &newscreenindex)
      const final;
00196
00197     void GeodesicFinished(largecounter index, std::vector<real> finalValues) final;
00198
00199     void EndCurrentLoop() final;
00200
00201     bool IsFinished() const final;
00202
00203     // Description string getter
00204     std::string getFullDescriptionStr() const final;
```

```
00205
00206 private:
00208     const largecounter m_InitialPixels;
00210     const int m_MaxSubdivide;
00212     const pixelcoord m_RowColumnSize;
00214     const largecounter m_IterationPixels;
00216     const largecounter m_MaxPixels;
00218     const bool m_InitialSubDividideToFinal;
00220     const bool m_InfinitePixels;
00221
00223     largecounter m_PixelsLeft;
00224
00228     struct PixelInfo
00229     {
00230         // Constructor with its ScreenIndex and current subdivision level
00231         PixelInfo(ScreenIndex ind, int subdiv) : Index{ind}, SubdivideLevel{subdiv} {}
00232
00234         ScreenIndex Index{};
00235
00237         int SubdivideLevel{};
00238
00240         real Weight{-1};
00241
00243         std::vector<real> DiagValue{};
00244
00246         largecounter LowerNbrIndex{0};
00248         largecounter RightNbrIndex{0};
00249     };
00251     std::vector<PixelInfo> m_CurrentPixelQueue{};
00253     std::vector<bool> m_CurrentPixelQueueDone{};
00255     std::vector<PixelInfo> m_AllPixels{};
00256
00257     // Initializes the first nxn screen in m_CurrentPixelQueue
00258     void InitializeFirstGrid();
00259
00260     // Updates the neighbors of all pixels (that should have neighbors and don't yet) in m_AllPixels
00261     void UpdateAllNeighbors();
00262
00263     // Updates all weights of the pixels in m_AllPixels with weight < 0 and subdiv > 0 and subdiv <
      m_MaxSubdivide
00264     // Assumes all squares have neighbors assigned correctly
00265     void UpdateAllWeights();
00266
00267     // This will take the pixel m_AllPixels[ind] and subdivide it,
00268     // adding up to <=5 pixels to the CurrentPixelQueue
00269     void SubdivideAndQueue(largecounter ind);
00270
00271     // Helper function to exponentiate an int to an int
00272     // Note: the result can be larger than fits in an int, but the base is always 2 and the exp is
      always
00273     // a number <=m_MaxSubdivide (which is an int)
00274     pixelcoord ExpInt(int base, int exp);
00275 };
00276
00286 class SquareSubdivisionMeshV2 : public Mesh
00287 {
00288 public:
00289     // default constructor not possible
00290     SquareSubdivisionMeshV2() = delete;
00291     // Constructor must be called with arguments:
00292     // - maxPixels: max. nr of pixels that can be integrated in TOTAL, over all iterations (if 0, then
      this is infinite,
00293     // i.e. we keep integrating until all squares are maximally subdivided or have weight 0)
00294     // - initialPixels: initial number of pixels to integrate (spaced equally over the screen)
00295     // - maxSubdivide: maximum number of times that we can subdivide squares (note: 1 denotes the
00296     // initial grid, so 2 would be subdividing the squares once)
00297     // - iterationPixels: maximum number of pixels to subdivide in each integration iteration
00298     // (max number of pixels that will be integrates is then 5*iterationPixels)
00299     // - initialSubToFinal: once we decide to subdivide a square, do we automatically keep subdividing
      it
00300     // until we reach maxSubdivision?
00301     // - valdiag: the "value" and "distance" Diagnostic to use
00302     SquareSubdivisionMeshV2(largecounter maxPixels, largecounter initialPixels, int maxSubdivide,
      largecounter iterationPixels, bool initialSubToFinal,
00303                             DiagBitflag valdiag)
00304         : m_InitialPixels{static_cast<pixelcoord>(sqrt(initialPixels)) *
      static_cast<pixelcoord>(sqrt(initialPixels))},
00305           m_MaxSubdivide{maxSubdivide},
00306           m_RowColumnSize{static_cast<pixelcoord>((sqrt(initialPixels) - 1) * ExpInt(2, maxSubdivide -
      1) + 1)},
00307           m_PixelsLeft{maxPixels}, m_MaxPixels{maxPixels}, m_InfinitePixels{maxPixels == 0},
      m_IterationPixels{iterationPixels},
00308           m_InitialSubDividideToFinal{initialSubToFinal}, Mesh(valdiag)
00309     {
00310         if constexpr (dimension != 4)
00311             ScreenOutput("SquareSubdivisionMeshV2 only defined in 4D!", OutputLevel::Level_0_WARNING);
00312
```

```
00313         // Initialize the initial square, equally spaced grid
00314         InitializeFirstGrid();
00315     }
00316
00317     // Declarations of overriding virtual functions
00318
00319     largecounter getCurNrGeodesics() const final;
00320
00321     void getNewInitConds(largecounter index, ScreenPoint &newunitpoint, ScreenIndex &newscreenindex)
      const final;
00322
00323     void GeodesicFinished(largecounter index, std::vector<real> finalValues) final;
00324
00325     void EndCurrentLoop() final;
00326
00327     bool IsFinished() const final;
00328
00329     // Description string getter
00330     std::string getFullDescriptionStr() const final;
00331
00332 private:
00334     const largecounter m_InitialPixels;
00336     const int m_MaxSubdivide;
00338     const pixelcoord m_RowColumnSize;
00340     const largecounter m_IterationPixels;
00342     const largecounter m_MaxPixels;
00344     const bool m_InitialSubDividideToFinal;
00346     const bool m_InfinitePixels;
00347
00349     largecounter m_PixelsLeft;
00350
00352     largecounter m_PixelsIntegrated{0};
00353
00357     struct PixelInfo
00358     {
00359         // Constructor with its ScreenIndex and current subdivision level
00360         PixelInfo(ScreenIndex ind, int subdiv) : Index{ind}, SubdivideLevel{subdiv} {}
00361
00363         const ScreenIndex Index{};
00364
00366         int SubdivideLevel{};
00367
00369         real Weight{-1};
00370
00372         std::vector<real> DiagValue{};
00373
00375         PixelInfo *LeftNbr{nullptr};
00377         PixelInfo *RightNbr{nullptr};
00379         PixelInfo *UpNbr{nullptr};
00381         PixelInfo *DownNbr{nullptr};
00383         PixelInfo *SEdiagNbr{nullptr};
00384     };
00385
00387     std::forward_list<std::unique_ptr<PixelInfo>> m_AllPixels{};
00388
00390     std::vector<PixelInfo *> m_ActivePixels{};
00392     std::vector<PixelInfo *> m_CurrentPixelQueue{};
00394     std::vector<bool> m_CurrentPixelQueueDone{};
00396     std::vector<PixelInfo *> m_CurrentPixelUpdating{};
00397
00398     // Initializes the first nxn screen and puts them in m_CurrentPixelQueue
00399     void InitializeFirstGrid();
00400
00401     // Updates all weights of the pixels in m_CurrentPixelUpdating;
00402     // these should have subdiv > 0 and subdiv < m_MaxSubdivide, and all their neigbors assigned
      correctly
00403     // All pixels with weight > 0 will be added to m_ActivePixels
00404     void UpdateAllWeights();
00405
00406     // Helper functions that return the appropriate neighbor of p, ONLY if this neighbor exists at the
      subdivision level specified
00407     // Returns nullptr otherwise; also return nullptr if p==nullptr
00408     PixelInfo *GetUp(PixelInfo *p, int subdiv) const;
00409     PixelInfo *GetDown(PixelInfo *p, int subdiv) const;
00410     PixelInfo *GetRight(PixelInfo *p, int subdiv) const;
00411     PixelInfo *GetLeft(PixelInfo *p, int subdiv) const;
00412
00413     // This will take the pixel m_AllPixels[ind] and subdivide it,
00414     // adding up to <=5 pixels to the CurrentPixelQueue
00415     void SubdivideAndQueue(largecounter ind);
00416
00417     // Helper function to exponentiate an int to an int
00418     // Note: the result can be larger than fits in an int, but the base is always 2 and the exp is
      always
00419     // a number <=m_MaxSubdivide (which is an int)
00420     pixelcoord ExpInt(int base, int exp) const;
00421 };
```

```
00422
00423 #endif
```

7.34 FOORT/src/Metric.cpp File Reference

```
#include "Metric.h"
#include "InputOutput.h"
#include "Integrators.h"
#include <cmath>
#include <algorithm>
#include <iomanip>
#include "Spline.h"
#include "Grid.h"
#include "Interpolator.h"
#include <sstream>
```

7.35 FOORT/src/Metric.h File Reference

Declarations of abstract base Metric class and all its descendants.

```
#include "Geometry.h"
#include "Spline.h"
#include "Grid.h"
#include "Interpolator.h"
#include <string>
#include <vector>
#include <memory>
```

Classes

- class Metric

  The abstract base class for all Metrics.
- class SphericalHorizonMetric

  Abstract base class for a metric with a spherical horizon (i.e. horizon at constant radius r)
- class KerrMetric

  The Kerr metric.
- class FlatSpaceMetric

  Flat space metric in 4D.
- class RasheedLarsenMetric

  Rasheed-Larsen black hole.
- class JohannsenMetric

  Johannsen black hole metric.
- class MankoNovikovMetric

  Mano-Novikov metric (with angular momentum and M3 parameter turned on)
- class KerrSchildMetric

  Kerr metric in Kerr-Schild coordinates.
- class SingularityMetric

  Abstract base class for a metric with an arbitrary number of singularities (of arbitrary codimension)
- class ST3CrMetric

  Ring fuzzball metric.
- class BosonStarMetric

  Boson star metric with solitonic potential.
- class RotatingBosonStarMetric

7.35.1 Detailed Description

Declarations of abstract base Metric class and all its descendants.

Author

Daniel R. Mayerson

Version

0.1

Date

2025-01-14

Copyright

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7.36 Metric.h

Go to the documentation of this file.

```
00001 #ifndef _FOORT_METRIC_H
00002 #define _FOORT_METRIC_H
00003
00004 #include "Geometry.h" // Needed for basic tensor objects etc.
00005
00006 #include "Spline.h"       // needed for spline interpolation, for the Boson star metric
00007 #include "Grid.h"       // needed for the grid class, for the Rotating Boson star metric
00008 #include "Interpolator.h" // needed for the Interpolator class, for the Rotating Boson star metric
00009 #include <string>         // for strings
00010 #include <vector>         // needed for the (non-fixed size) vector of symmetries in the metric
00011 #include <memory>         // for std::unique_ptr, for the Rotating Boson star metric's
      grids/interpolators
00012
00021
00026 class Metric
00027 {
00028 public:
00029     // Virtual destructor to ensure correct destruction of descendants
00030     virtual ~Metric() = default;
00031
00032     Metric(bool rlogscale = false);
00033
00034     // Basic functions that return the metric with indices down or up:
00035     // pure virtual as they must be defined in the descendant classes.
00036     //
00037     // Get the metric at Point p, indices down
00038     virtual TwoIndex getMetric_dd(const Point &p) const = 0;
00039     // Get the metric at Point p, indices up
00040     virtual TwoIndex getMetric_uu(const Point &p) const = 0;
00041
00042     // The following functions return the Christoffel and other derivative quantities of the metric.
00043     // They are implemented for this base class, BUT are left as virtual functions to allow for
00044     // other metrics to implement their own (more efficient)
00045     // way of calculating them, if so desired.
00046     //
00047     // Get the Christoffel symbol, indices up-down-down
00048     virtual ThreeIndex getChristoffel_udd(const Point &p) const;
00049     // Get the Riemann tensor, indices up-down-down-down
00050     virtual FourIndex getRiemann_uddd(const Point &p) const;
00051     // Get the Kretschmann scalar
00052     virtual real getKretschmann(const Point &p) const;
00053
00054     // Function to get the description of the metric
00055     // (used for outputting to the screen while running and possibly to the output files)
00056     // There is a base class implementation of this function returning an undescriptive string
```

```
00057     virtual std::string getFullDescriptionStr() const;
00058
00059     bool getrLogScale() const;
00060
00061 protected:
00063     std::vector<int> m_Symmetries{};
00065     const bool m_rLogScale;
00066 };
00067
00071 class SphericalHorizonMetric : public Metric
00072 {
00073 public:
00074     // No default construction allowed, must specify horizon radius
00075     SphericalHorizonMetric() = delete;
00076     // Constructor that initializes horizon radius
00077     SphericalHorizonMetric(real HorizonRadius, bool rLogScale);
00078
00079     // Getter functions for the two member variables
00080     real getHorizonRadius() const;
00081
00082 protected:
00084     const real m_HorizonRadius;
00085 };
00086
00090 class KerrMetric final : public SphericalHorizonMetric
00091 {
00092 private:
00094     const real m_aParam;
00095
00097     const real m_mParam;
00098
00099 public:
00100     // No default constructor allowed, must specify a
00101     KerrMetric() = delete;
00102
00103     // Constructor setting parameter a
00104     KerrMetric(real aParam, bool rLogScale = false, real mParam = 1.);
00105
00106     // The override of the basic metric getter functions
00107     TwoIndex getMetric_dd(const Point &p) const final;
00108     TwoIndex getMetric_uu(const Point &p) const final;
00109
00110     // The override of the description string getter
00111     std::string getFullDescriptionStr() const final;
00112 };
00113
00117 class FlatSpaceMetric final : public Metric
00118 {
00119 public:
00120     // Simple (default) constructor is all that is needed
00121     FlatSpaceMetric(bool rlogscale = false);
00122
00123     // The override of the basic metric getter functions
00124     TwoIndex getMetric_dd(const Point &p) const final;
00125     TwoIndex getMetric_uu(const Point &p) const final;
00126
00127     // The override of the description string getter
00128     std::string getFullDescriptionStr() const final;
00129 };
00130
00134 class RasheedLarsenMetric final : public SphericalHorizonMetric
00135 {
00136 private:
00138     const real m_aParam;
00140     const real m_mParam;
00142     const real m_pParam;
00144     const real m_qParam;
00145
00146 public:
00147     // No default constructor allowed, must specify parameters
00148     RasheedLarsenMetric() = delete;
00149
00150     // Constructor setting parameter a
00151     RasheedLarsenMetric(real mParam, real aParam, real pParam, real qParam, bool rLogScale = false);
00152
00153     // The override of the basic metric getter functions
00154     TwoIndex getMetric_dd(const Point &p) const final;
00155     TwoIndex getMetric_uu(const Point &p) const final;
00156
00157     // The override of the description string getter
00158     std::string getFullDescriptionStr() const final;
00159 };
00160
00165 class JohannsenMetric final : public SphericalHorizonMetric
00166 {
00167 private:
00168     // Johannsen up to first order in deviation function is specified by five parameters (if M=1)
```

```
00170     const real m_aParam;
00172     const real m_alpha13Param;
00174     const real m_alpha22Param;
00176     const real m_alpha52Param;
00178     const real m_eps3Param;
00179
00180 public:
00181     // No default constructor allowed, must specify parameters
00182     JohannsenMetric() = delete;
00183
00184     // Constructor setting parameter a
00185     JohannsenMetric(real aParam, real alpha13Param, real alpha22Param, real alpha52Param, real
      eps3Param, bool rLogScale = false);
00186
00187     // The override of the basic metric getter functions
00188     TwoIndex getMetric_dd(const Point &p) const final;
00189     TwoIndex getMetric_uu(const Point &p) const final;
00190
00191     // The override of the description string getter
00192     std::string getFullDescriptionStr() const final;
00193 };
00194
00199 class MankoNovikovMetric final : public SphericalHorizonMetric
00200 {
00201 private:
00202     // Manko-Novikov metric with only alpha3 as symmetry breaking parameter
00204     const real m_aParam;
00206     const real m_alpha3Param;
00207
00208     // These are convenient derived quantities from a
00210     const real m_alphaParam;
00212     const real m_kParam;
00213
00214 public:
00215     // No default constructor allowed, must specify parameters
00216     MankoNovikovMetric() = delete;
00217
00218     // Constructor setting parameter a and alpha3
00219     MankoNovikovMetric(real aParam, real alpha3Param, bool rLogScale = false);
00220
00221     // The override of the basic metric getter functions
00222     TwoIndex getMetric_dd(const Point &p) const final;
00223     TwoIndex getMetric_uu(const Point &p) const final;
00224
00225     // The override of the description string getter
00226     std::string getFullDescriptionStr() const final;
00227 };
00228
00233 class KerrSchildMetric final : public SphericalHorizonMetric
00234 {
00235 private:
00237     const real m_aParam;
00238
00239 public:
00240     // No default constructor allowed, must specify a
00241     KerrSchildMetric() = delete;
00242
00243     // Constructor setting parameter a
00244     KerrSchildMetric(real aParam, bool rLogScale = false);
00245
00246     // The override of the basic metric getter functions
00247     TwoIndex getMetric_dd(const Point &p) const final;
00248     TwoIndex getMetric_uu(const Point &p) const final;
00249
00250     // The override of the description string getter
00251     std::string getFullDescriptionStr() const final;
00252 };
00253
00257 class SingularityMetric : public Metric
00258 {
00259 public:
00260     // Constructor that initializes singularities
00261     SingularityMetric(std::vector<Singularity> thesings, bool rLogScale);
00262
00263     // Getter functions for the two member variables
00264     std::vector<Singularity> getSingularities() const;
00265
00266 protected:
00268     const std::vector<Singularity> m_AllSingularities;
00269 };
00270
00275 class ST3CrMetric final : public SingularityMetric
00276 {
00277 public:
00278     // Constructor which will be called to initialize all parameters of the metric
00279     ST3CrMetric(real P, real q0, real lambda, bool rlogscale = false);
00280
```

```
00281     // The basic getter functions
00282     TwoIndex getMetric_dd(const Point &p) const final;
00283     TwoIndex getMetric_uu(const Point &p) const final;
00284 
00285     // The description string getter
00286     std::string getFullDescriptionStr() const final;
00287 
00288 private:
00289     const real m_P;
00290     const real m_q0;
00291     const real m_lambda;
00292 
00293     real get_omega(real r, real theta, real l) const;
00294     real f_phi(real phi, real r, real theta, real l, real R) const;
00295     real f_om_phi(real phi, real r, real theta, real l, real R) const;
00296 };
00297 
00303 class BosonStarMetric final : public Metric
00304 {
00305 public:
00306     // Simple (default) constructor is all that is needed
00307     BosonStarMetric(double Phi_infinity, int num_lines, bool rLogScale = false,
00308                     std::string Phi_filename = "Phi.dat", std::string m_filename = "m.dat");
00309     // The override of the basic metric getter functions
00310     TwoIndex getMetric_dd(const Point &p) const final;
00311     TwoIndex getMetric_uu(const Point &p) const final;
00312     // The override of the description string getter
00313     std::string getFullDescriptionStr() const final;
00314 
00315 protected:
00317     const double m_Phi_infinity;
00320     const int m_num_lines;
00321 
00323     std::string m_Phi_filename;
00325     std::string m_m_filename;
00326 
00328     tk::spline m_PhiSpline;
00330     tk::spline m_mSpline;
00331 
00333     void read_data();
00334 };
00335 
00336 class RotatingBosonStarMetric final : public Metric
00337 {
00338 public:
00339     // Simple (default) constructor is all that is needed
00340     RotatingBosonStarMetric(bool rLogScale = false, std::string MetricFolder =
      "RotatingBosonStar/data_Will/", int num_x = 500, int num_th = 399, bool FlipAngularMomentum = false);
00341     ~RotatingBosonStarMetric() = default;
00342 
00343     // The override of the basic metric getter functions
00344     TwoIndex getMetric_dd(const Point &p) const final;
00345     TwoIndex getMetric_uu(const Point &p) const final;
00346     // The override of the description string getter
00347     std::string getFullDescriptionStr() const final;
00348 
00349 private:
00351     std::unique_ptr<Grid> m_grid_f;
00352     std::unique_ptr<Grid> m_grid_l;
00353     std::unique_ptr<Grid> m_grid_g;
00354     std::unique_ptr<Grid> m_grid_Omega;
00355 
00356 public:
00358     std::unique_ptr<BicubicSplineInterpolator> m_fInterpolator;
00359     std::unique_ptr<BicubicSplineInterpolator> m_lInterpolator;
00360     std::unique_ptr<BicubicSplineInterpolator> m_gInterpolator;
00361     std::unique_ptr<BicubicSplineInterpolator> m_OmegaInterpolator;
00362 
00363     // Sign multiplier for Omega: +1 for normal rotation, -1 for flipped rotation
00364     const int m_OmegaSign;
00365 };
00366 
00368 // Declare your new Metric class here, publically inheriting from the base class Metric
00369 // (or SphericalHorizonMetric if your Metric has a horizon, or SingularityMetric if your Metric has
      other, arbitrary singularities)
00370 // Give definitions (implementation) of these functions in Metric.cpp (or other source code file)
00371 // Don't forget to set m_Symmetries appropriately (in the constructor),
00372 // if your metric has any symmetry (e.g. stationarity, axisymmetry)!
00373 // Sample code:
00374 /*
00375 class MyMetric final : public Metric // good practice to make the class final unless descendant
      classes are possible
00376 {
00377 public:
00378     // Constructor which will be called to initialize all parameters of the metric
00379     MyMetric(args...);
00380 
```

```
00381     // The basic getter functions
00382     // These MUST be implemented
00383     TwoIndex getMetric_dd(const Point& p) const final;
00384     TwoIndex getMetric_uu(const Point& p) const final;
00385
00386     // The description string getter
00387     // This is optional (but recommended) to implement; if not implemented,
00388     // the base class Metric::getFullDescriptionStr() will be called instead
00389     std::string getFullDescriptionStr() const final;
00390
00391 private:
00392     // good practice to have all const params (initialized in the constructor)
00393     // since the metric cannot change after initialization
00394     // const params...;
00395
00396 };
00397 */
00399
00400 #endif
```

7.37 FOORT/src/Spline.cpp File Reference

Simple cubic spline interpolation library without external dependencies.

```
#include "Spline.h"
#include <cstdio>
#include <cassert>
#include <cmath>
#include <vector>
#include <algorithm>
```

7.37.1 Detailed Description

Simple cubic spline interpolation library without external dependencies.

This library is not made by the authors of FOORT, but implemented as found in the following link: https://kluge.in-chemnitz.de/opensource/spline/. The main body has been separated into Spline.cpp, however.

Author

Tino Kluge

Date

2021

Copyright

GNU General Public License

7.38 FOORT/src/Spline.h File Reference

Simple cubic spline interpolation library without external dependencies.

```
#include <cstdio>
#include <cassert>
#include <cmath>
#include <vector>
#include <algorithm>
```

Classes

- class tk::spline
- class tk::internal::band_matrix

Namespaces

- namespace tk
- namespace tk::internal

7.38.1 Detailed Description

Simple cubic spline interpolation library without external dependencies.

This library is not made by the authors of FOORT, but implemented as found in the following link: https://kluge.in-chemnitz.de/opensource/spline/. The main body has been separated into Spline.cpp, however.

Author

Tino Kluge

Date

2021

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GNU General Public License

7.39 Spline.h

Go to the documentation of this file.

```
00001 /*
00002  * spline.h
00003  *
00004  * simple cubic spline interpolation library without external
00005  * dependencies
00006  *
00007  * ---------------------------------------------------------------------
00008  * Copyright (C) 2011, 2014, 2016, 2021 Tino Kluge (ttk448 at gmail.com)
00009  *
00010  *  This program is free software; you can redistribute it and/or
00011  *  modify it under the terms of the GNU General Public License
00012  *  as published by the Free Software Foundation; either version 2
00013  *  of the License, or (at your option) any later version.
00014  *
00015  *  This program is distributed in the hope that it will be useful,
00016  *  but WITHOUT ANY WARRANTY; without even the implied warranty of
00017  *  MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  See the
00018  *  GNU General Public License for more details.
00019  *
00020  *  You should have received a copy of the GNU General Public License
00021  *  along with this program.  If not, see <http://www.gnu.org/licenses/>.
00022  * ---------------------------------------------------------------------
00023  *
00024  */
00025
00035
00036 #ifndef TK_SPLINE_H
```

```
00037 #define TK_SPLINE_H
00038
00039 #include <cstdio>
00040 #include <cassert>
00041 #include <cmath>
00042 #include <vector>
00043 #include <algorithm>
00044 #ifdef HAVE_SSTREAM
00045 #include <sstream>
00046 #include <string>
00047 #endif // HAVE_SSTREAM
00048
00049 // not ideal but disable unused-function warnings
00050 // (we get them because we have implementations in the header file,
00051 // and this is because we want to be able to quickly separate them
00052 // into a cpp file if necessary)
00053 #pragma GCC diagnostic push
00054 #pragma GCC diagnostic ignored "-Wunused-function"
00055 namespace tk
00056 {
00057     // spline interpolation
00058     class spline
00059     {
00060     public:
00061         // spline types
00062         enum spline_type
00063         {
00064             linear = 10,         // linear interpolation
00065             cspline = 30,        // cubic splines (classical C^2)
00066             cspline_hermite = 31 // cubic hermite splines (local, only C^1)
00067         };
00068
00069         // boundary condition type for the spline end-points
00070         enum bd_type
00071         {
00072             first_deriv = 1,
00073             second_deriv = 2
00074         };
00075
00076     protected:
00077         std::vector<double> m_x, m_y; // x,y coordinates of points
00078         // interpolation parameters
00079         // f(x) = a_i + b_i*(x-x_i) + c_i*(x-x_i)^2 + d_i*(x-x_i)^3
00080         // where a_i = y_i, or else it won't go through grid points
00081         std::vector<double> m_b, m_c, m_d; // spline coefficients
00082         double m_c0;                       // for left extrapolation
00083         spline_type m_type;
00084         bd_type m_left, m_right;
00085         double m_left_value, m_right_value;
00086         bool m_made_monotonic;
00087         void set_coeffs_from_b();            // calculate c_i, d_i from b_i
00088         size_t find_closest(double x) const; // closest idx so that m_x[idx]<=x
00089
00090     public:
00091         // default constructor: set boundary condition to be zero curvature
00092         // at both ends, i.e. natural splines
00093         spline();
00094
00095         spline(const std::vector<double> &X, const std::vector<double> &Y,
00096                spline_type type = cspline,
00097                bool make_monotonic = false,
00098                bd_type left = second_deriv, double left_value = 0.0,
00099                bd_type right = second_deriv, double right_value = 0.0);
00100
00101         // modify boundary conditions: if called it must be before set_points()
00102         void set_boundary(bd_type left, double left_value,
00103                           bd_type right, double right_value);
00104
00105         // set all data points (cubic_spline=false means linear interpolation)
00106         void set_points(const std::vector<double> &x,
00107                         const std::vector<double> &y,
00108                         spline_type type = cspline);
00109
00110         // adjust coefficients so that the spline becomes piecewise monotonic
00111         // where possible
00112         //   this is done by adjusting slopes at grid points by a non-negative
00113         //   factor and this will break C^2
00114         //   this can also break boundary conditions if adjustments need to
00115         //   be made at the boundary points
00116         // returns false if no adjustments have been made, true otherwise
00117         bool make_monotonic();
00118
00119         // evaluates the spline at point x
00120         double operator()(double x) const;
00121         double deriv(int order, double x) const;
00122
00123         // returns the input data points
```

```
00124         std::vector<double> get_x() const;
00125         std::vector<double> get_y() const;
00126         double get_x_min() const;
00127         double get_x_max() const;
00128 
00129 #ifdef HAVE_SSTREAM
00130         // spline info string, i.e. spline type, boundary conditions etc.
00131         std::string info() const;
00132 #endif // HAVE_SSTREAM
00133     };
00134 
00135     namespace internal
00136     {
00137 
00138         // band matrix solver
00139         class band_matrix
00140         {
00141         private:
00142             std::vector<std::vector<double> m_upper; // upper band
00143             std::vector<std::vector<double> m_lower; // lower band
00144         public:
00145             band_matrix() {};                            // constructor
00146             band_matrix(int dim, int n_u, int n_l); // constructor
00147             ~band_matrix() {};                           // destructor
00148             void resize(int dim, int n_u, int n_l); // init with dim,n_u,n_l
00149             int dim() const;                             // matrix dimension
00150             int num_upper() const;
00151             int num_lower() const;
00152             // access operator
00153             double &operator()(int i, int j);      // write
00154             double operator()(int i, int j) const; // read
00155             // we can store an additional diagonal (in m_lower)
00156             double &saved_diag(int i);
00157             double saved_diag(int i) const;
00158             void lu_decompose();
00159             std::vector<double> r_solve(const std::vector<double> &b) const;
00160             std::vector<double> l_solve(const std::vector<double> &b) const;
00161             std::vector<double> lu_solve(const std::vector<double> &b,
00162                                          bool is_lu_decomposed = false);
00163         };
00164 
00165     } // namespace internal
00166 } // namespace tk
00167 #pragma GCC diagnostic pop
00168 
00169 #endif /* TK_SPLINE_H */
```

7.40 FOORT/src/Terminations.cpp File Reference

```
#include "Terminations.h"
#include "Geodesic.h"
#include "InputOutput.h"
#include <cmath>
```

Functions

- TerminationUniqueVector CreateTerminationVector (TermBitflag termflags, Geodesic *const theGeodesic)

  Create a Termination Vector object.

7.40.1 Detailed Description

Author

Daniel R. Mayerson

Version

0.1

Date

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Copyright

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7.40.2 Function Documentation

7.40.2.1 CreateTerminationVector()

```
TerminationUniqueVector CreateTerminationVector (
            TermBitflag termflags,
            Geodesic *const theGeodesic)
```

Create a Termination Vector object.

Parameters

termflags	bitflags for the terminations
theGeodesic	the geodesic

Returns

TerminationUniqueVector

7.41 FOORT/src/Terminations.h File Reference

Declarations of abstract base Termination class and all its descendants.

```
#include "Geometry.h"
#include <cstdint>
#include <memory>
#include <vector>
#include <utility>
```

Classes

- class Termination

  Abstract base class for all Terminations.
- class HorizonTermination

  HorizonTermination: Terminate geodesics if they get too close to the horizon.
- class BoundarySphereTermination

  BoundarySphereTermination: Terminate geodesics if they reach outside of a boundary sphere.
- class TimeOutTermination

  TimeOutTermination: Terminate geodesics if they take too many steps.
- class ThetaSingularityTermination

  ThetaSingularityTermination: Terminate geodesics if they get too close to a polar coordinate singularity.
- class NaNTermination

  NaNTermination: Terminate geodesics if they contain a NaN in position or velocity.
- class GeneralSingularityTermination

  GeneralSingularityTermination: Terminate geodesics if they get too close to one of a number of singularities (of arbitrary codimension)
- struct TerminationOptions

  Base class for all TerminationOptions structs. Other TerminationOptions can inherit from here if they require more options.
- struct HorizonTermOptions

  Options for HorizonTermination.
- struct BoundarySphereTermOptions

  Options for BoundarySphereTermination.
- struct TimeOutTermOptions

  Options for TimeOutTermination.
- struct ThetaSingularityTermOptions

  Options for ThetaSingularityTermination.
- struct NaNTermOptions

  Options for NaNTermination.
- struct GeneralSingularityTermOptions

  Options for GeneralSingularityTermination.

Typedefs

- using TermBitflag = std::uint16_t

  Used for constructing vector of Terminations. Note that this means every Termination is either "on" or "off"; it is not possible to have a Termination "on" more than once.
- using TerminationUniqueVector = std::vector<std::unique_ptr<Termination>>

  The owner vector of derived Termination classes.

Enumerations

- enum class Term {
  Uninitialized = -1 , Continue = 0 , Horizon , BoundarySphere ,
  TimeOut , ThetaSingularity , NaN , GeneralSingularity ,
  Maxterms }

  Possible termination conditions that can be set by Terminations.

Functions

- TerminationUniqueVector CreateTerminationVector (TermBitflag termflags, Geodesic ∗const theGeodesic)
  Create a Termination Vector object.

Variables

- constexpr TermBitflag Term_None {0b0000'0000'0000'0000}
  No terminations bitflag.
- constexpr TermBitflag Term_BoundarySphere {0b0000'0000'0000'0001}
  Boundary Sphere termination bitflag.
- constexpr TermBitflag Term_TimeOut {0b0000'0000'0000'0010}
  Time Out termination bitflag.
- constexpr TermBitflag Term_Horizon {0b0000'0000'0000'0100}
  Horizon termination bitflag.
- constexpr TermBitflag Term_ThetaSingularity {0b0000'0000'0000'1000}
  Theta Singularity termination bitflag.
- constexpr TermBitflag Term_NaN {0b0000'0000'0001'0000}
  NaN termination bitflag.
- constexpr TermBitflag Term_GeneralSingularity {0b0000'0000'0010'0000}
  General Singularity termination bitflag.

7.41.1 Detailed Description

Declarations of abstract base Termination class and all its descendants.

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Copyright

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7.41.2 Typedef Documentation

7.41.2.1 TermBitflag

```
using TermBitflag = std::uint16_t
```

Used for constructing vector of Terminations. Note that this means every Termination is either "on" or "off"; it is not possible to have a Termination "on" more than once.

7.41.2.2 TerminationUniqueVector

```
using TerminationUniqueVector = std::vector<std::unique_ptr<Termination>>
```

The owner vector of derived Termination classes.

7.41.3 Enumeration Type Documentation

7.41.3.1 Term

```
enum class Term  [strong]
```

Possible termination conditions that can be set by Terminations.

Enumerator

Uninitialized	Geoedesic has not been properly initialized yet with initial position/velocity.
Continue	All is right, continue integrating geodesic.
Horizon	STOP, encountered horizon (set by HorizonTermination)
BoundarySphere	STOP, encountered boundary sphere (set by BoundarySphereTermination)
TimeOut	STOP, taken too many steps (set by TimeOutTermination)
ThetaSingularity	STOP, too close to polar coordinate singularity (theta = 0 or theta = pi/2) (set by ThetaSingularityTermination)
NaN	STOP, NaN encountered in geodesic position or velocity (set by NaNTermination)
GeneralSingularity	STOP, singularity encountered (of any codimension) (set by GeneralSingularityTermination)
Maxterms	

7.41.4 Function Documentation

7.41.4.1 CreateTerminationVector()

```
TerminationUniqueVector CreateTerminationVector (
            TermBitflag termflags,
            Geodesic *const theGeodesic)
```

Create a Termination Vector object.

Parameters

termflags	bitflags for the terminations
theGeodesic	the geodesic

Returns

TerminationUniqueVector

7.41.5 Variable Documentation

7.41.5.1 Term_BoundarySphere

```
TermBitflag Term_BoundarySphere {0b0000'0000'0000'0001}  [constexpr]
```

Boundary Sphere termination bitflag.

7.41.5.2 Term_GeneralSingularity

```
TermBitflag Term_GeneralSingularity {0b0000'0000'0010'0000}  [constexpr]
```

General Singularity termination bitflag.

7.41.5.3 Term_Horizon

```
TermBitflag Term_Horizon {0b0000'0000'0000'0100}  [constexpr]
```

Horizon termination bitflag.

7.41.5.4 Term_NaN

```
TermBitflag Term_NaN {0b0000'0000'0001'0000}  [constexpr]
```

NaN termination bitflag.

7.41.5.5 Term_None

```
TermBitflag Term_None {0b0000'0000'0000'0000}  [constexpr]
```

No terminations bitflag.

7.41.5.6 Term_ThetaSingularity

```
TermBitflag Term_ThetaSingularity {0b0000'0000'0000'1000}  [constexpr]
```

Theta Singularity termination bitflag.

7.41.5.7 Term_TimeOut

```
TermBitflag Term_TimeOut {0b0000'0000'0000'0010}  [constexpr]
```

Time Out termination bitflag.

7.42 Terminations.h

Go to the documentation of this file.

```
00001 #ifndef _FOORT_TERMINATIONS_H
00002 #define _FOORT_TERMINATIONS_H
00003
00012
00013 #include "Geometry.h" // for basic tensor objects
00014
00015 #include <cstdint> // for std::uint16_t
00016 #include <memory>  // for std::unique_ptr
00017 #include <vector>  // for std::vector
00018 #include <utility> // for std::pair
00019
00020 // Forward declaration of Geodesic class needed here, since Diagnostics are passed a pointer to their
      owner Geodesic
00021 // (note "Geodesic.h" is NOT included to avoid header loop, and we do not need Geodesic member
      functions here!)
00022 class Geodesic;
00023
00024 // TERMINATION BITFLAGS
00026 using TermBitflag = std::uint16_t;
00027
00028 // Define a bitflag per existing Termination
00030 constexpr TermBitflag Term_None{0b0000'0000'0000'0000};
00032 constexpr TermBitflag Term_BoundarySphere{0b0000'0000'0000'0001};
00034 constexpr TermBitflag Term_TimeOut{0b0000'0000'0000'0010};
00036 constexpr TermBitflag Term_Horizon{0b0000'0000'0000'0100};
00038 constexpr TermBitflag Term_ThetaSingularity{0b0000'0000'0000'1000};
00040 constexpr TermBitflag Term_NaN{0b0000'0000'0001'0000};
00042 constexpr TermBitflag Term_GeneralSingularity{0b0000'0000'0010'0000};
00043
00045 // Add a TermBitflag for your new Termination. Make sure you use a bitflag that has not been used
      before!
00046 // Sample code:
00047 /*
00048 constexpr TermBitflag Term_MyTerm               { 0b0000'0000'0000'1000 };
00049 */
00051
00052 // TERMINATION CONDITIONS
00053
00057 enum class Term
00058 {
00060     Uninitialized = -1,
00062     Continue = 0,
00064     Horizon,
00066     BoundarySphere,
00068     TimeOut,
00070     ThetaSingularity,
00072     NaN,
00074     GeneralSingularity,
00075
00077     // Add a new Termination condition that your new Termination can set
00078     // Sample code:
00079     /*
00080     MyTermCond,         // STOP, encountered (...)
00081     */
00083
00084     Maxterms // Number of termination conditions that exist
00085 };
00086
00087 // GENERAL DECLARATIONS OF ABSTRACT BASE CLASS
00088
00092 class Termination
00093 {
00094 public:
00095     // Constructor must initialize the pointer to its owner Geodesic
00096     Termination() = delete;
00097     Termination(Geodesic *const theGeodesic) : m_OwnerGeodesic{theGeodesic}
00098     {
00099     }
00100
00101     // Resets Termination object. This is called when the owner Geodesic is reset in order to start
      integrating
00102     // a new geodesic.
00103     // The base class implementation only resets m_StepsSinceUpdated
00104     // Descendants can override this if they need to reset additional internal variables
00105     virtual void Reset();
00106
00107     // virtual destructor to ensure correct destruction of descendants
00108     virtual ~Termination() = default;
00109
00110     // Function that is called to determine whether Termination wants to
00111     // terminate the Geodesic. Returns Term::Continue if no termination wanted,
00112     // otherwise it returns the appropriate Term condition
```

```
00113     virtual Term CheckTermination() = 0;
00114
00115     // This returns the full description of the Termination
00116     virtual std::string getFullDescriptionStr() const = 0;
00117
00118 protected:
00120     Geodesic *const m_OwnerGeodesic;
00121
00122     // Helper function to decide if the Termination should indeed update its status, based on
00123     // UpdateNSteps (which is set to 0 if we always update)
00124     bool DecideUpdate(largecounter UpdateNSteps);
00125
00127     largecounter m_StepsSinceUpdated{};
00128 };
00129
00131 using TerminationUniqueVector = std::vector<std::unique_ptr<Termination>>;
00132
00133 // Helper to create a new vector of Termination options, based on the bitflag
00134 TerminationUniqueVector CreateTerminationVector(TermBitflag termflags, Geodesic *const theGeodesic);
00135
00136 // DECLARATIONS FOR DERIVED TERMINATION CLASSES (DESCENDANTS)
00137
00138 // Forward declaration needed before Termination
00139 struct HorizonTermOptions;
00140
00144 class HorizonTermination final : public Termination
00145 {
00146 public:
00147     // Basic constructor only passes on Geodesic pointer to base class constructor
00148     HorizonTermination(Geodesic *const theGeodesic) : Termination(theGeodesic) {}
00149
00150     // Check if we are too close to the horizon
00151     Term CheckTermination() final;
00152
00153     // Description string
00154     std::string getFullDescriptionStr() const final;
00155
00157     static std::unique_ptr<HorizonTermOptions> TermOptions;
00158 };
00159
00160 // Forward declaration needed before Termination
00161 struct BoundarySphereTermOptions;
00162
00166 class BoundarySphereTermination final : public Termination
00167 {
00168 public:
00169     // Basic constructor only passes on Geodesic pointer to base class constructor
00170     BoundarySphereTermination(Geodesic *const theGeodesic) : Termination(theGeodesic) {}
00171
00172     // Check if we have passed the boundary sphere
00173     Term CheckTermination() final;
00174
00175     // Description string
00176     std::string getFullDescriptionStr() const final;
00177
00179     static std::unique_ptr<BoundarySphereTermOptions> TermOptions;
00180 };
00181
00182 // Forward declaration needed before Termination
00183 struct TimeOutTermOptions;
00184
00188 class TimeOutTermination final : public Termination
00189 {
00190 public:
00191     // Basic constructor only passes on Geodesic pointer to base class constructor
00192     TimeOutTermination(Geodesic *const theGeodesic) : Termination(theGeodesic) {}
00193
00194     // This descendant needs to override Reset in order to also reset m_CurNrSteps
00195     void Reset() final;
00196
00197     // Check if we have already taken too many steps
00198     Term CheckTermination() final;
00199
00200     // Description string
00201     std::string getFullDescriptionStr() const final;
00202
00204     static std::unique_ptr<TimeOutTermOptions> TermOptions;
00205
00206 private:
00208     largecounter m_CurNrSteps{0};
00209 };
00210
00211 // Forward declaration needed before Termination
00212 struct ThetaSingularityTermOptions;
00213
00217 class ThetaSingularityTermination final : public Termination
00218 {
```

```
00219 public:
00220     ThetaSingularityTermination(Geodesic *const theGeodesic) : Termination(theGeodesic) {}
00221
00222     // No override of Reset() necessary
00223
00224     // Check the specific termination condition
00225     Term CheckTermination() final;
00226
00227     // Description string
00228     std::string getFullDescriptionStr() const final;
00229
00231     static std::unique_ptr<ThetaSingularityTermOptions> TermOptions;
00232 };
00233
00234 // Forward declaration needed before Termination
00235 struct NaNTermOptions;
00236
00240 class NaNTermination final : public Termination
00241 {
00242 public:
00243     // Basic constructor only passes on Geodesic pointer to base class constructor
00244     NaNTermination(Geodesic *const theGeodesic) : Termination(theGeodesic) {}
00245
00246     // Check if we are too close to the horizon
00247     Term CheckTermination() final;
00248
00249     // Description string
00250     std::string getFullDescriptionStr() const final;
00251
00253     static std::unique_ptr<NaNTermOptions> TermOptions;
00254 };
00255
00256 // Forward declaration needed before Termination
00257 struct GeneralSingularityTermOptions;
00258
00262 class GeneralSingularityTermination final : public Termination
00263 {
00264 public:
00265     // Basic constructor only passes on Geodesic pointer to base class constructor
00266     GeneralSingularityTermination(Geodesic *const theGeodesic) : Termination(theGeodesic) {}
00267
00268     // Check if we are too close to the horizon
00269     Term CheckTermination() final;
00270
00271     // Description string
00272     std::string getFullDescriptionStr() const final;
00273
00275     static std::unique_ptr<GeneralSingularityTermOptions> TermOptions;
00276
00277 private:
00278     std::string SingularityToString(int singnr) const;
00279 };
00280
00282 // Declare your Termination class here, inheriting from Termination.
00283 // Sample code:
00284 /*
00285 // Forward declaration needed before Termination
00286 struct TerminationOptions; // possibly will instead need to declare descendant options struct
00287 class MyTermination final : public Termination // good practice to make the class final unless
    descendant classes are possible
00288 {
00289 public:
00290     // Constructor must at least pass on Geodesic pointer to base class constructor
00291     MyTermination(Geodesic* const theGeodesic) : Termination(theGeodesic) {}
00292
00293     // Do you need to reset any internal variables specific to MyTermination? If so, override Reset()
    (This is not mandatory)
00294     // Note: make sure to call the base class implementation Termination::Reset()
00295     // from within your implementation of MyTermination::Reset(), so that the base class internal
    variable is also reset!
00296     void Reset() final;
00297
00298     // Check the specific termination condition
00299     Term CheckTermination() final;
00300
00301     // Description string
00302     std::string getFullDescriptionStr() const final;
00303
00304     // The options that the Termination keeps (will probably be a descendant struct instead, which
    specifies
00305     // any additional options the Termination needs)
00306     static std::unique_ptr<TerminationOptions> TermOptions;
00307
00308 private:
00309     // any private member variables that are needed to keep track of things
00310 };
00311 */
```

```
00313
00314 // ALL TERMINATIONOPTIONS STRUCTS
00315
00319 struct TerminationOptions
00320 {
00321 public:
00322     // Basic constructor only sets the number of steps between updates
00323     TerminationOptions(largecounter Nsteps) : UpdateEveryNSteps{Nsteps}
00324     {
00325     }
00326
00327     // virtual destructor to ensure correct destruction of descendants
00328     virtual ~TerminationOptions() = default;
00329
00331     const largecounter UpdateEveryNSteps;
00332 };
00333
00338 struct HorizonTermOptions : public TerminationOptions
00339 {
00340 public:
00341     HorizonTermOptions(real theHorizonRadius, bool therLogScale, real theAtHorizonEps, largecounter
      Nsteps) : HorizonRadius{theHorizonRadius}, AtHorizonEps{theAtHorizonEps}, rLogScale{therLogScale},
      TerminationOptions(Nsteps)
00342     {
00343     }
00344
00346     const real HorizonRadius;
00348     const real AtHorizonEps;
00350     const bool rLogScale;
00351 };
00352
00357 struct BoundarySphereTermOptions : public TerminationOptions
00358 {
00359 public:
00360     BoundarySphereTermOptions(real theRadius, bool therLogScale, largecounter Nsteps) :
      SphereRadius{theRadius}, rLogScale{therLogScale},
00361
      TerminationOptions(Nsteps)
00362     {
00363     }
00364
00366     const real SphereRadius;
00368     const bool rLogScale;
00369 };
00370
00375 struct TimeOutTermOptions : public TerminationOptions
00376 {
00377 public:
00378     TimeOutTermOptions(largecounter MaxStepsAllowed, largecounter Nsteps) : MaxSteps{MaxStepsAllowed},
      TerminationOptions(Nsteps)
00379     {
00380     }
00381
00383     const largecounter MaxSteps;
00384 };
00385
00386 // Options class for ThetaSingularityTermination
00391 struct ThetaSingularityTermOptions : public TerminationOptions
00392 {
00393 public:
00394     ThetaSingularityTermOptions(real epsilon, largecounter Nsteps) : ThetaSingEpsilon{epsilon},
      TerminationOptions(Nsteps)
00395     {
00396     }
00397
00399     const real ThetaSingEpsilon;
00400 };
00401
00405 struct NaNTermOptions : public TerminationOptions
00406 {
00407 public:
00408     NaNTermOptions(bool consoleoutputon, largecounter Nsteps) : OutputToConsole{consoleoutputon},
      TerminationOptions(Nsteps)
00409     {
00410     }
00411
00413     const bool OutputToConsole;
00414 };
00415
00420 struct GeneralSingularityTermOptions : public TerminationOptions
00421 {
00422 public:
00423     GeneralSingularityTermOptions(std::vector<Singularity> sings,
00424                                   real eps, bool consoleoutputon, bool therlogscale, largecounter
      Nsteps)
00425         : Singularities{std::move(sings)}, Epsilon{eps}, OutputToConsole{consoleoutputon},
00426           rLogScale{therlogscale},
```

```
00427           TerminationOptions(Nsteps) {}
00428
00430     const std::vector<Singularity> Singularities;
00432     const real Epsilon;
00434     const bool OutputToConsole;
00436     const bool rLogScale;
00437 };
00438
00440 // Add your new TerminationOptions struct here, inheriting from TerminationOptions (if needed)
00441 // Sample code:
00442 /*
00443 struct MyTermOptions : public TerminationOptions
00444 {
00445 public:
00446     MyTermOptions(..., largecounter Nsteps) : TerminationOptions(Nsteps) //, other initialization
00447     {}
00448
00449     // member variables (const!) here
00450 };
00451 */
00453
00454 #endif
```

7.43 FOORT/src/Utilities.cpp File Reference

```
#include "Utilities.h"
#include <sstream>
#include <iomanip>
```

7.43.1 Detailed Description

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Version

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Date

2025-01-14

Copyright

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7.44 FOORT/src/Utilities.h File Reference

Declarations of various utility functions, in a Utilities namespace.

```
#include "Metric.h"
#include "Diagnostics.h"
#include "Terminations.h"
#include "Geodesic.h"
#include "ViewScreen.h"
#include "Integrators.h"
#include <chrono>
#include <string>
#include <vector>
```

Classes

- class Utilities::Timer

  Timer class to keep track of elapsed time.

Namespaces

- namespace Utilities

  Namespace that contains our utility functions.

Functions

- std::string Utilities::GetTimeStampString ()

  Get a string of the current time (in a format that can be used to append to file names)
- std::vector< std::string > Utilities::GetDiagNameStrings (DiagBitflag alldiags, DiagBitflag valdiag)

  Helper function to get all Diagnostic Names (for outputting to files)
- std::string Utilities::GetFirstLineInfoString (const Metric *theMetric, const Source *theSource, DiagBitflag alldiags, DiagBitflag valdiag, TermBitflag allterms, const ViewScreen *theView)

  This returns the full string to be written to every output file as its first line.

7.44.1 Detailed Description

Declarations of various utility functions, in a Utilities namespace.

Author

Daniel R. Mayerson

Version

0.1

Date

2025-01-14

Copyright

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7.45 Utilities.h

Go to the documentation of this file.

```
00001 #ifndef _FOORT_UTILITIES_H
00002 #define _FOORT_UTILITIES_H
00003
00012
00013 // We use Metric, Diagnostic, Termination, Geodesic, ViewScreen, and Integrator declarations here
00014 #include "Metric.h"
00015 #include "Diagnostics.h"
00016 #include "Terminations.h"
00017 #include "Geodesic.h"
00018 #include "ViewScreen.h"
00019 #include "Integrators.h"
00020
00021 #include <chrono> // for timer functionality
00022 #include <string> // for strings
00023 #include <vector> // for std::vector
00024
00026 namespace Utilities
00027 {
00031     class Timer
00032     {
00033     private:
00035         using Clock = std::chrono::steady_clock;
00037         using Second = std::chrono::duration<double, std::ratio<1>>;
00038
00040         std::chrono::time_point<Clock> m_beg{Clock::now()};
00041
00042     public:
00043         // reset begin time
00044         void reset();
00045
00046         // returns time elapsed since begin time
00047         double elapsed() const;
00048     };
00049
00050     // Returns a string of the current time (in a format that can be used to append to file names)
00051     std::string GetTimeStampString();
00052
00053     // Helper function to get all Diagnostic Names (for outputting to files)
00054     std::vector<std::string> GetDiagNameStrings(DiagBitflag alldiags, DiagBitflag valdiag);
00055
00056     // This returns the full string to be written to every output file as its first line
00057     // It contains information about all the settings used to produce the output
00058     std::string GetFirstLineInfoString(const Metric *theMetric, const Source *theSource,
00059                                        DiagBitflag alldiags, DiagBitflag valdiag, TermBitflag
      allterms, const ViewScreen *theView);
00060 }
00061
00062 #endif
```

7.46 FOORT/src/ViewScreen.cpp File Reference

```
#include "ViewScreen.h"
#include <algorithm>
```

7.46.1 Detailed Description

Author

Daniel R. Mayerson

Version

0.1

Date

2025-01-14

Copyright

Copyright (c) 2025

7.47 FOORT/src/ViewScreen.h File Reference

Declarations of ViewScreen class.

```
#include "Geometry.h"
#include "Metric.h"
#include "Mesh.h"
#include <memory>
#include <utility>
#include <array>
#include <string>
```

Classes

- class ViewScreen

  This class is in charge of converting a pixel on the screen into physical initial conditions for a geodesic.

Enumerations

- enum class GeodesicType { Null = 0 , Timelike = -1 , Spacelike = 1 }

  Type of geodesic being integrated.

7.47.1 Detailed Description

Declarations of ViewScreen class.

Author

Daniel R. Mayerson

This class is in charge of converting a pixel on the screen (which the Mesh wants to integrate) into physical initial conditions for the position and velocity of a geodesic.

Version

0.1

Date

2025-01-14

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7.47.2 Enumeration Type Documentation

7.47.2.1 GeodesicType

```
enum class GeodesicType  [strong]
```

Type of geodesic being integrated.

Null is the default, Timelike is -1, Spacelike is 1. Only Null is supported/implemented at the moment.

Enumerator

Null	
Timelike	
Spacelike	

7.48 ViewScreen.h

Go to the documentation of this file.

```
00001 #ifndef _FOORT_VIEWSCREEN_H
00002 #define _FOORT_VIEWSCREEN_H
00003
00013
00014 #include "Geometry.h" // For basic tensor objects
00015 #include "Metric.h"   // For the Metric object
00016 #include "Mesh.h"   // For the Mesh object
00017
00018 #include <memory>  // std::unique_ptr
00019 #include <utility> // std::move
00020 #include <array>   // std::array
00021 #include <string>  // strings
00022
00027 enum class GeodesicType
00028 {
00029     Null = 0,
00030     Timelike = -1,
00031     Spacelike = 1,
00032 };
00033
00039 class ViewScreen
00040 {
00041 public:
00042     // No default constructor possible
00043     ViewScreen() = delete;
00044     // constructor must pass following arguments along:
00045     // - physical position and looking direction;
00046     // - screen dimensions (in dimensions of length)
00047     // - the Mesh used (ViewScreen must become a owner of this object!)
00048     // - the Metric used (ViewScreen is NOT the owner of the Metric)
00049     // - the geodesic type to be integrated (null, timelike, spacelike)
00050     ViewScreen(Point pos, OneIndex dir, ScreenPoint screensize, ScreenPoint screencenter,
00051                std::unique_ptr<Mesh> theMesh, const Metric *const theMetric, GeodesicType thegeodtype
      = GeodesicType::Null)
00052         : m_Pos{pos}, m_Direction{dir}, m_ScreenSize{screensize}, m_ScreenCenter{screencenter},
00053           m_theMesh{std::move(theMesh)},
00054           m_theMetric{theMetric}, m_GeodType{thegeodtype},
00055           m_rLogScale{theMetric->getrLogScale()}
00056     {
00057         // At the moment, we don't even use the direction; we are always pointed towards the origin
00058         if (m_Direction != Point{0, -1, 0, 0})
00059         {
00060             ScreenOutput("ViewScreen is only supported pointing inwards at the moment; Direction = {0,
      -1, 0, 0} will be used",
00061                          OutputLevel::Level_0_WARNING);
00062         }
00063         // At the moment, we are only integrating null geodesics
00064         if (m_GeodType != GeodesicType::Null)
00065         {
00066             ScreenOutput("ViewScreen only supports null geodesics at the moment; geodesics integrated
      will be null.",
```

```
00067                           OutputLevel::Level_0_WARNING);
00068         }
00069
00070         ScreenOutput("ViewScreen initialized with position " + toString(m_Pos) + ", looking direction
      " + toString(m_Direction) +
00071                           ", screen size " + toString(m_ScreenSize) + ", screen center " +
      toString(m_ScreenCenter) +
00072                           ", geodesic type " + (m_GeodType == GeodesicType::Null ? "Null" : (m_GeodType
      == GeodesicType::Timelike ? "Timelike" : "Spacelike")) +
00073                           ", and metric: " + m_theMetric->getFullDescriptionStr(),
00074                       OutputLevel::Level_4_DEBUG);
00075         // Construct the vielbein now
00076         ConstructVielbein();
00077     }
00078
00079     // Heart of the ViewScreen: here, the ViewScreen is asked to provide initial conditions
00080     // for the geodesic nr index of the current iteration; based on the screen index
00081     // that the Mesh gives, it sets up these physical initial conditions.
00082     void SetNewInitialConditions(largecounter index, Point &pos, OneIndex &vel, ScreenIndex &scrIndex)
      const;
00083
00084     // These member functions essentially pass on information to/from the Mesh
00085     bool IsFinished() const;              // Does the ViewScreen (i.e. the Mesh) want to integrate
      more geodesics or not?
00086     largecounter getCurNrGeodesics() const; // Current number of geodesics in this iteration
00087     void EndCurrentLoop();                    // The current iteration of geodesics is finished;
      prepare the next one
00088     // NOTE: despite not being const, this function has been designed to be threadsafe!
00089     void GeodesicFinished(largecounter index, std::vector<real> finalValues); // This geodesic has
      been integrated, returning its final "values"
00090
00091     // Description string getter (spaces allowed), also will contain information about the Mesh
00092     std::string getFullDescriptionStr() const;
00093
00094 private:
00096     TwoIndex m_Metric_dd{};
00098     TwoIndex m_Vielbein{};
00099
00100     // Helper function to construct the vielbein given the metric
00101     void ConstructVielbein();
00102
00104     const Point m_Pos;
00106     const OneIndex m_Direction;
00108     const ScreenPoint m_ScreenSize;
00110     const ScreenPoint m_ScreenCenter;
00111
00113     const bool m_rLogScale;
00114
00116     const Metric *const m_theMetric;
00117
00119     const GeodesicType m_GeodType{GeodesicType::Null};
00120
00122     const std::unique_ptr<Mesh> m_theMesh;
00123 };
00124
00125 #endif
```

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